

Display CMC's

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00415	155S00391	ALT_PARTS	ALL	CMC, 350SM, 7HGz, MUR

NAND Ultimate

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00340	1	HYNIX, 3DV4, ULTIMATE	U2600	CRITICAL	ULTIMATE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00359	335S00340	ALT_PARTS	U2600	TOSHIBA, BICS3, ULT
335S00286	335S00340	ALT_PARTS	U2600	SANDISK, BICS3, ULT
335S00288	335S00340	ALT_PARTS	U2600	SAMSUNG, 3DV4, ULT

Extreme

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00342	1	HYNIX, 3DV4, Extreme	U2600	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00247	335S00342	ALT_PARTS	U2600	SANDISK, BISC3, SUPREME
335S00276	335S00342	ALT_PARTS	U2600	SAMSUNG, 3DV4, SUPREME
335S00358	335S00342	ALT_PARTS	U2600	TOSHIBA, 3DV4, SUPREME

Max

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00343	1	HYNIX, 3DV4, MAX	U2600	CRITICAL	MAX

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00339	335S00343	ALT_PARTS	U2600	SAMSUNG, 3DV4, MAX

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	ALT_PARTS	ALL	0402-3T, 10.5uF@1V, Kyocera
138S00150	138S00149	ALT_PARTS	ALL	0402-3T, 10.5uF@1V, SENC0
138S00151	138S00149	ALT_PARTS	ALL	0402-3T, 10.5uF@1V, TY

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	ALT_PARTS	ALL	0402, 16uF@1V, Kyocera
138S00163	138S00144	ALT_PARTS	ALL	0402, 16uF@1V, TY

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	ALT_PARTS	ALL	0201, 3uF@1V, Kyocera
138S00164	138S00139	ALT_PARTS	ALL	0201, 3uF@1V, TY

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00221	138S00146	ALT_PARTS	ALL	0402, 5.1uF@3V, Kyocera

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	ALT_PARTS	ALL	0201, 1.1uF@3V, Kyocera
138S00142	138S00141	ALT_PARTS	ALL	0201, 1.1uF@3V, SENC0
138S00166	138S00141	ALT_PARTS	ALL	0201, 1.1uF@3V, Taiyo

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00648	138S00652	ALT_PARTS	ALL	CAP, X5R, 4.7UF, 5. 3V, 0. 60PM, 0402, TAIYO
138S0739	138S0706	ALT_PARTS	ALL	CAP, X5R, X5R, 0. 22UF, 20V, 0. 3V, 20V
138S00049	138S0031	ALT_PARTS	ALL	CAP, C0G, X5R, 0. 22UF, 20V, 0. 3V, 500V

Yangtze Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00872	152S00918	ALT_PARTS	ALL	IND, MLD, 0. 47UH, TDK
152S00847	152S00918	ALT_PARTS	ALL	IND, MLD, 0. 47UH, CYN

CRITICAL PART#	COMMENT
152S00918	IND, MLD, 0. 22UH, 20V, 5. 0A, 40MOHM, H=-. 65, 1608

Denali Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00878	152S00831	ALT_PARTS	ALL	IND, MLD, 0. 22UH, 20V, 5. 0A, 30MOHM, H=-. 65, 1608
152S00818	152S00831	ALT_PARTS	ALL	IND, MLD, 0. 22UH, 20V, 5. 0A, 30MOHM, H=-. 65, 1608, CYN
152S00835	152S00822	ALT_PARTS	ALL	IND, MLD, 0. 47UH, 20V, 4. 5A, 40MOHM, H=-. 80, 2012
152S00827	152S00822	ALT_PARTS	ALL	IND, MLD, 0. 47UH, 20V, 4. 5A, 50MOHM, H=-. 80, 2012
152S00877	152S00817	ALT_PARTS	ALL	IND, MLD, 0. 1UH, 20V, 0A, 20MOHM, H=-. 0, 2012
152S00829	152S00817	ALT_PARTS	ALL	IND, MLD, 0. 1UH, 20V, 0. 0A, 20MOHM, H=-. 0, 2012
152S00825	152S00823	ALT_PARTS	ALL	IND, MLD, 1UH, 20V, 0A, 60MO, H=-. 80, 2012
152S00833	152S00819	ALT_PARTS	ALL	IND, MLD, 1UH, 20V, 2. 1A, 60MO, H=-. 65, 2012
152S00824	152S00819	ALT_PARTS	ALL	IND, MLD, 1UH, 20V, 2A, 60MO, H=-. 80, 2012
152S00834	152S00820	ALT_PARTS	ALL	IND, MLD, 0. 47UH, 20V, 3. 0A, 270MO, H=-. 80, 2012
152S00828	152S00820	ALT_PARTS	ALL	IND, MLD, 0. 47UH, 20V, 3. 2A, 400MO, H=-. 80, 2012
152S00826	152S00821	ALT_PARTS	ALL	IND, MLD, 1UH, 20V, 2. 1A, 60MO, H=-. 80, 2012
152S00866	152S00821	ALT_PARTS	ALL	IND, MLD, 1UH, 20V, 2. 1A, 60MO, H=-. 80, 2012

CRITICAL PART#	COMMENT
152S00831	IND, MLD, 0. 22UH, 20V, 5. 0A, 40MOHM, H=-. 65, 1608
152S00822	IND, MLD, 0. 47UH, 20V, 4. 5A, 47MOHM, H=-. 8, 2012
152S00817	IND, MLD, 0. 1UH, 20V, 9. 4A, 22MOHM, H=0. 65, 1608
152S00823	IND, MLD, 1UH, 20V, 3. 0A, 60MO, H=-. 65, 2016
152S00819	IND, MLD, 1UH, 20V, 1. 7A, 60MO, H=-. 65, 2012
152S00820	IND, MLD, 0. 47UH, 20V, 3. 2A, 42MO, H=-. 80, 2012
152S00821	IND, MLD, 1UH, 20V, 2. 2A, 60MO, H=-. 80, 2012

XTAL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S00612	197S00118	ALT_PARTS	Y1000	XTAL, 24M, 1612
197S00120	197S00118	ALT_PARTS	Y1000	XTAL, 24M, 1612

NEON Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00721	152S00876	ALT_PARTS	L41200, L41210	TY, 1M0

ANSEL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00716	152S00875	ALT_PARTS	L3700	TY, 1M0

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB_TOP, 639-03991, ULTIMATE)	EEEE_HMW1	CRITICAL	ULTIMATE
825-7691	1	EEEE FOR (MLB_TOP, 639-03992, EXTREME)	EEEE_HMW2	CRITICAL	EXTREME
825-7691	1	EEEE FOR (MLB_TOP, 639-03996, MAX)	EEEE_HMW0	CRITICAL	MAX

Cyprus OMIT

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00510	1	CYPRUS 4GB Micron	U1000	CRITICAL	SOC

Cyprus ALTs

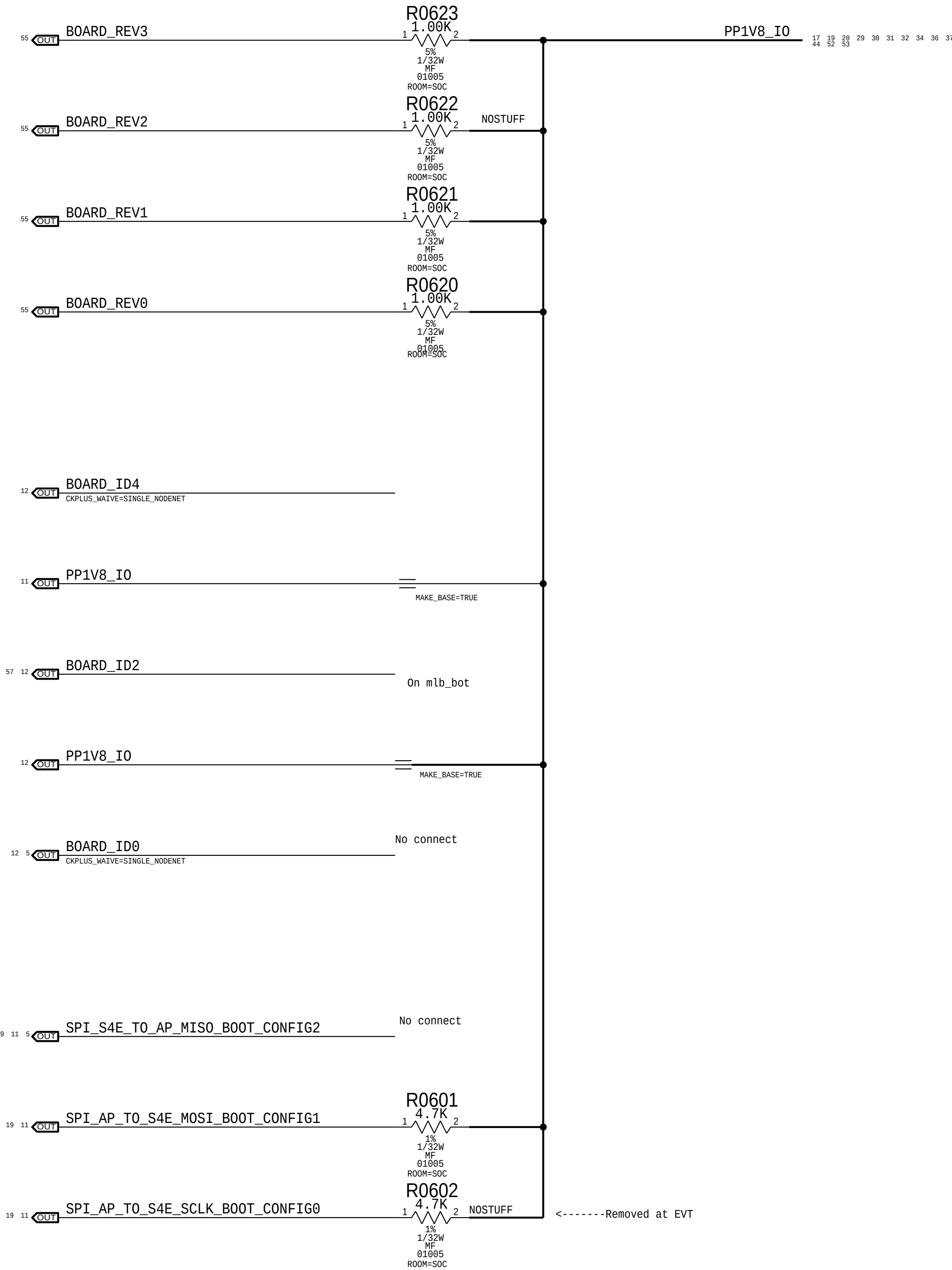
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00511	339S00510	ALT_PARTS	U1000	CYPRUS 4GB Hynix
339S00512	339S00510	ALT_PARTS	U1000	CYPRUS 4GB Samsung

Combo Stiffener

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
604-19651	1	Combo Stiffener	ST0401	CRITICAL	ALL

TOP BOARD ONLY CONFIGURATION IS D33 MLB MAV
BOTTOM BOARD SELECTS ICE/MAV and D32/D33

BOOTSTRAPPING:BOARD REV
BOARD ID
BOOT CONFIG



SELECTED -->

DEFAULT -->

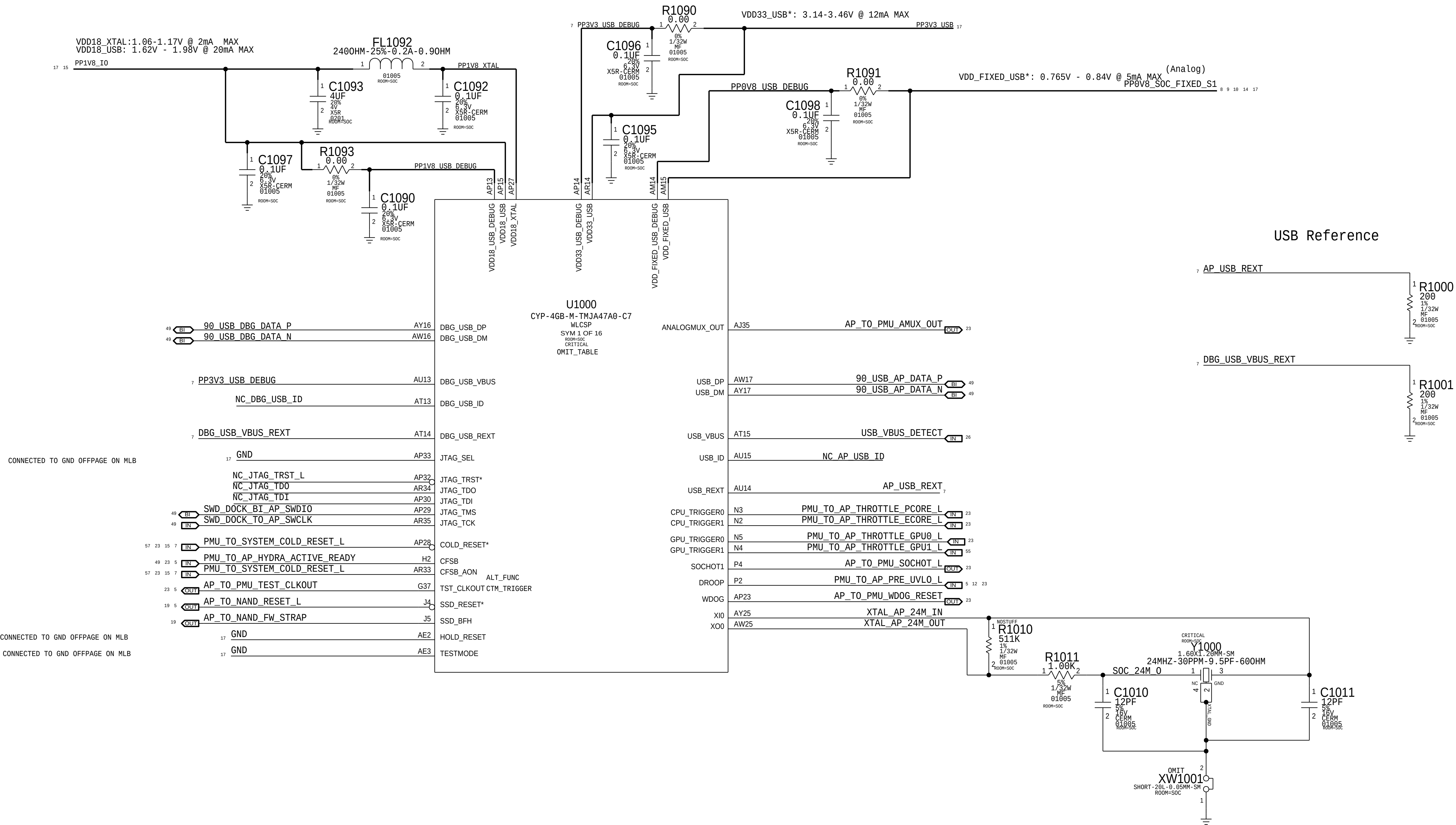
POR -->

Board Rev[3:0]				
Float = Low PU = High				
	3	2	1	0
Build Major		Build Minor		
Proto 1	1	1	1	1
(Spare)	1	1	1	0
Proto 2	1	1	0	1
(Spare)	1	1	0	0
EVT	1	0	1	1
(Spare)	1	0	1	0
Carrier	0	1	1	1
(Spare)	0	1	1	0
DVT	0	0	1	1
(Spare)	0	0	1	0
(Spare)	0	0	0	1
PVT	0	0	0	0

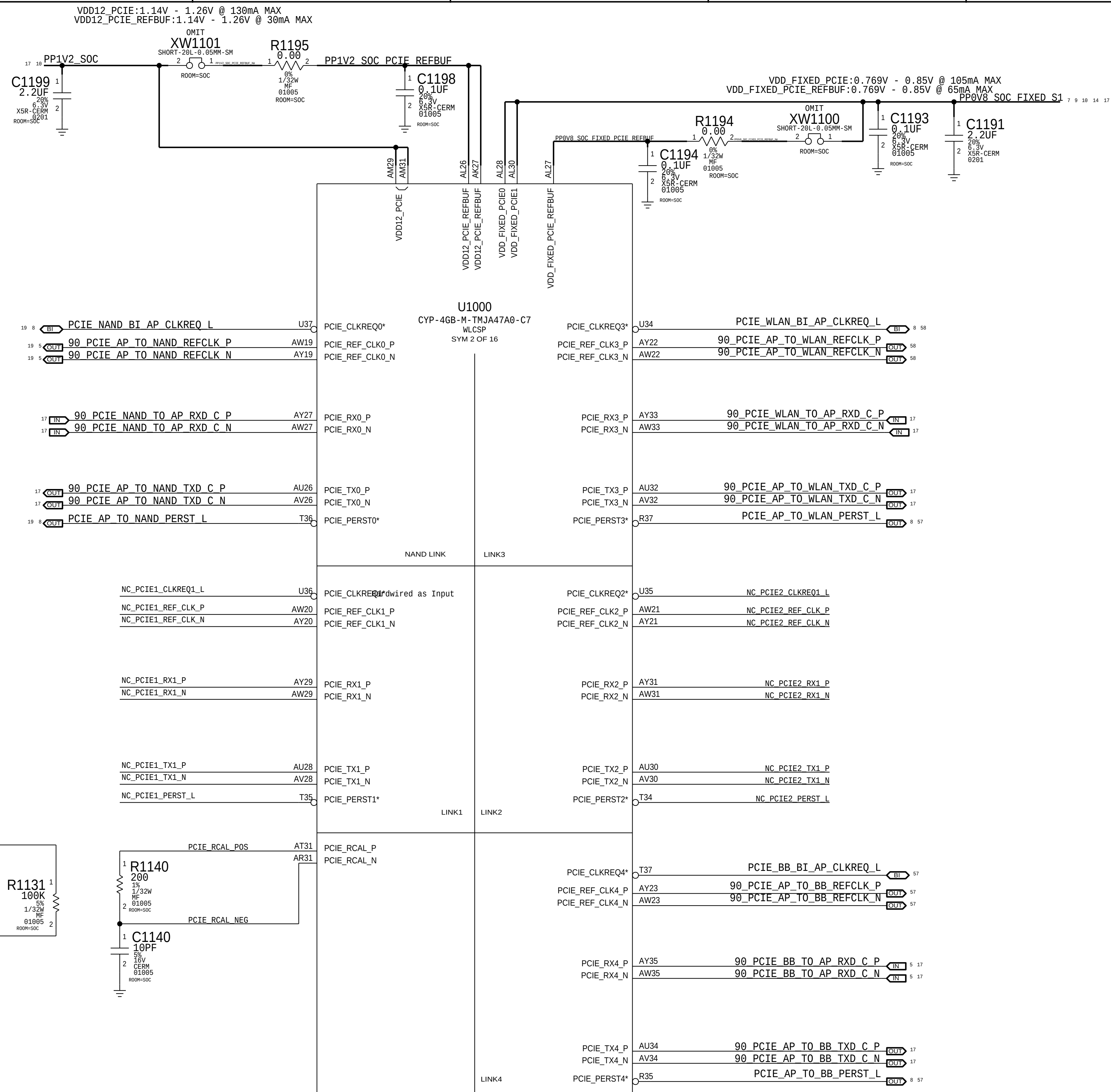
Board ID[4:0]					
Float = Low PU = High					
	4	3	2	1	0
D32 MLB	0	0	1	1	0
D32p MLB	1	0	1	1	0
D32 Dev	0	0	1	1	1
D32p Dev	1	0	1	1	1
D321 MLB	0	1	1	1	0
D321p MLB	1	1	1	1	0
D321 Dev	0	1	1	1	1
D321p Dev	1	1	1	1	1
D33 MLB	0	0	0	1	0
D33p MLB	1	0	0	1	0
D33 Dev	0	0	0	1	1
D33p Dev	1	0	0	1	1
D331 MLB	0	1	0	1	0
D331p MLB	1	1	0	1	0
D331 Dev	0	1	0	1	1
D331p Dev	1	1	0	1	1

Boot Config [2:0]			
Float = Low PU = High			
	2	1	0
SPI NOR on SPI0 12 MHz	0	0	0
SPI NOR on SPI0 12 MHz Test	0	0	1
SPI NAND on SPI0 12 MHz	0	1	0
SPI NAND on SPI0 12 MHz Test	0	1	1
SPI NOR on SPI0 40 MHz	1	0	0
SPI NOR on SPI0 40 MHz Test	1	0	1
SPI NOR on SPI0 6 MHz	1	1	0
SPI NOR on SPI0 6 MHz Test	1	1	1

SOC - USB, JTAG, XTAL



SOC - PCIE

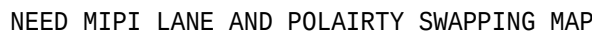


PCIE LINK 0

PCIE LINK 3

PCIE LINK 4

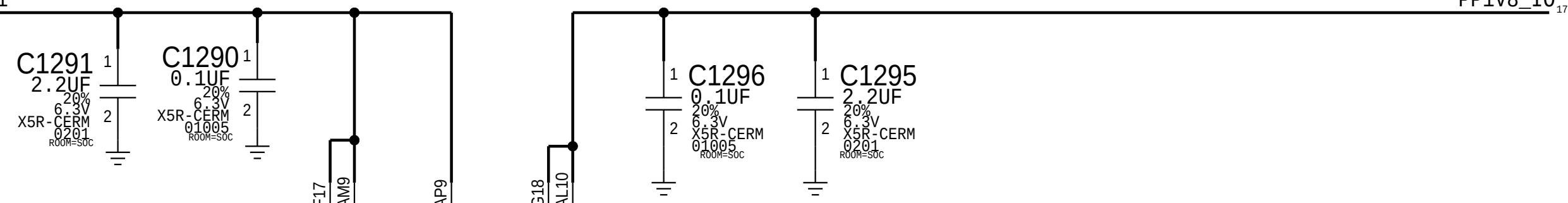
SOC - MIPI



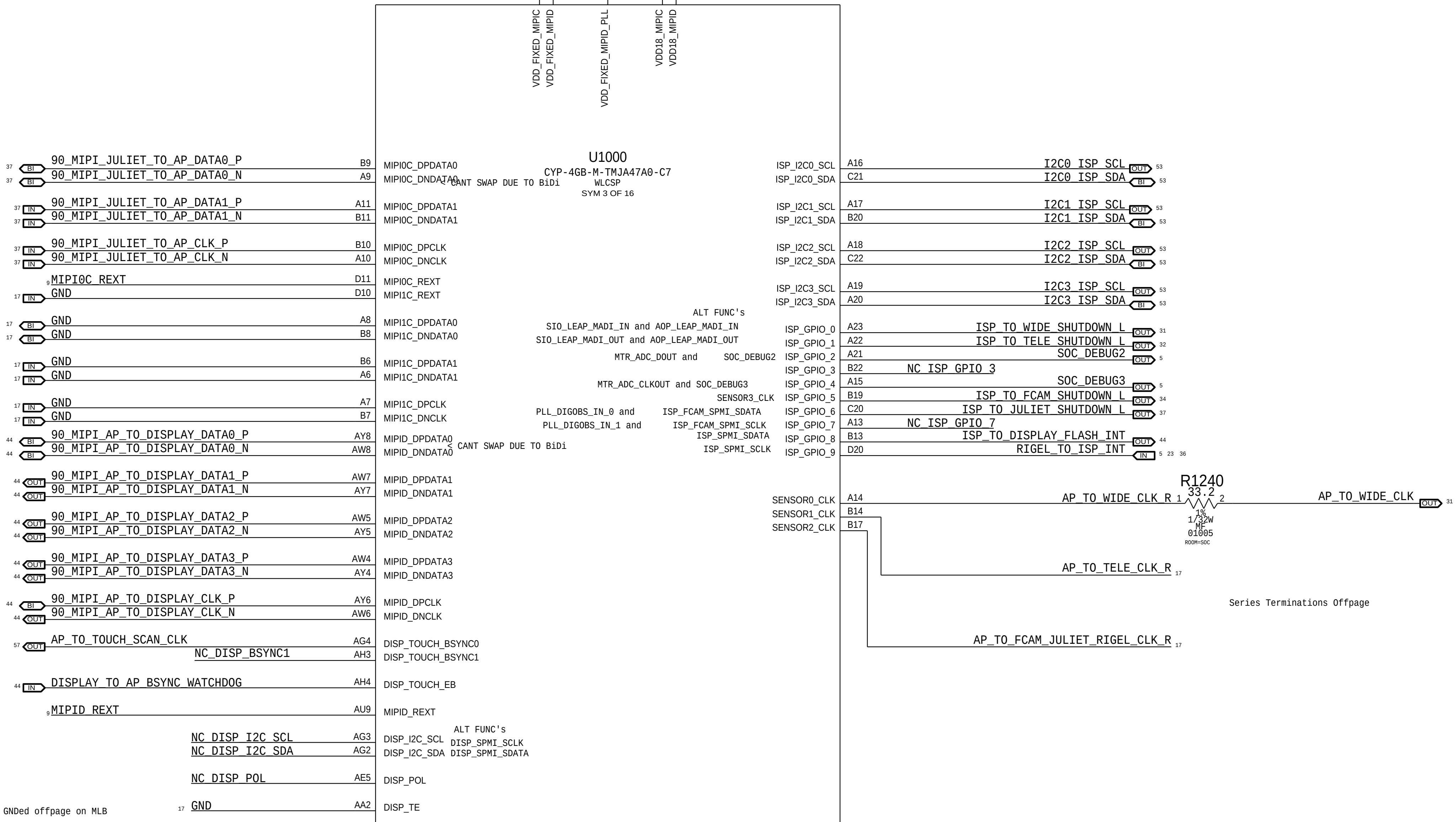
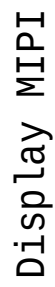
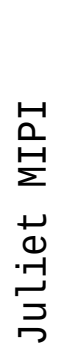
(Analog)

VDD_FIXED_MIPID	0.769V	-	0.85V	@	TBDmA	MAX
VDD_FIXED_MIPIC	0.769V	-	0.85V	@	TBDmA	MAX
VDD_FIXED_MIPID_PLL	0.769V	-	0.85V	@	TBDmA	MAX

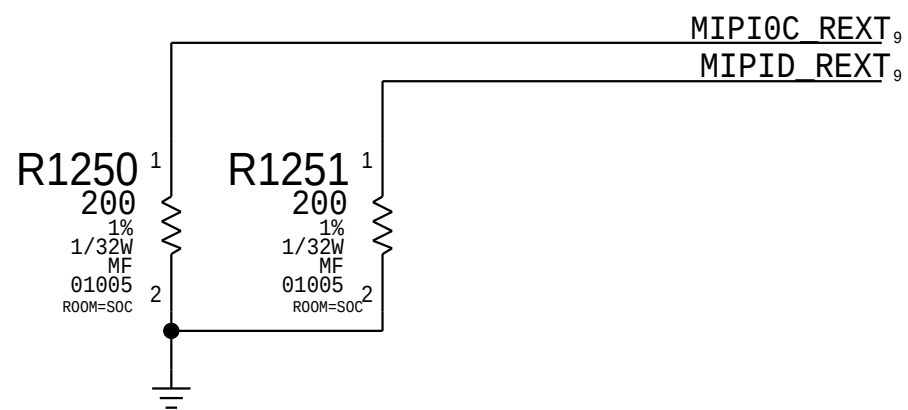
VDD18_MIPI*:1.62V - 1.98V @ TBDmA MAX



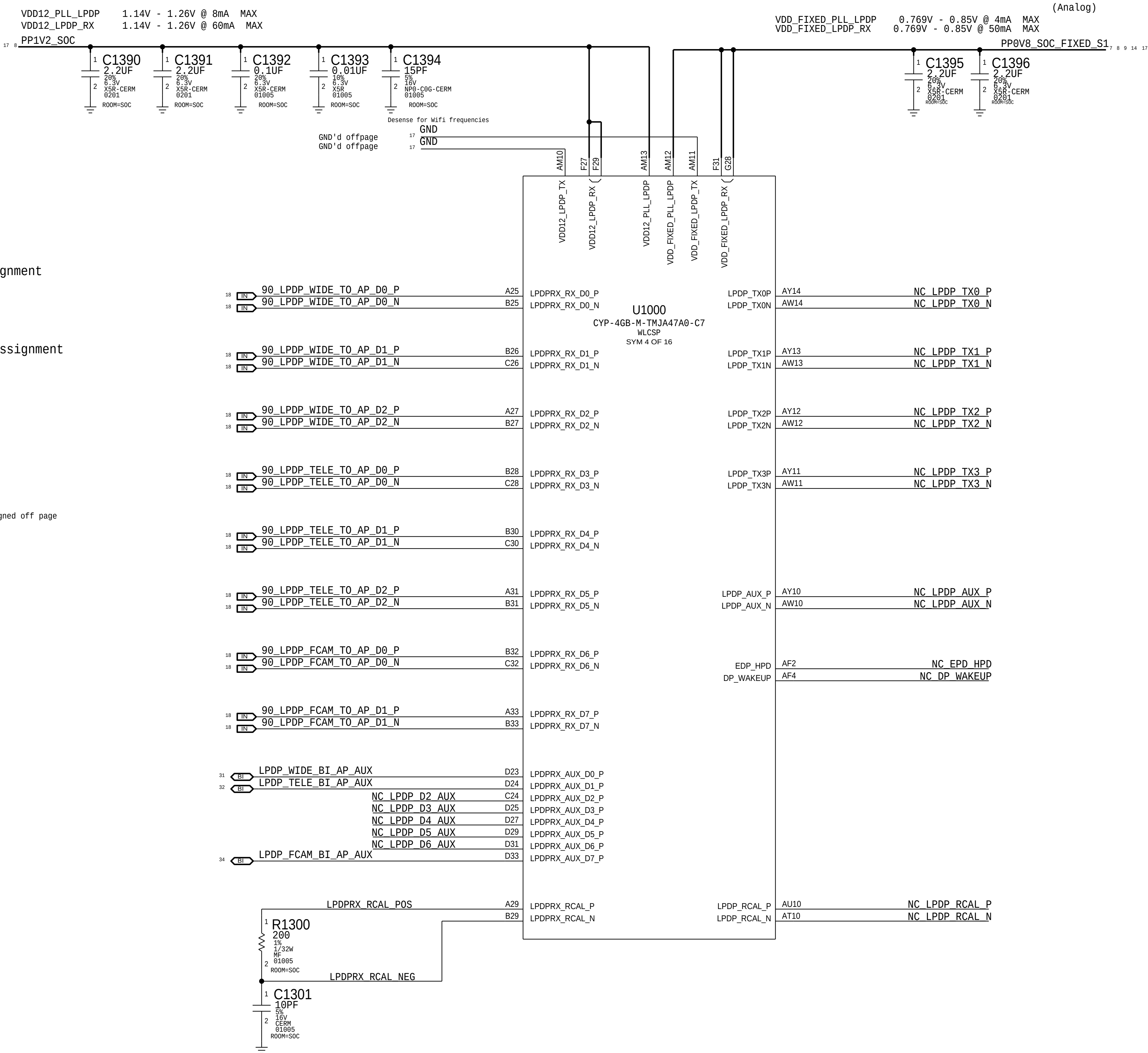
MIPI lanes can all flip polarity for routing purposes



MIPI Reference



SOC - LPDP



Dan LPDP Lane Assignment

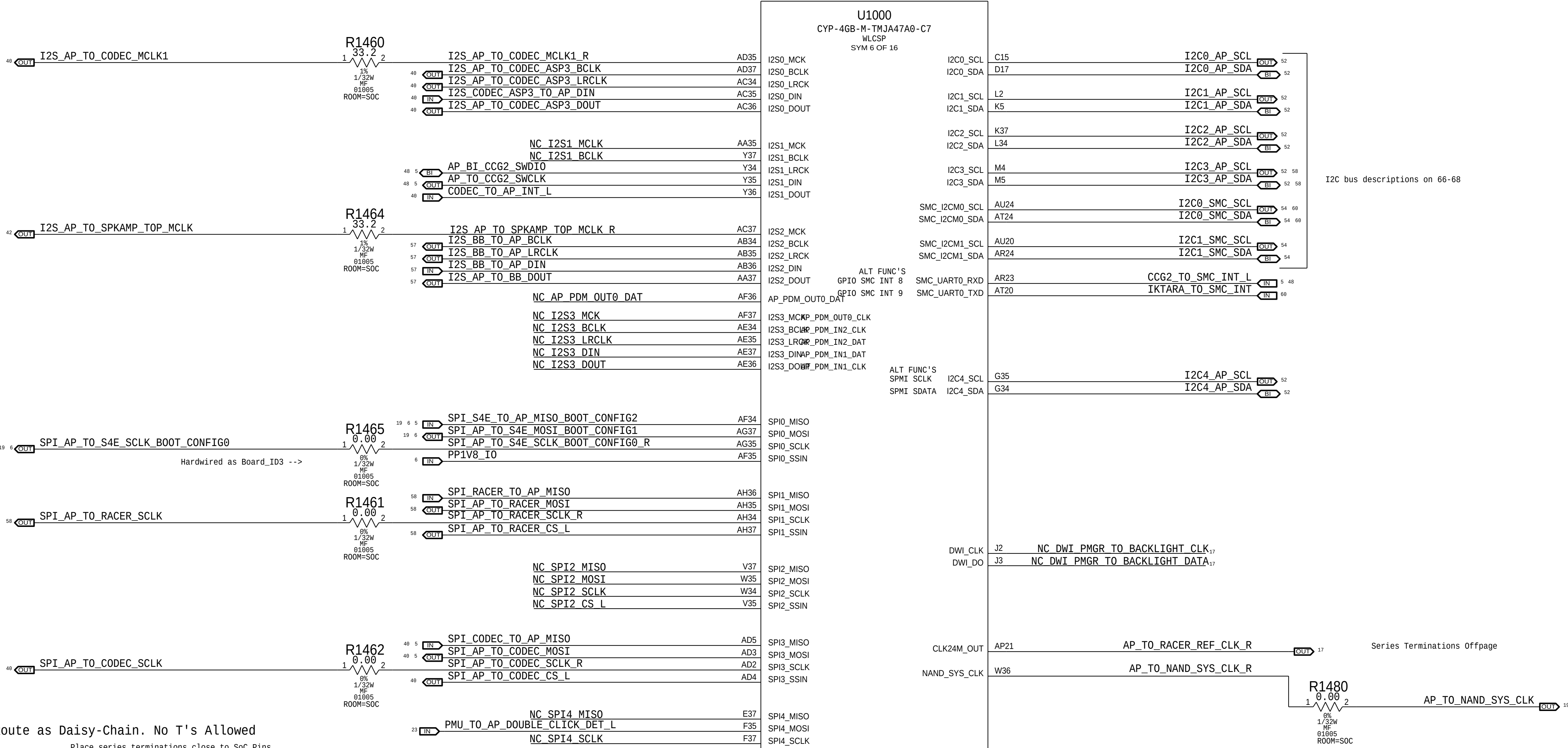
Wide: 0-2
Tele: 3-5
Fcam: 6-7

Justin LPDP Lane Assignment

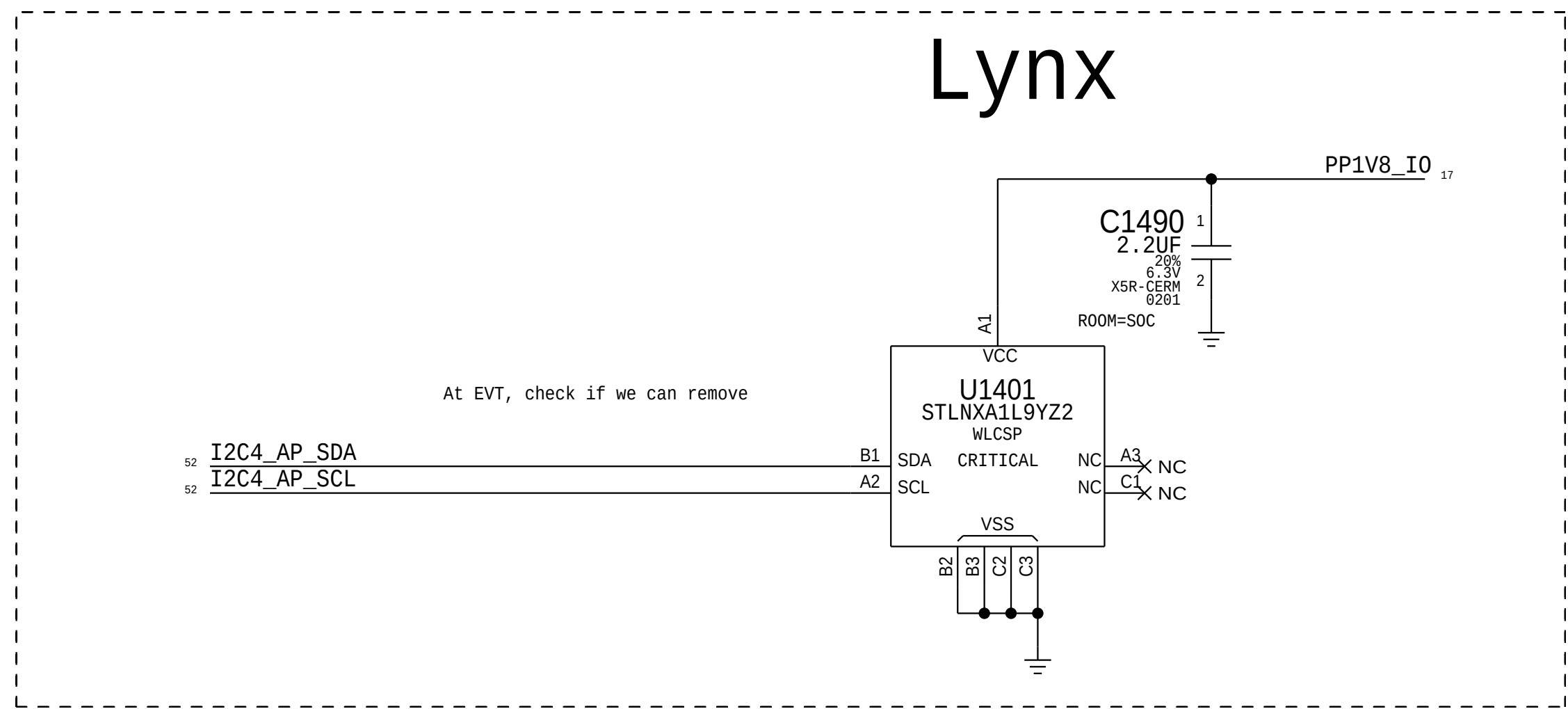
Wide: 2-4
Tele: 5-7
Fcam: 0-1

LPD Assigned off page

SOC - SERIAL INTERFACES

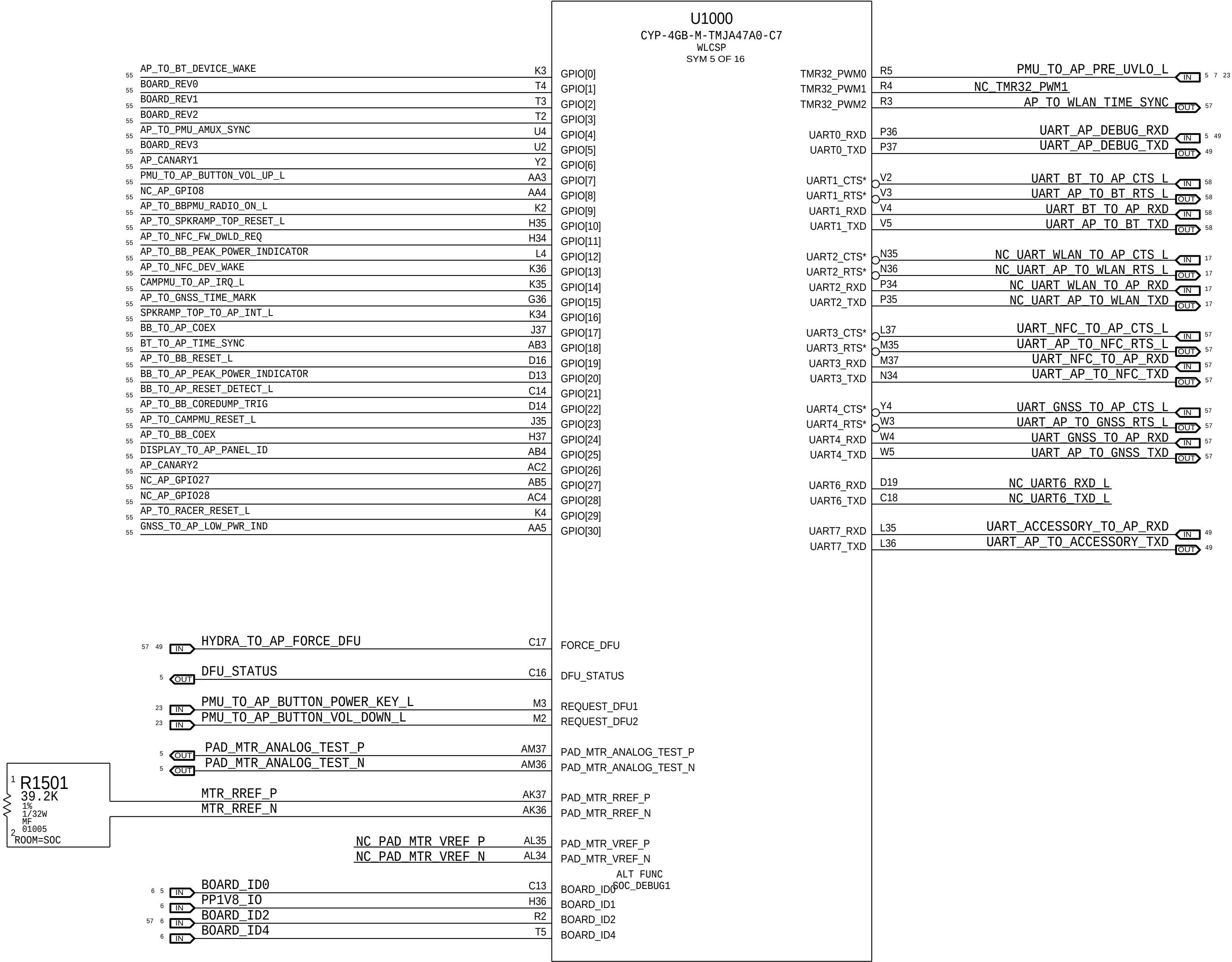


SPI: Route as Daisy-Chain. No T's Allowed
Place series terminations close to SoC Pins



SOC - GPIO INTERFACES

GPIOs are wired on page 70



SOC - AOP

D

C

B

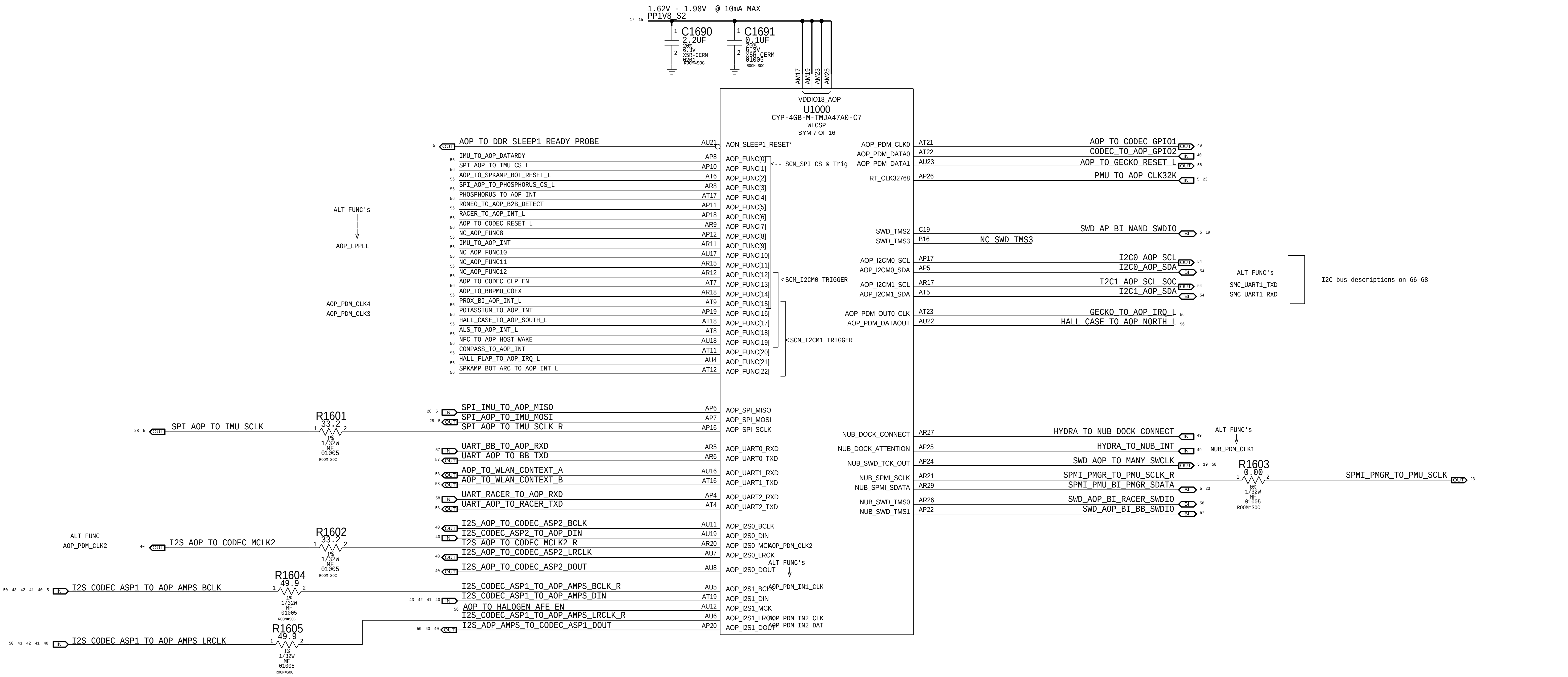
A

D

C

B

A



D

C

B

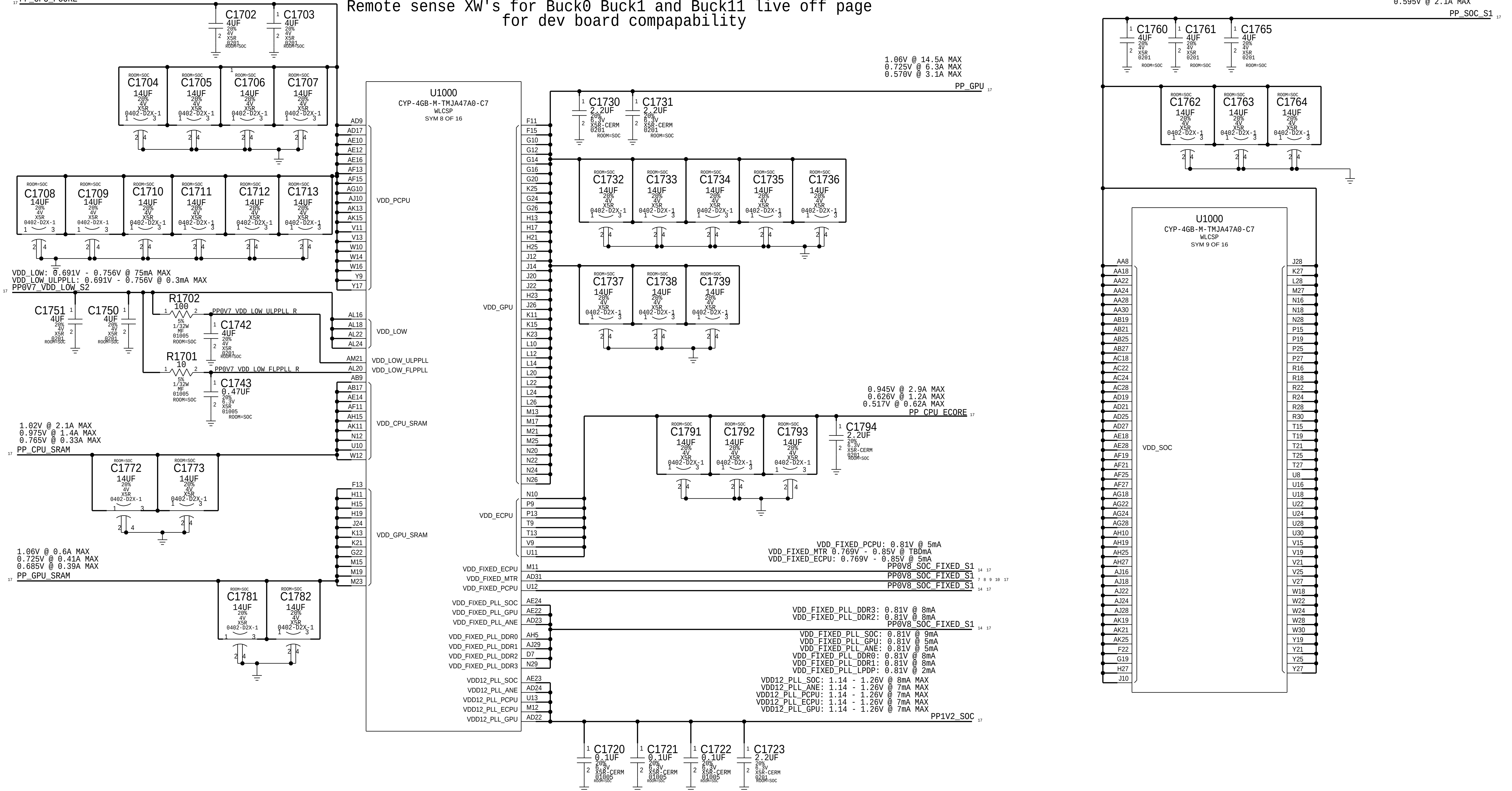
A

D

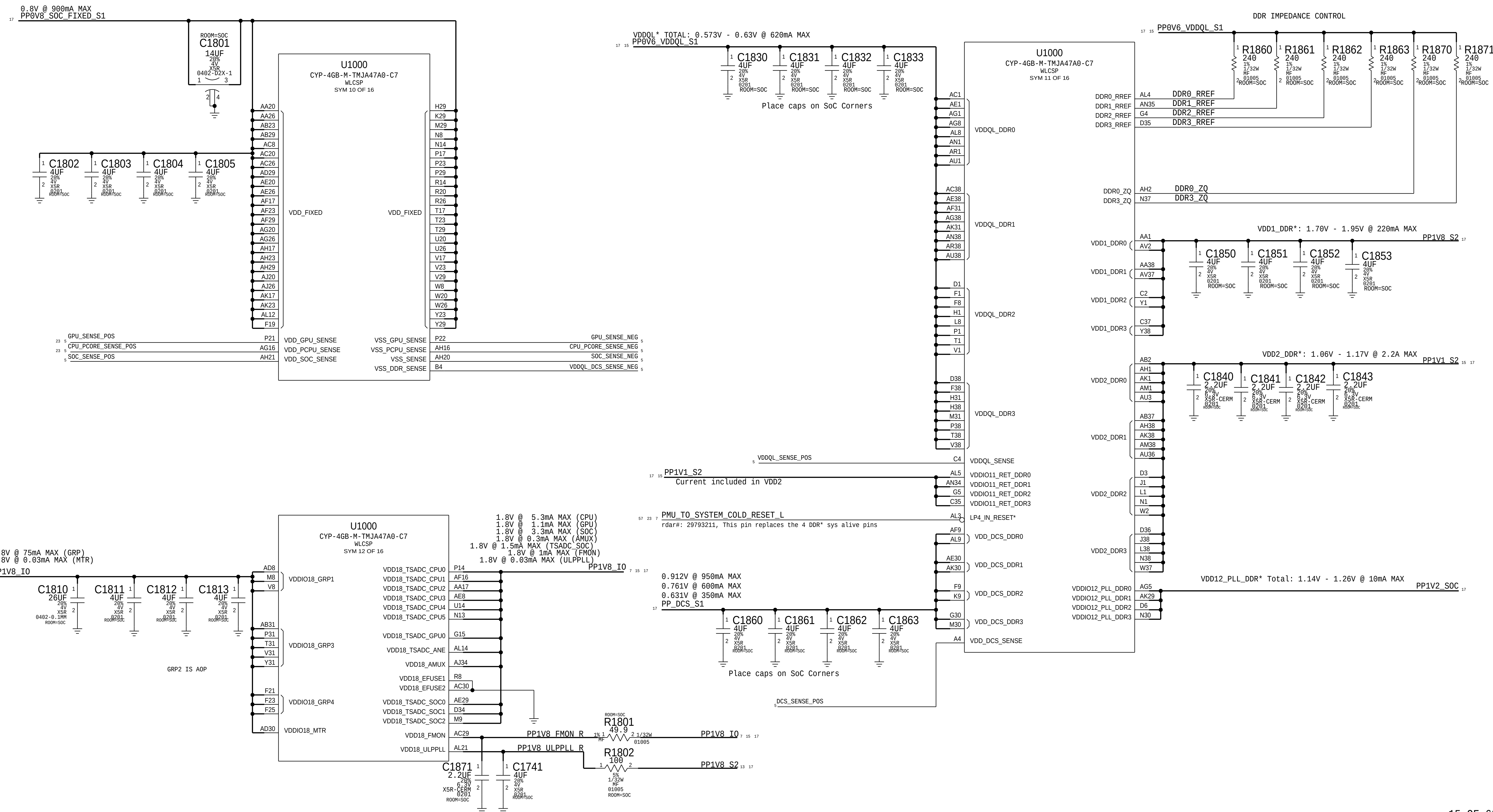
C

B

A



SOC - CPU, GPU & SOC RAILS



SOC - POWER SUPPLIES

D

C

B

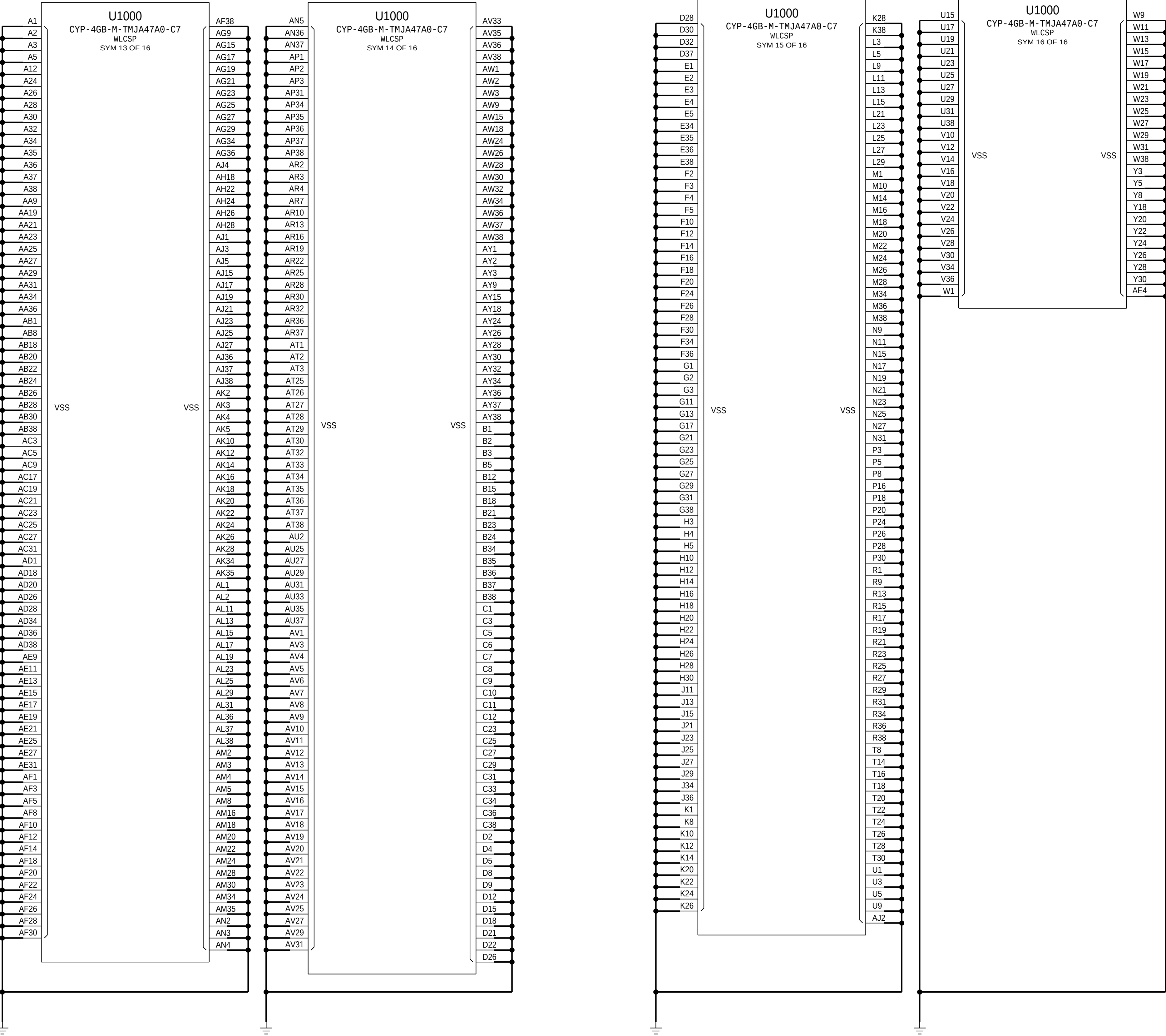
A

D

C

B

A

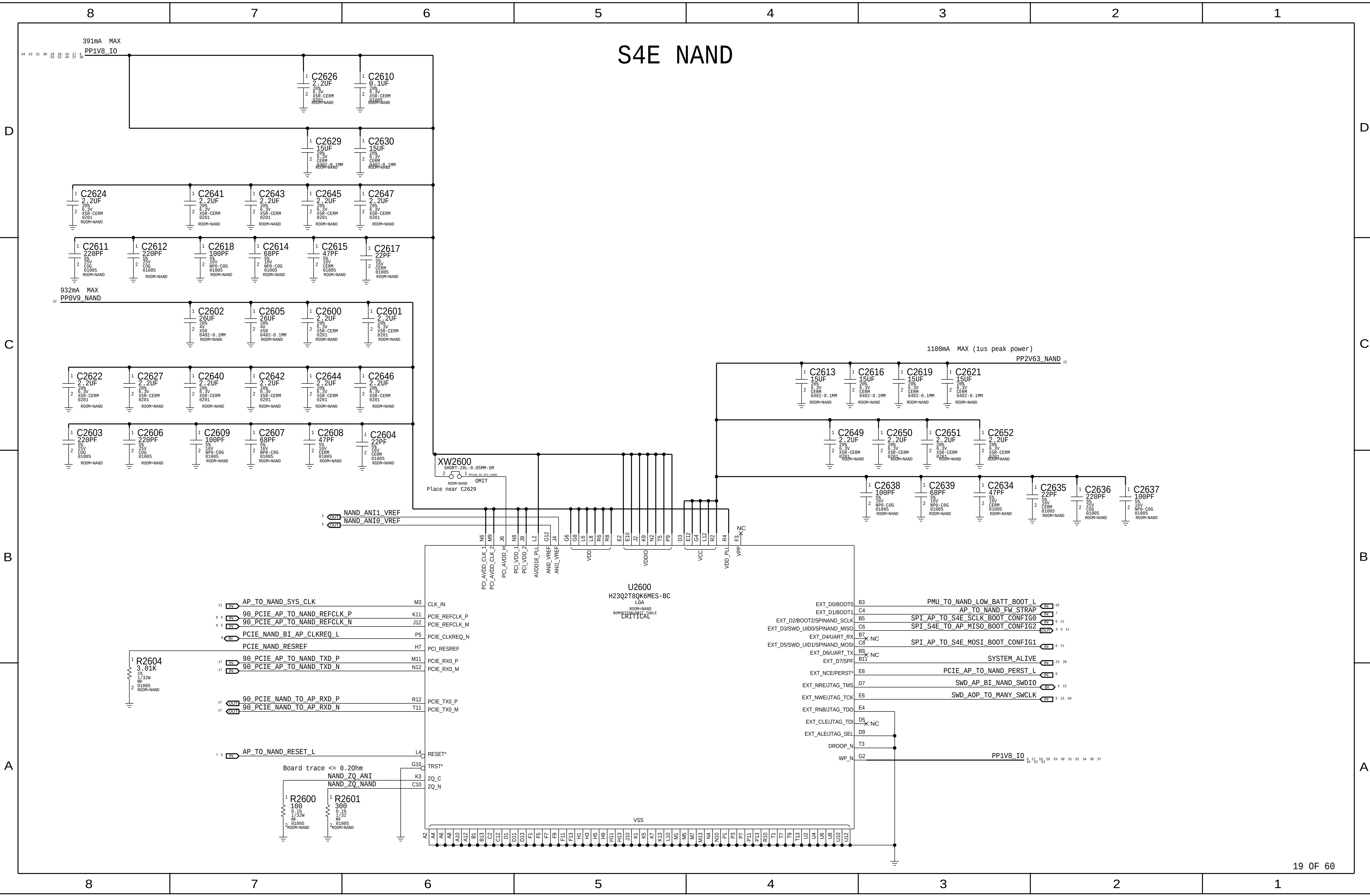


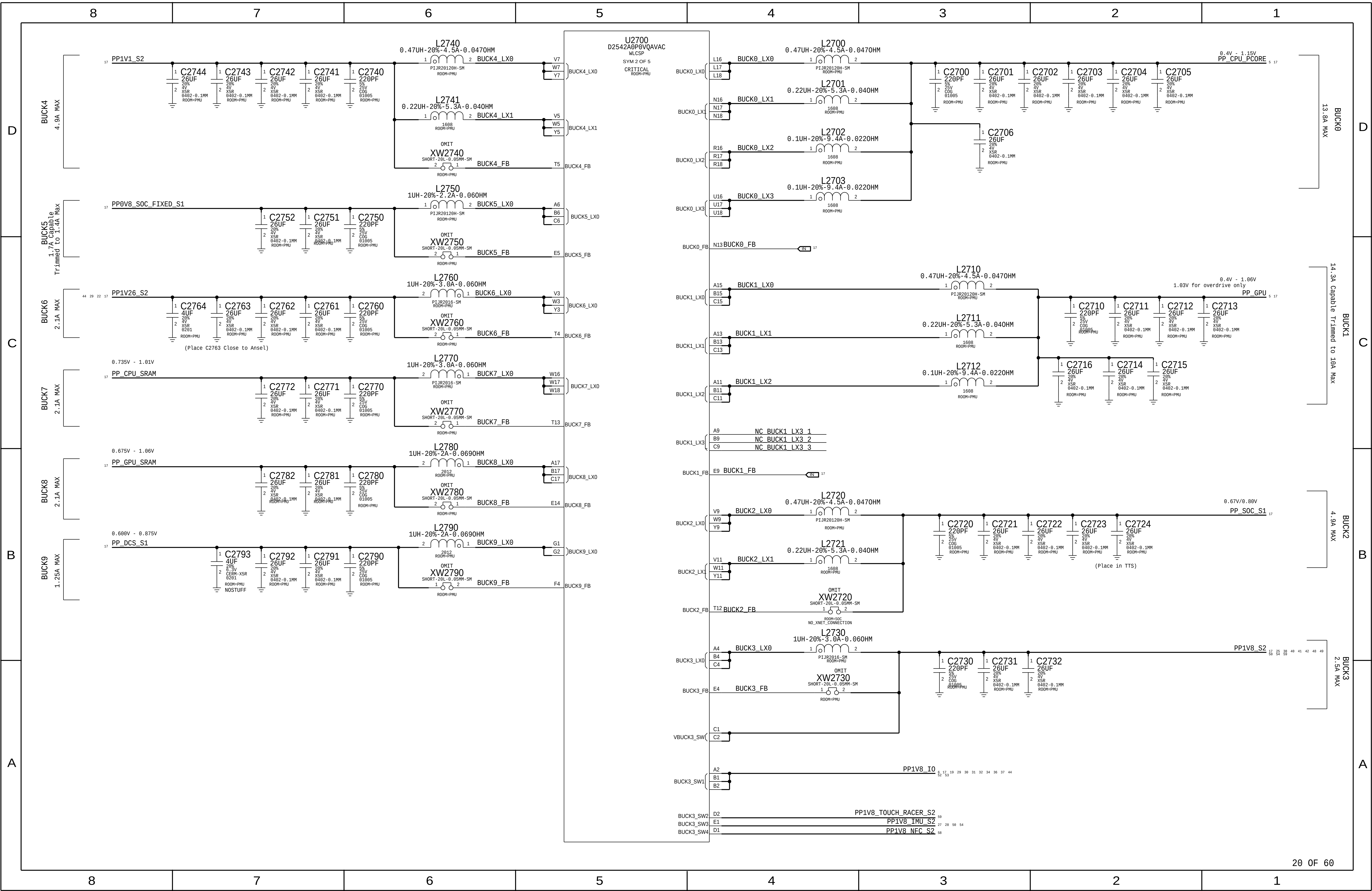
8		7		6		5		4		3		2		1													
D														D													
C														C													
		31 REV 90_LPDP_WIDE_TO_AP_D0_P 31 REV 90_LPDP_WIDE_TO_AP_D0_N MAKE_BASE=TRUE														10 90_LPDP_WIDE_TO_AP_D0_P 10 90_LPDP_WIDE_TO_AP_D0_N											
		31 REV 90_LPDP_WIDE_TO_AP_D1_P 31 REV 90_LPDP_WIDE_TO_AP_D1_N MAKE_BASE=TRUE														10 90_LPDP_WIDE_TO_AP_D1_P 10 90_LPDP_WIDE_TO_AP_D1_N											
		31 REV 90_LPDP_WIDE_TO_AP_D2_P 31 REV 90_LPDP_WIDE_TO_AP_D2_N MAKE_BASE=TRUE														10 90_LPDP_WIDE_TO_AP_D2_P 10 90_LPDP_WIDE_TO_AP_D2_N											
		32 REV 90_LPDP_TELE_TO_AP_D0_P 32 REV 90_LPDP_TELE_TO_AP_D0_N MAKE_BASE=TRUE														10 90_LPDP_TELE_TO_AP_D0_P 10 90_LPDP_TELE_TO_AP_D0_N											
		32 REV 90_LPDP_TELE_TO_AP_D1_P 32 REV 90_LPDP_TELE_TO_AP_D1_N MAKE_BASE=TRUE														10 90_LPDP_TELE_TO_AP_D1_P 10 90_LPDP_TELE_TO_AP_D1_N											
		32 REV 90_LPDP_TELE_TO_AP_D2_P 32 REV 90_LPDP_TELE_TO_AP_D2_N MAKE_BASE=TRUE														10 90_LPDP_TELE_TO_AP_D2_P 10 90_LPDP_TELE_TO_AP_D2_N											
		34 REV 90_LPDP_FCAM_TO_AP_D0_P 34 REV 90_LPDP_FCAM_TO_AP_D0_N MAKE_BASE=TRUE														10 90_LPDP_FCAM_TO_AP_D0_P 10 90_LPDP_FCAM_TO_AP_D0_N											
		34 REV 90_LPDP_FCAM_TO_AP_D1_P 34 REV 90_LPDP_FCAM_TO_AP_D1_N MAKE_BASE=TRUE														10 90_LPDP_FCAM_TO_AP_D1_P 10 90_LPDP_FCAM_TO_AP_D1_N											
A														A													

18 OF 60

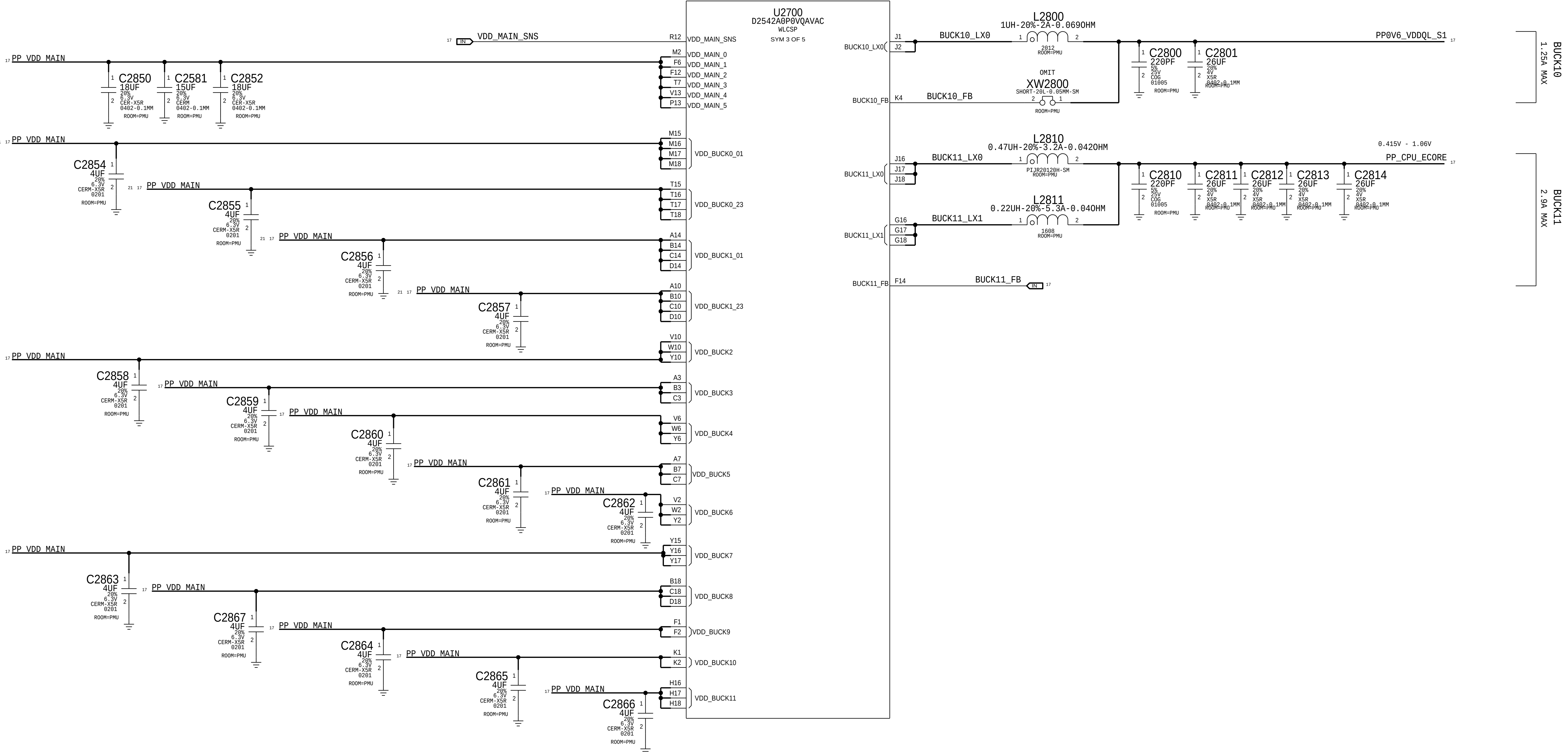
31	IN	90_LPDP_WIDE_TO_AP_D0_P	==	90_LPDP_WIDE_TO_AP_D0_P	10
		90_LPDP_WIDE_TO_AP_D0_N	MAKE_BASE=TRUE	90_LPDP_WIDE_TO_AP_D0_N	10
31	IN	90_LPDP_WIDE_TO_AP_D1_P	==	90_LPDP_WIDE_TO_AP_D1_P	10
		90_LPDP_WIDE_TO_AP_D1_N	MAKE_BASE=TRUE	90_LPDP_WIDE_TO_AP_D1_N	10
31	IN	90_LPDP_WIDE_TO_AP_D2_P	==	90_LPDP_WIDE_TO_AP_D2_P	10
		90_LPDP_WIDE_TO_AP_D2_N	MAKE_BASE=TRUE	90_LPDP_WIDE_TO_AP_D2_N	10
32	IN	90_LPDP_TELE_TO_AP_D0_P	==	90_LPDP_TELE_TO_AP_D0_P	10
		90_LPDP_TELE_TO_AP_D0_N	MAKE_BASE=TRUE	90_LPDP_TELE_TO_AP_D0_N	10
32	IN	90_LPDP_TELE_TO_AP_D1_P	==	90_LPDP_TELE_TO_AP_D1_P	10
		90_LPDP_TELE_TO_AP_D1_N	MAKE_BASE=TRUE	90_LPDP_TELE_TO_AP_D1_N	10
32	IN	90_LPDP_TELE_TO_AP_D2_P	==	90_LPDP_TELE_TO_AP_D2_P	10
		90_LPDP_TELE_TO_AP_D2_N	MAKE_BASE=TRUE	90_LPDP_TELE_TO_AP_D2_N	10
34	IN	90_LPDP_FCAM_TO_AP_D0_P	==	90_LPDP_FCAM_TO_AP_D0_P	10
		90_LPDP_FCAM_TO_AP_D0_N	MAKE_BASE=TRUE	90_LPDP_FCAM_TO_AP_D0_N	10
34	IN	90_LPDP_FCAM_TO_AP_D1_P	==	90_LPDP_FCAM_TO_AP_D1_P	10
		90_LPDP_FCAM_TO_AP_D1_N	MAKE_BASE=TRUE	90_LPDP_FCAM_TO_AP_D1_N	10

S4E NAND





PMU - BUCKS



PMU - LD0s

D

D

C

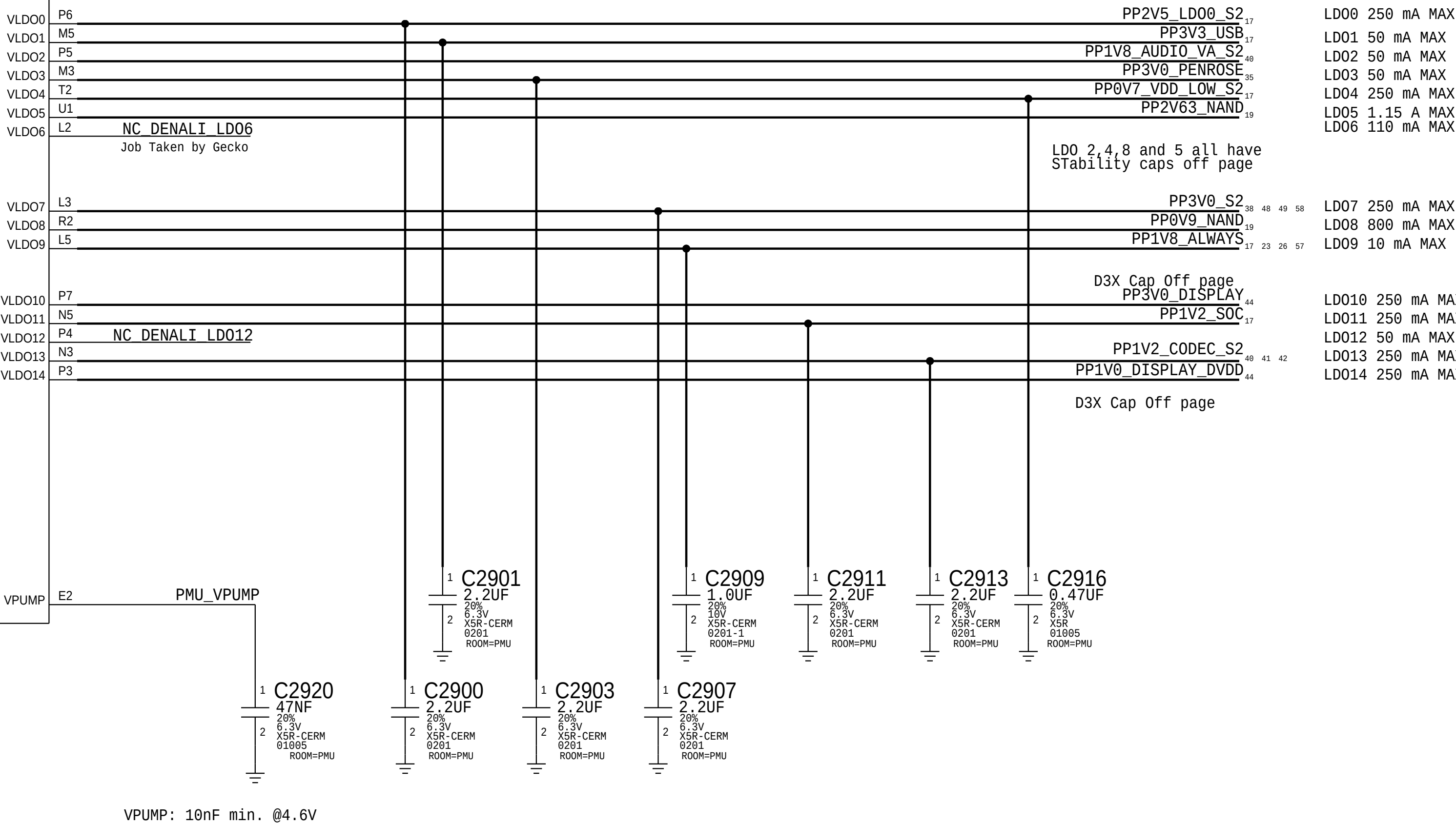
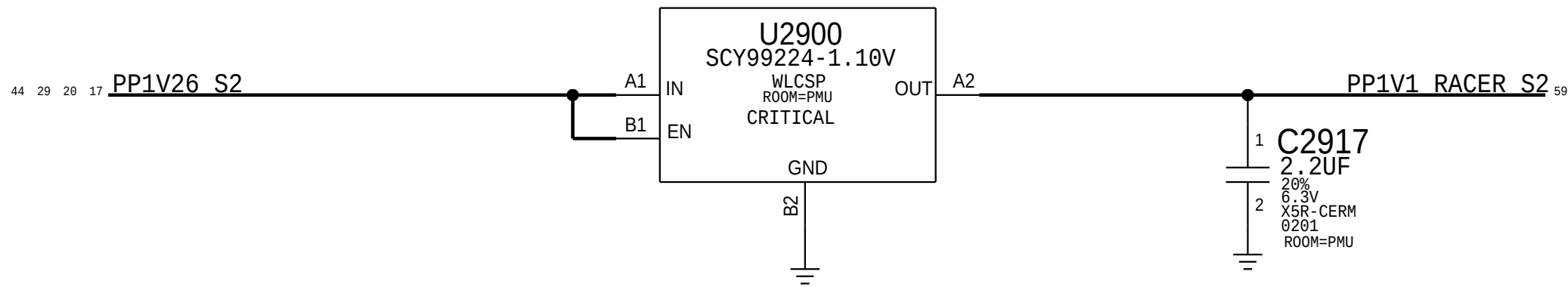
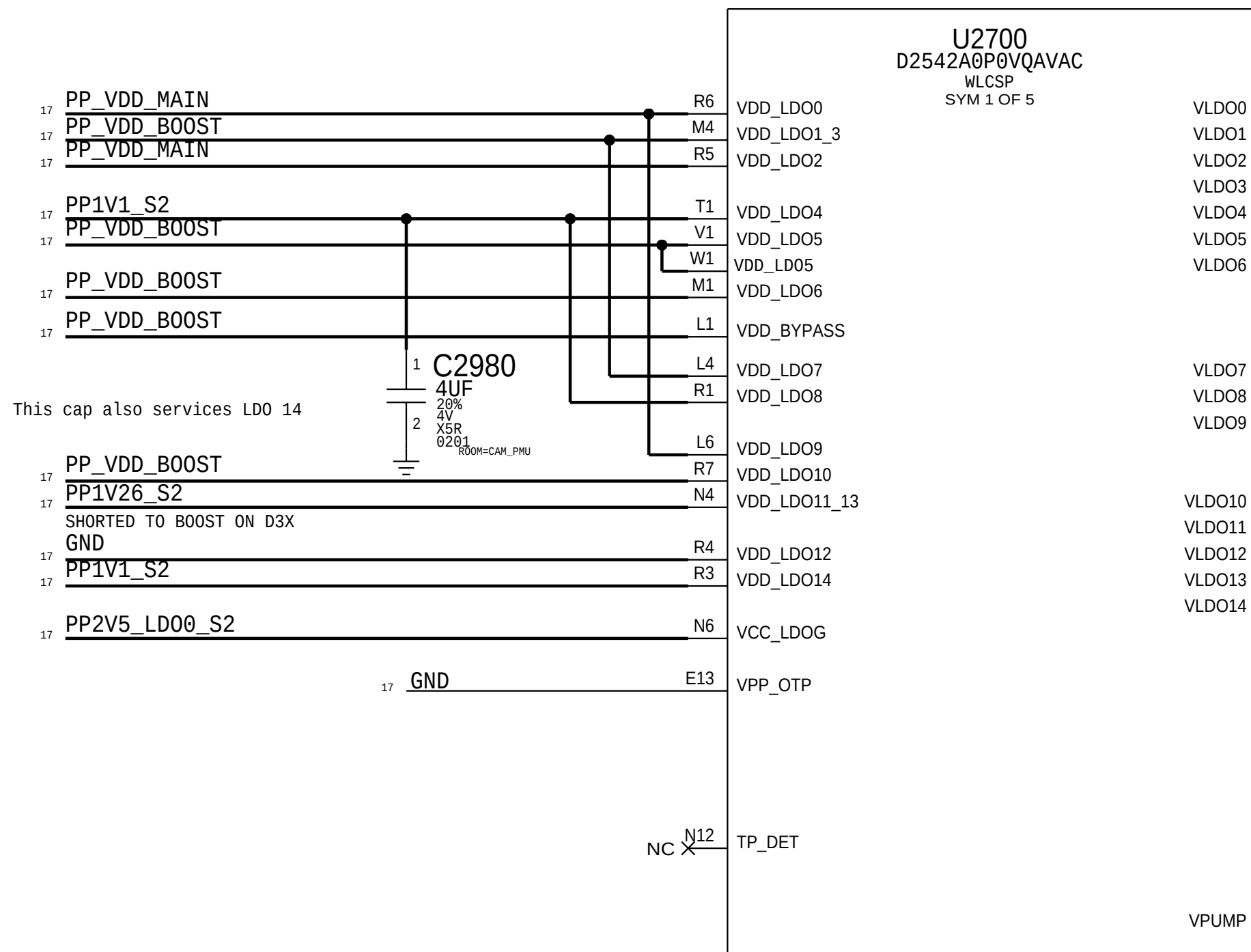
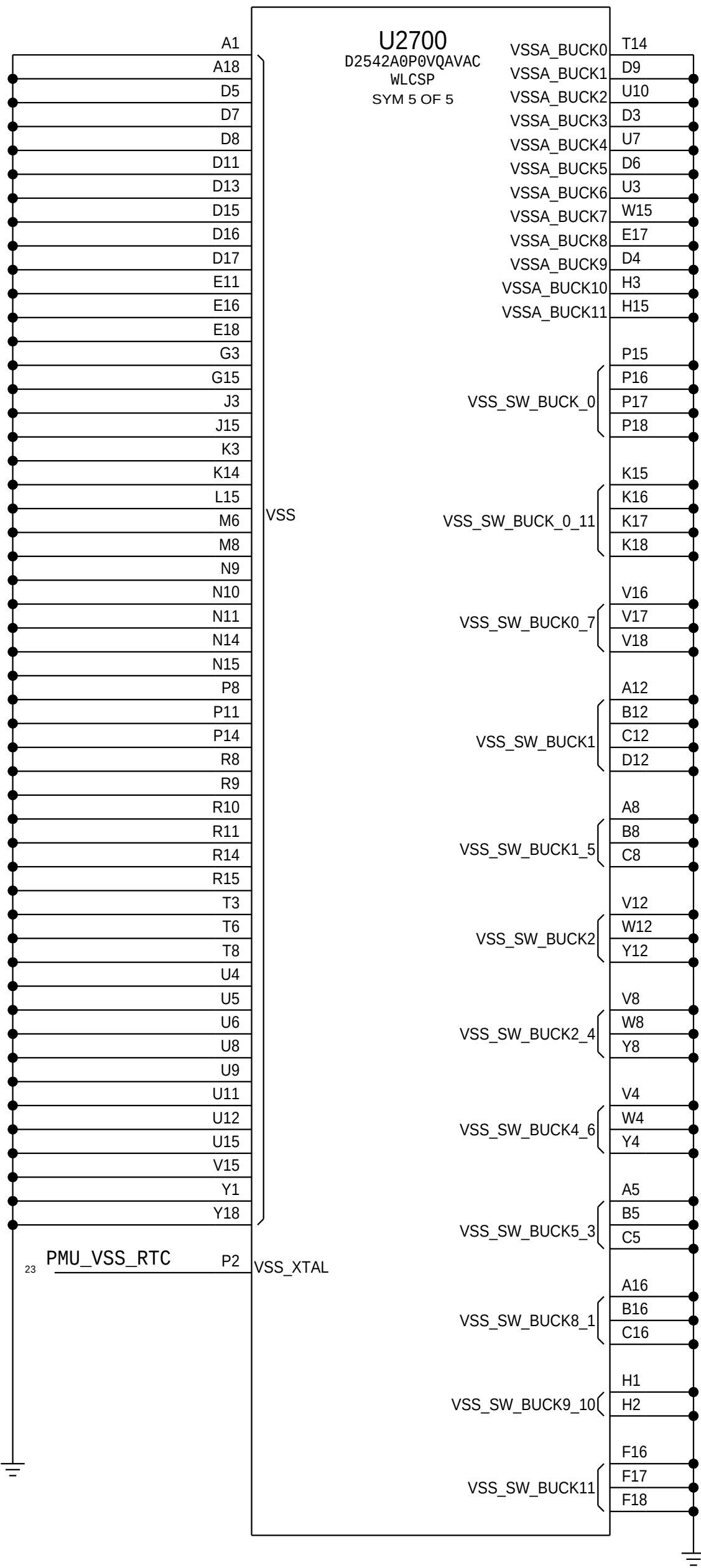
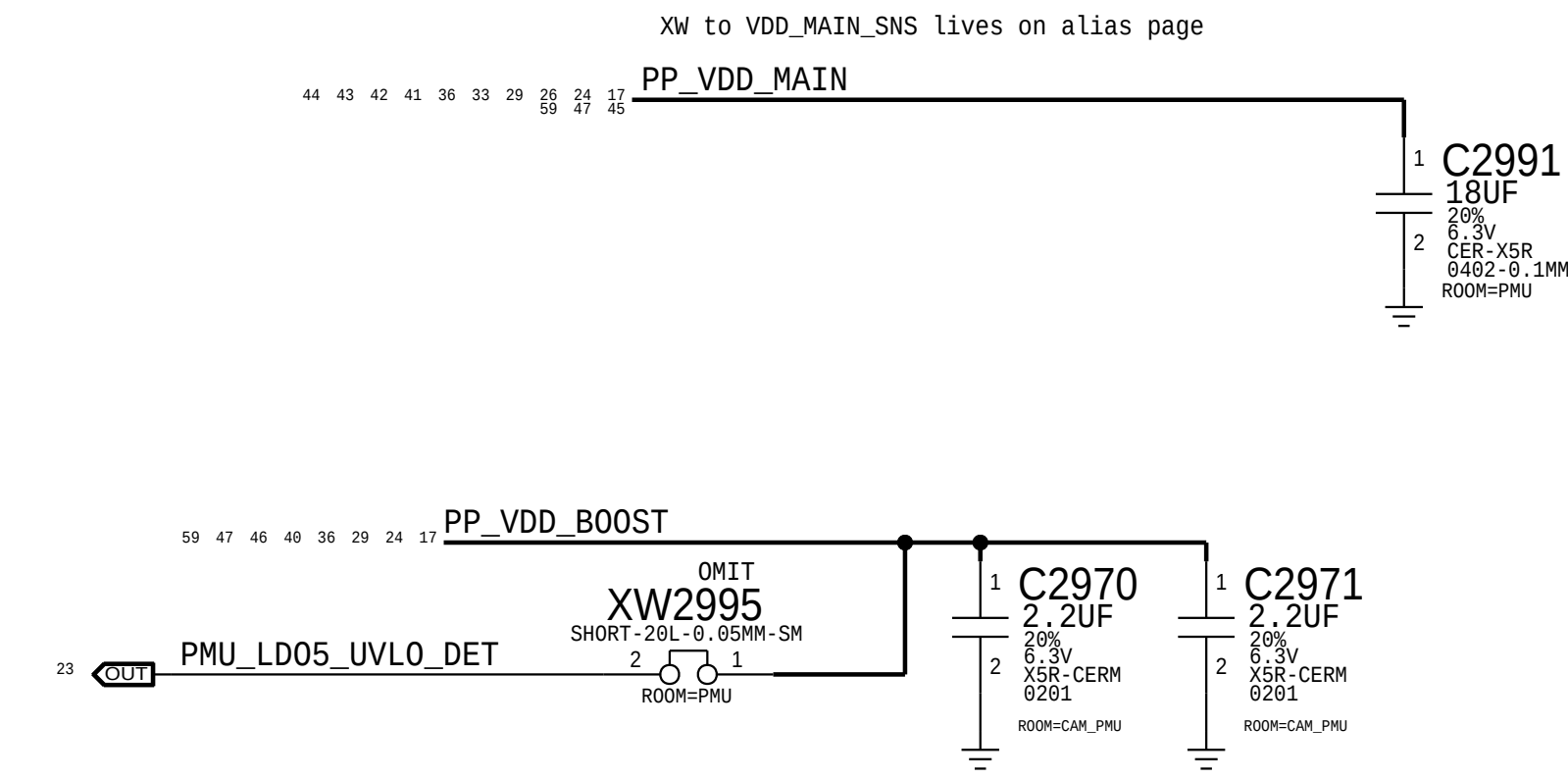
C

B

B

A

A

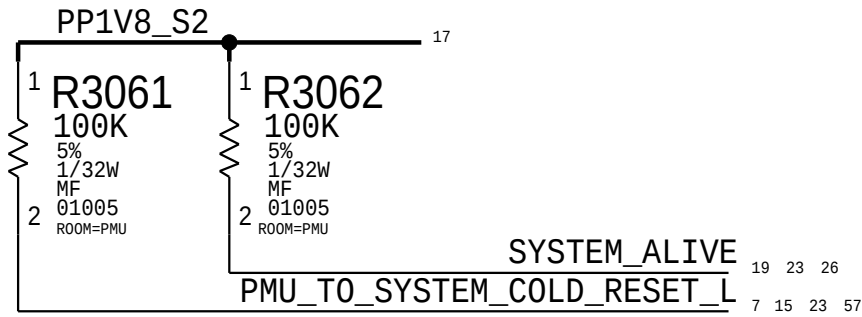


PMU - GPIOs

TODO: Update
CONTROL PIN NOTES:

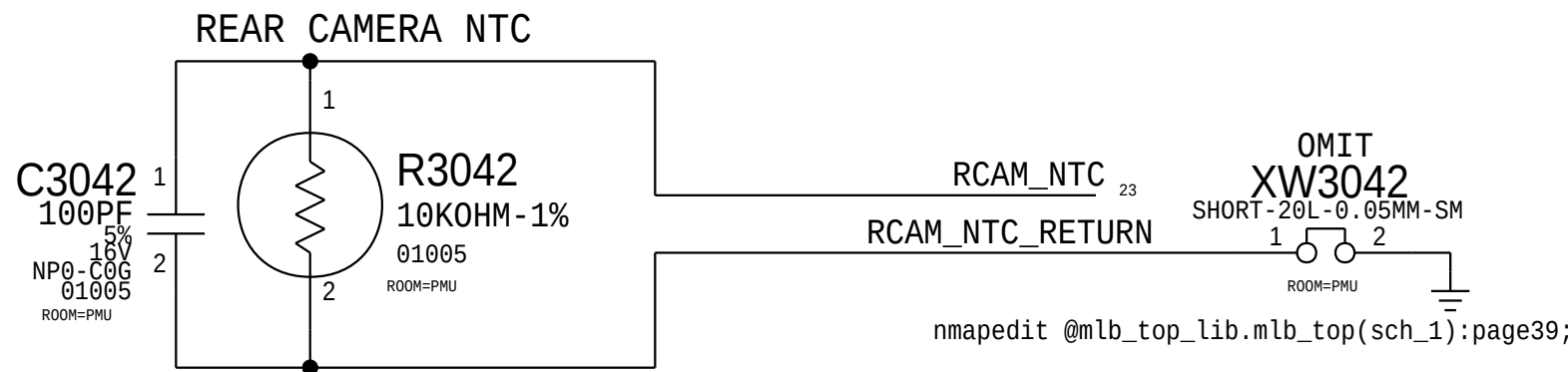
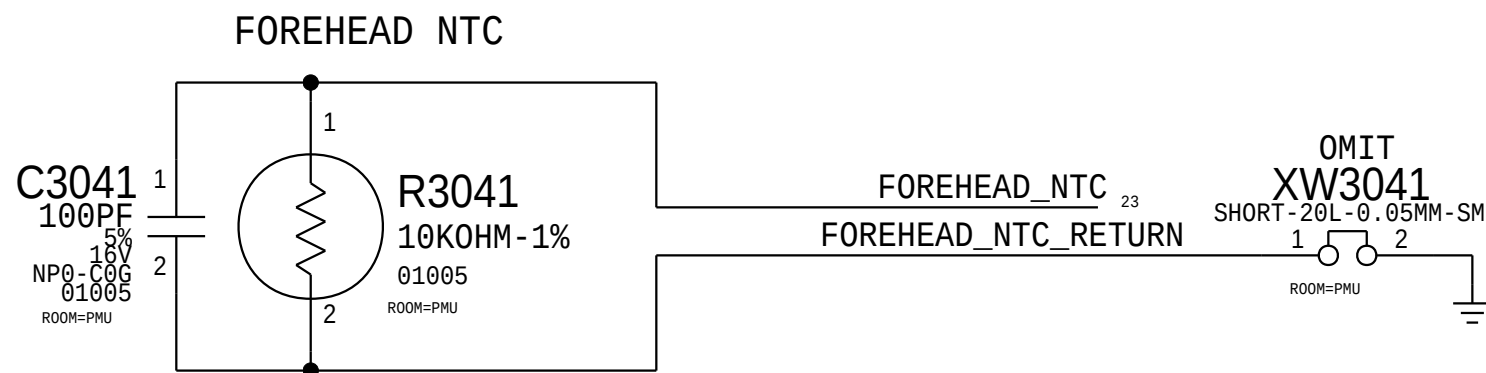
- NOTE (1):INPUT PULL-DOWN 100-300k
NOTE (2):INPUT PULL-DOWN 1M
NOTE (3):INPUT PULL-UP OR DOWN 100k-300k
NOTE (4):OUTPUT OPEN-DRAIN, REQUIRES PULL-UP

COLD_RESET & SYSTEM_ALIVE



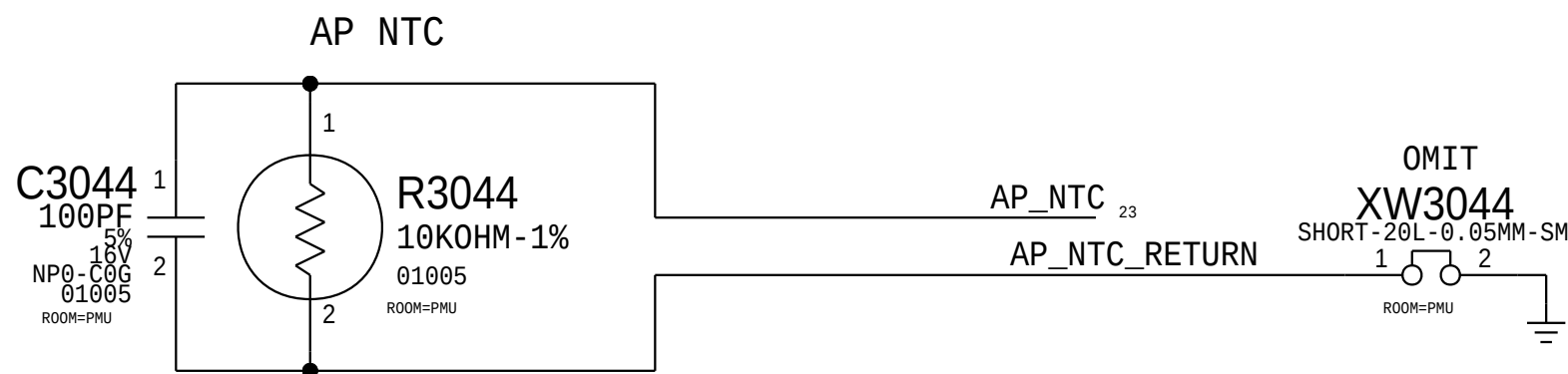
Only has DS control when powered by VBUCK3
Only has DS control when powered by VBUCK3

NTCs



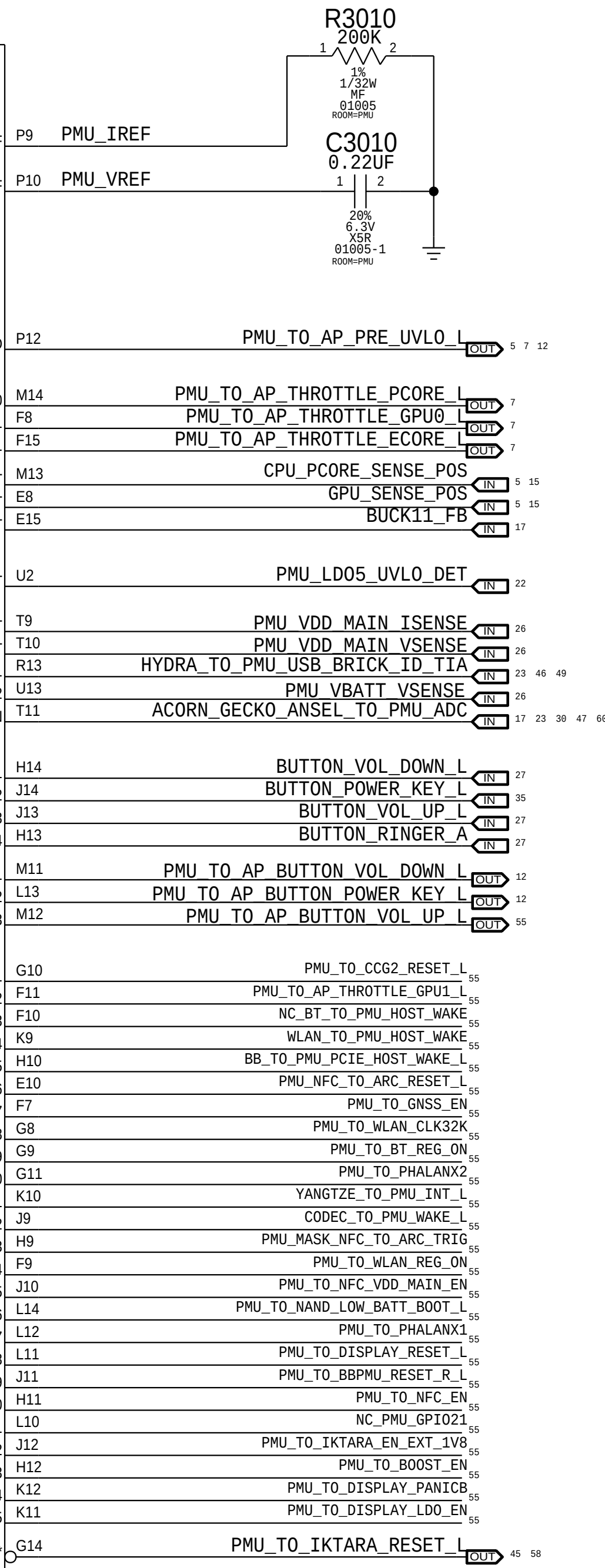
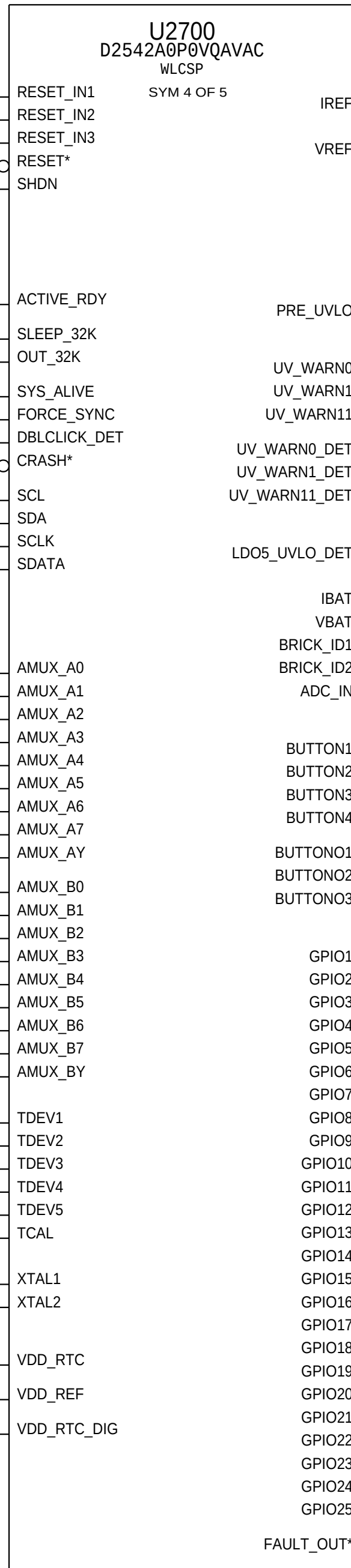
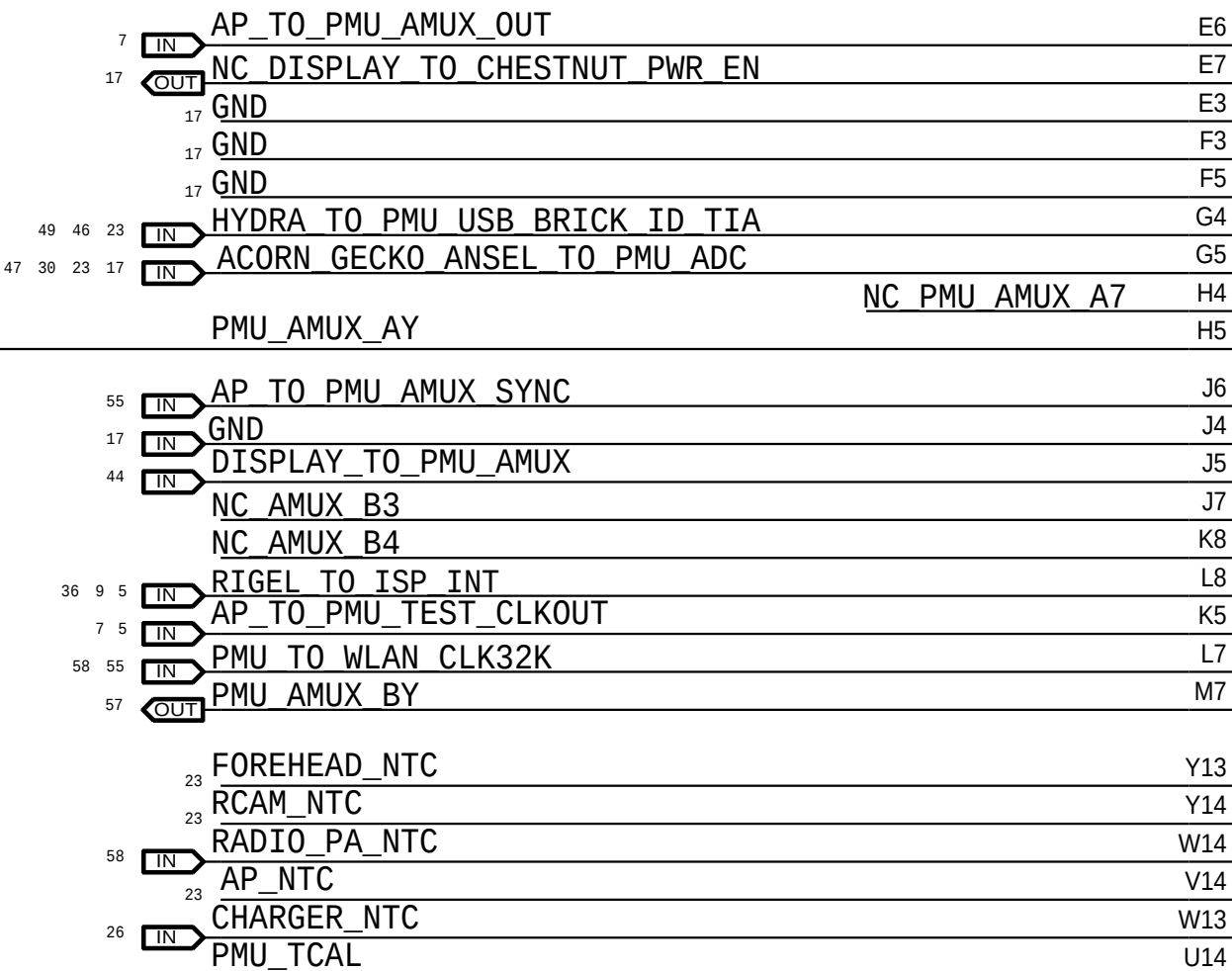
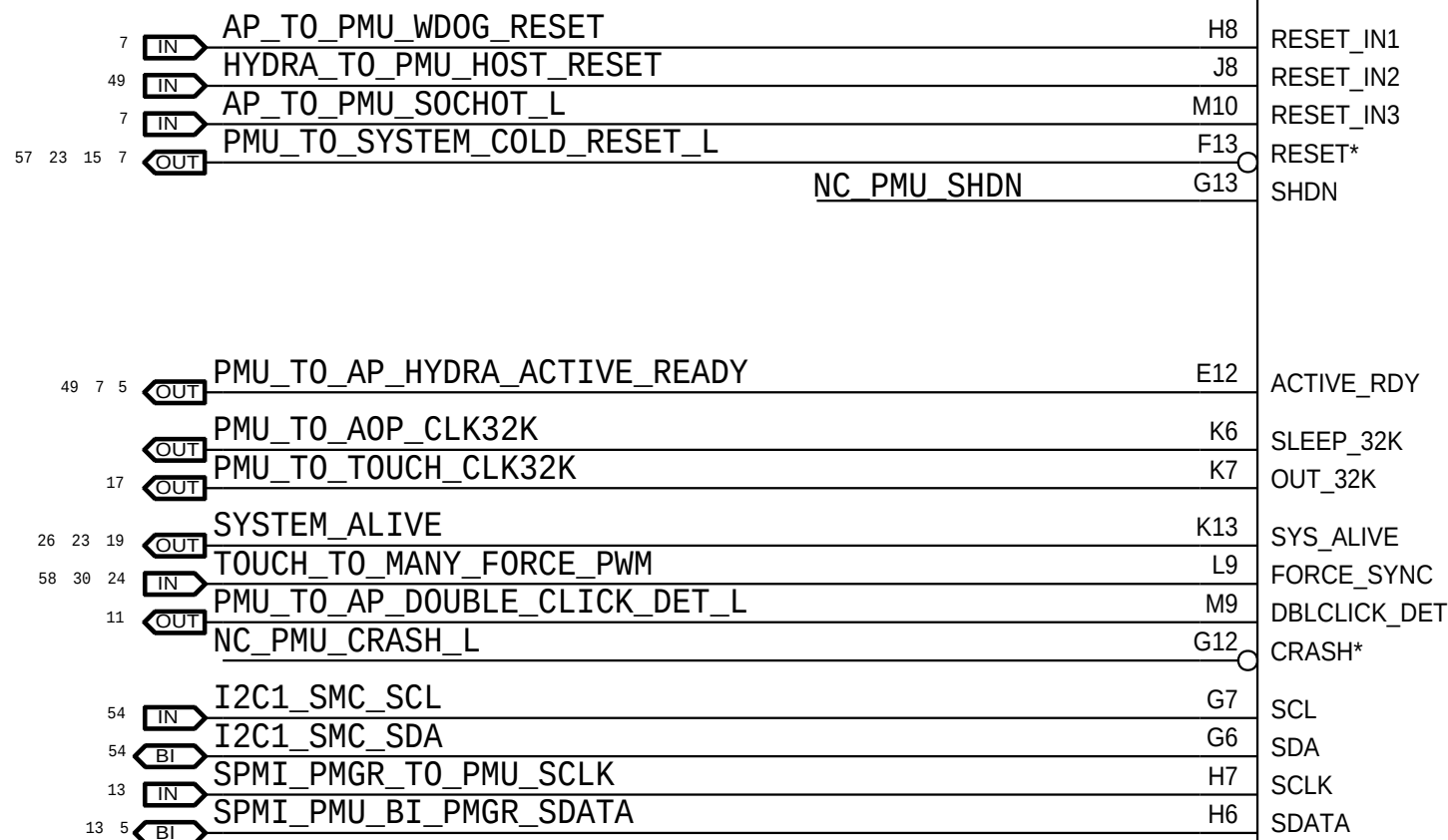
nmapedit @mlb_top_lib.mlb_top(sch_1):page39;

RADIO PA NTC on MLB Bottom

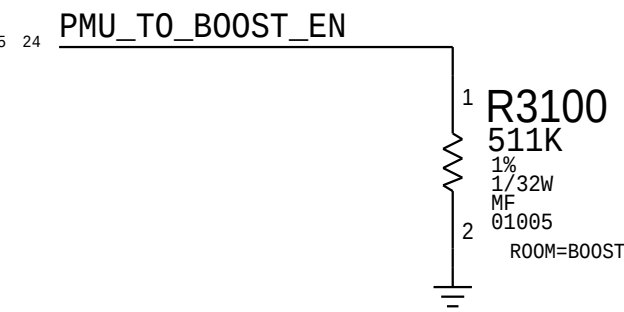


CHARGER NTC on Chrager Page

NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC



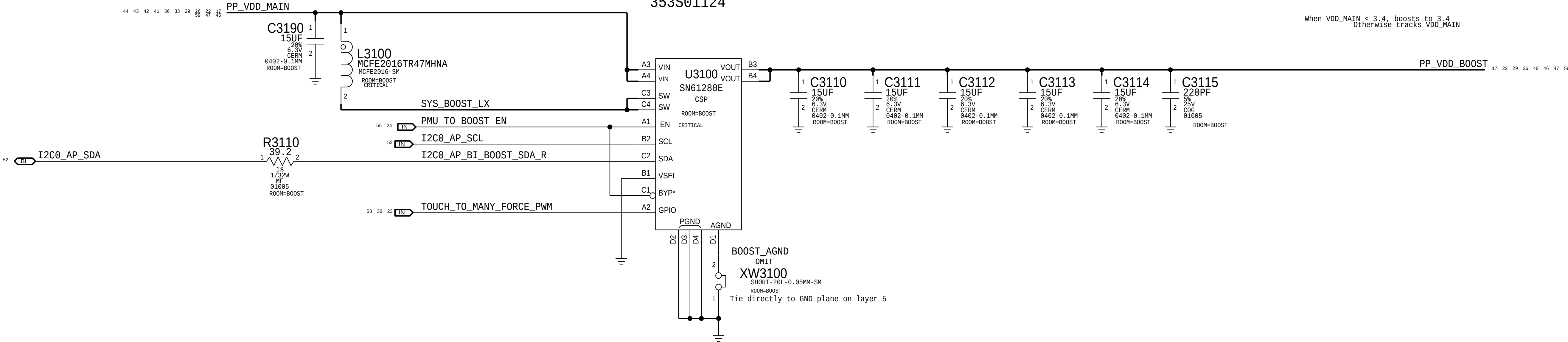
Boost Enable Pull



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00871	152S00869	ALT_PARTS	L3100	BOOST IND ALT, CYN
152S00873	152S00869	ALT_PARTS	L3100	BOOST IND ALT, YDK

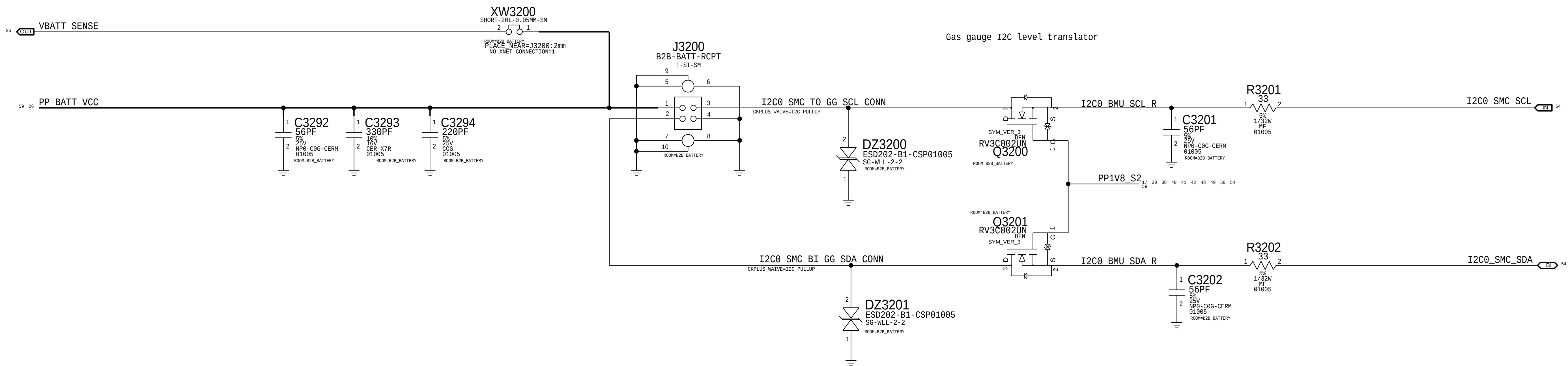
BOOST

353S01124



BATTERY CONNECTOR

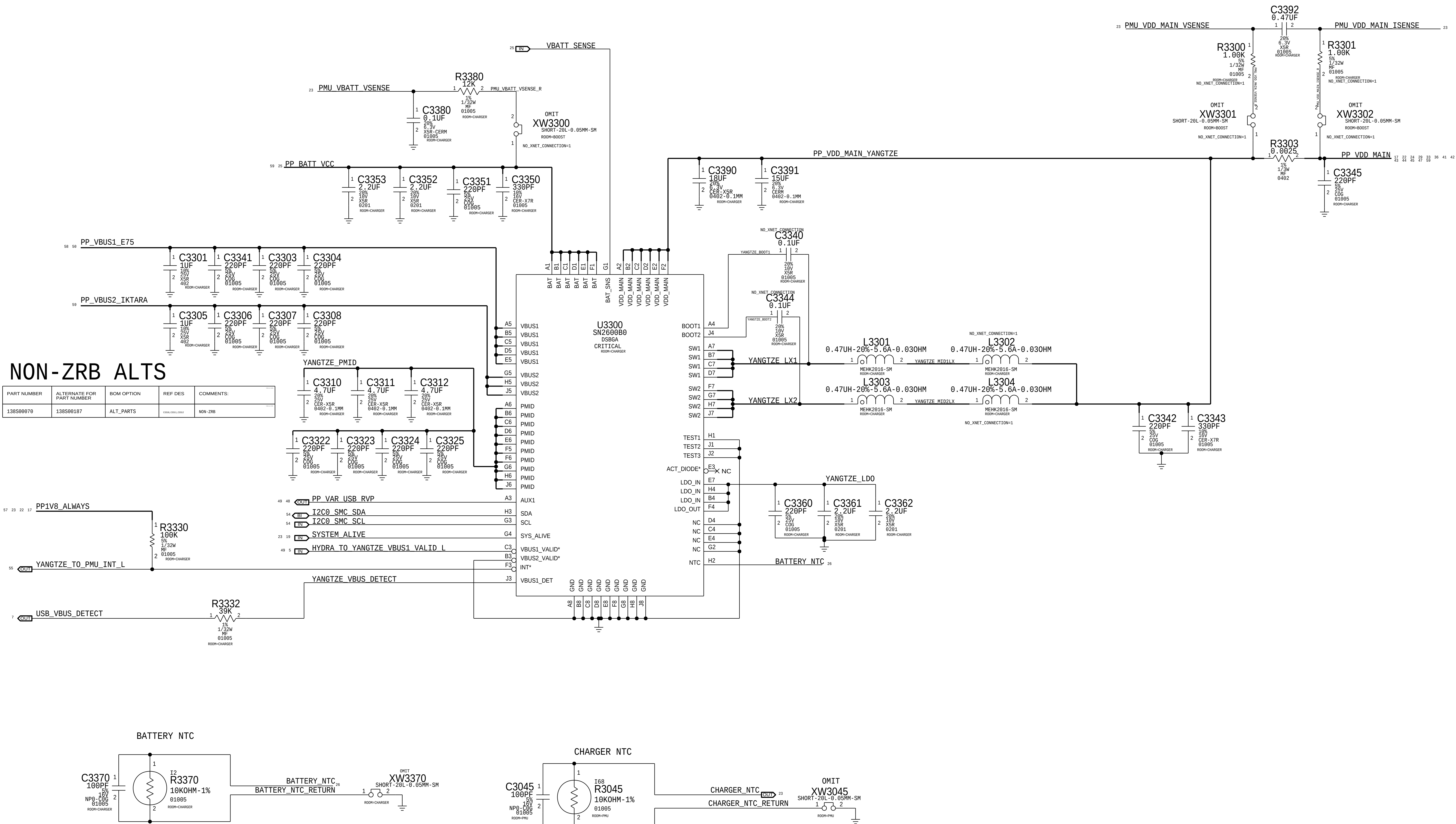
Rcpt: 516S00232
Plug: 516S00233

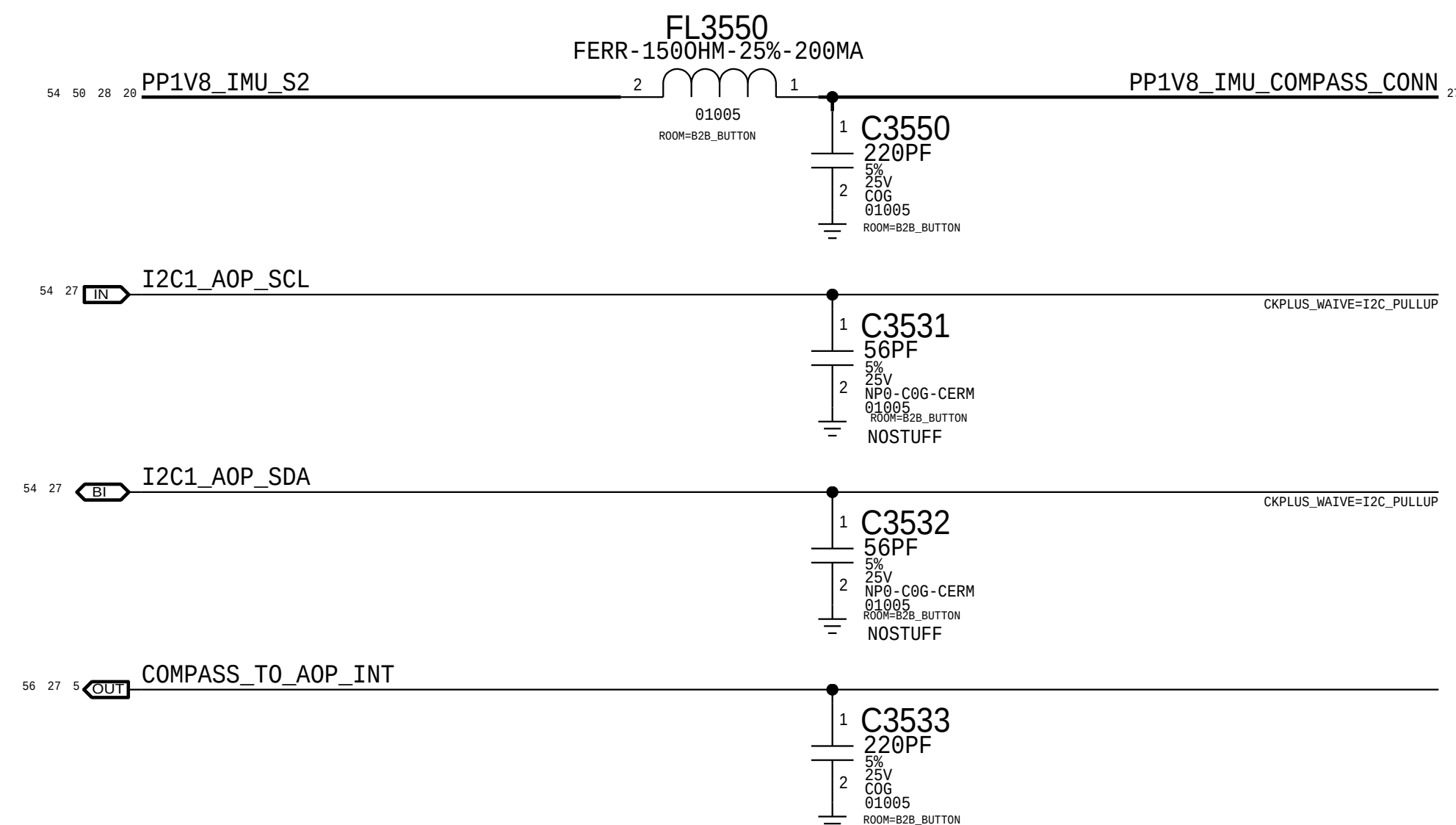
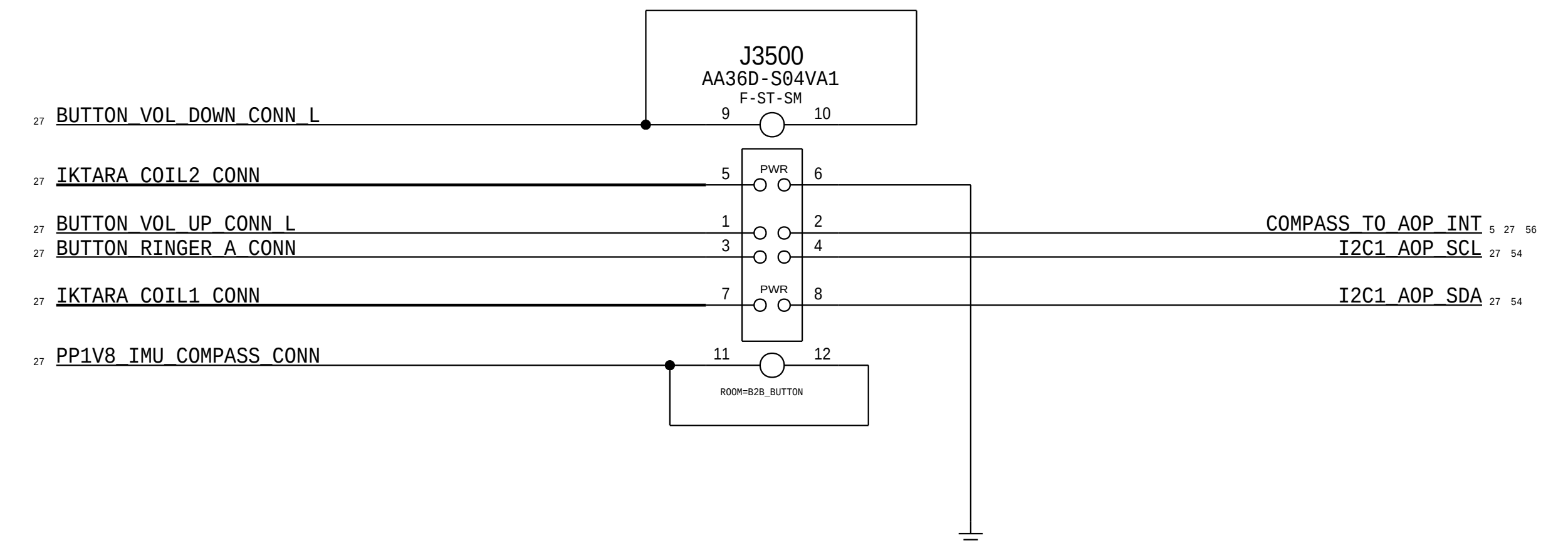
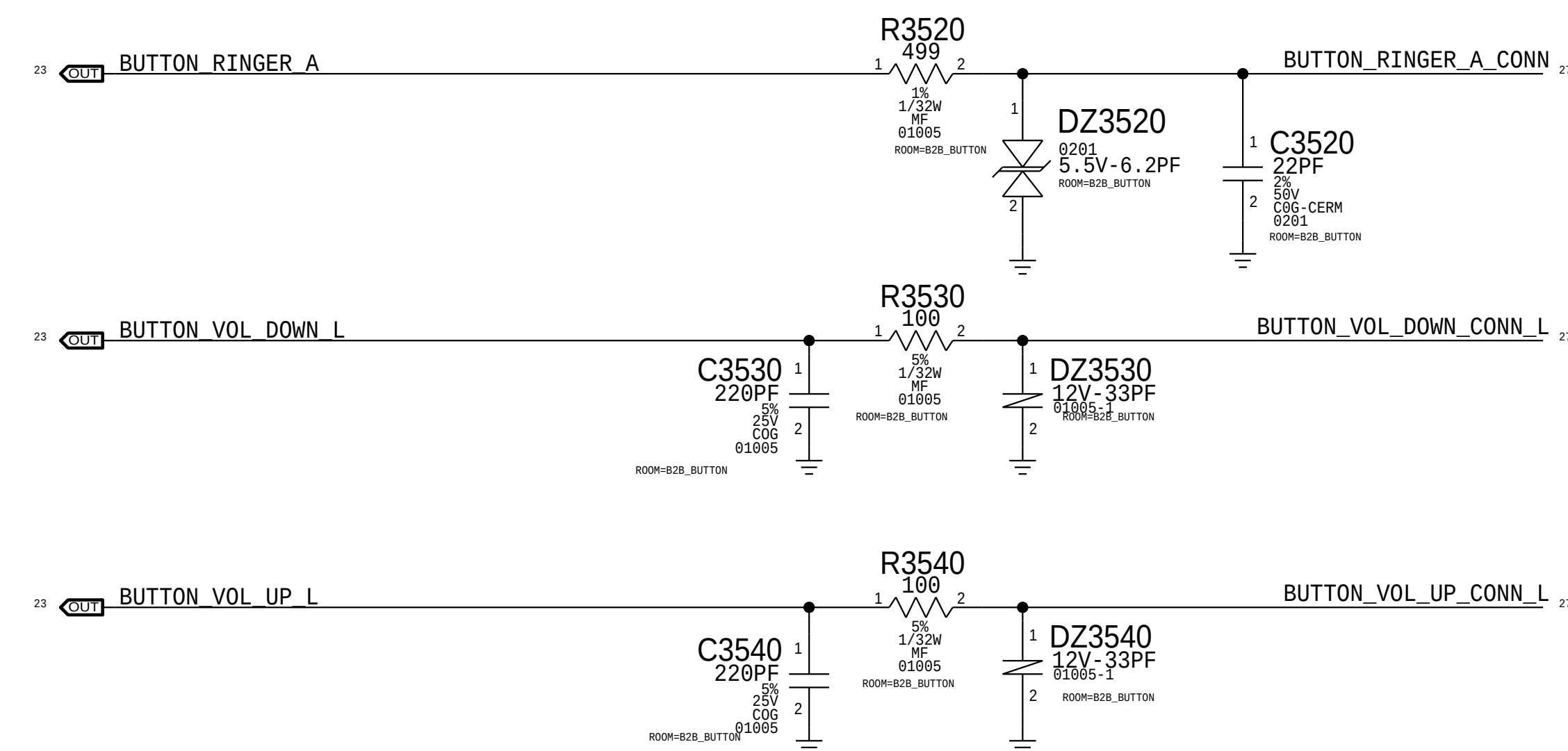
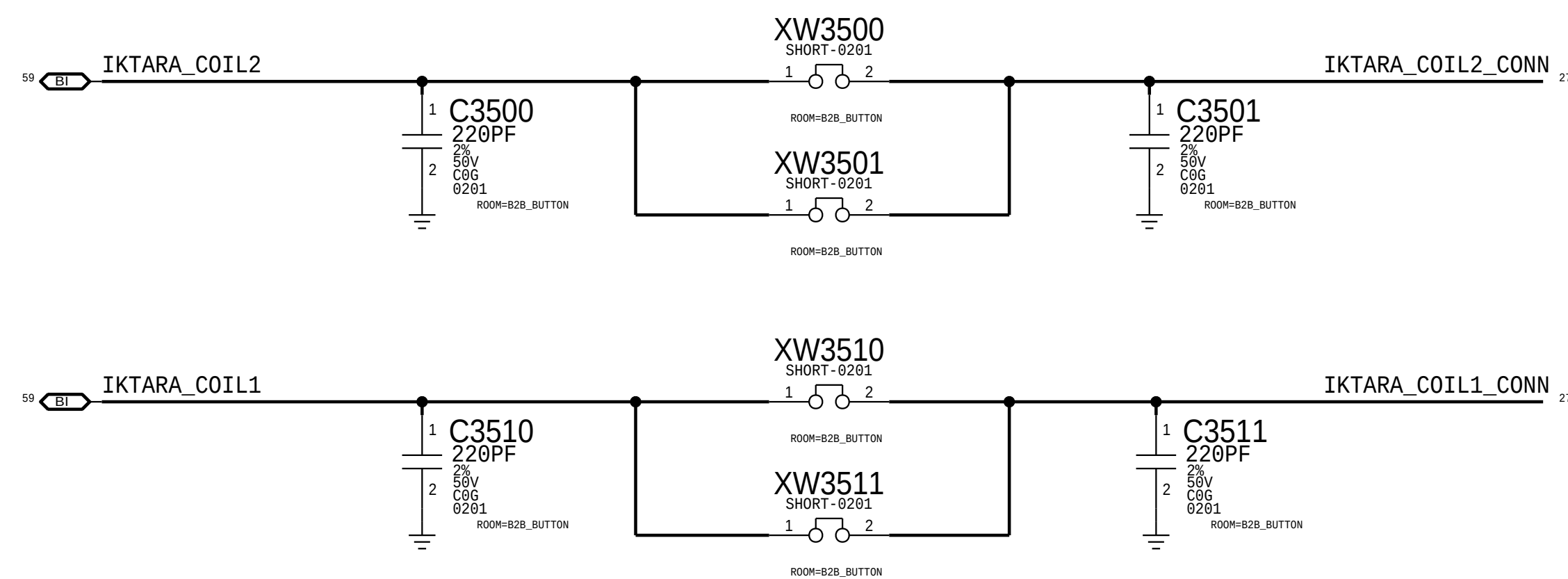


YANGTZE CHARGER

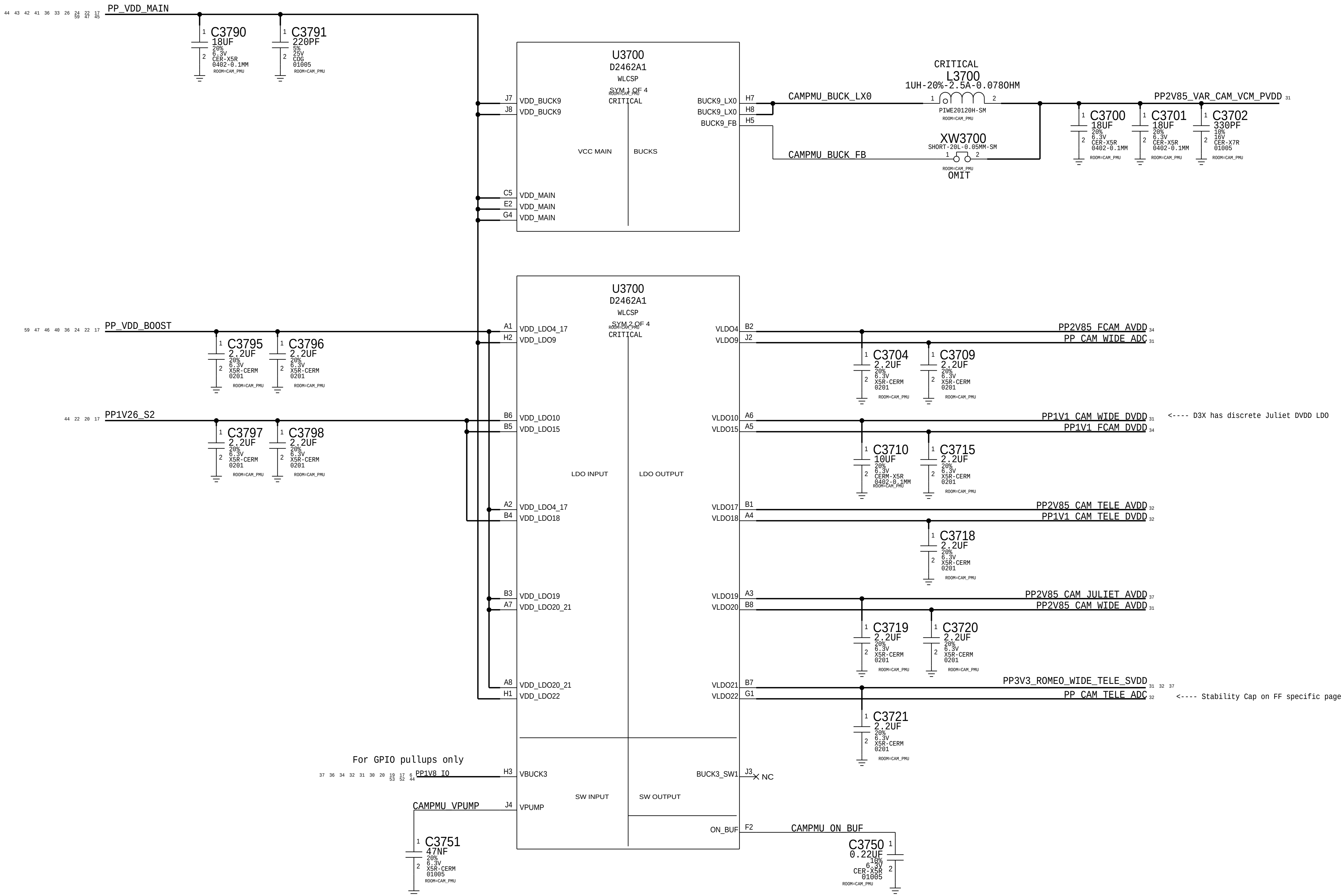
NON-ZRB ALTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00070	138S00187	ALT_PARTS	CHRG CHRG1,CHRG2	NON-ZRB

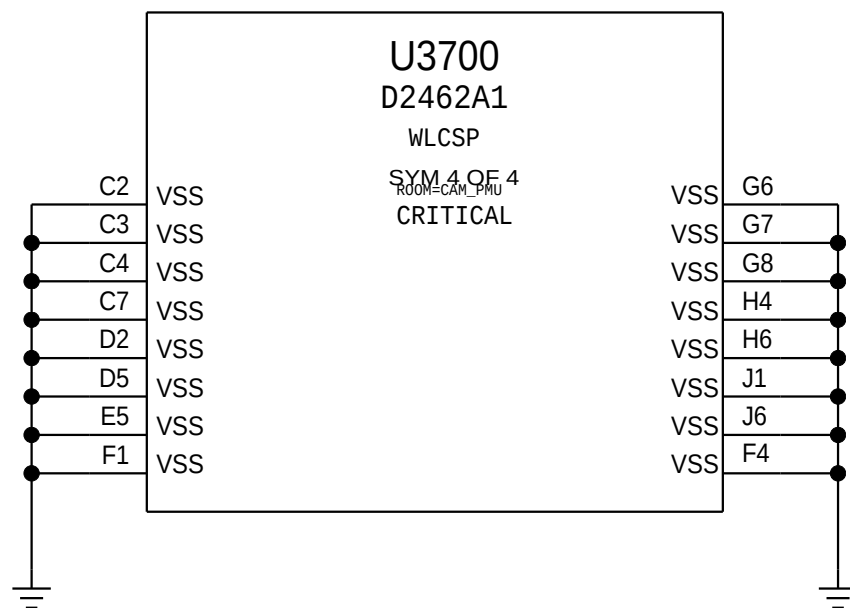
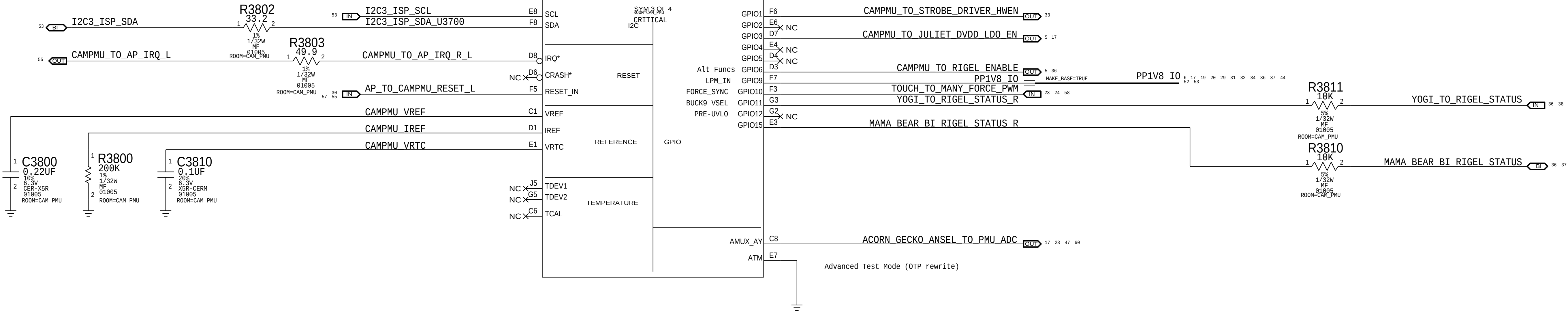
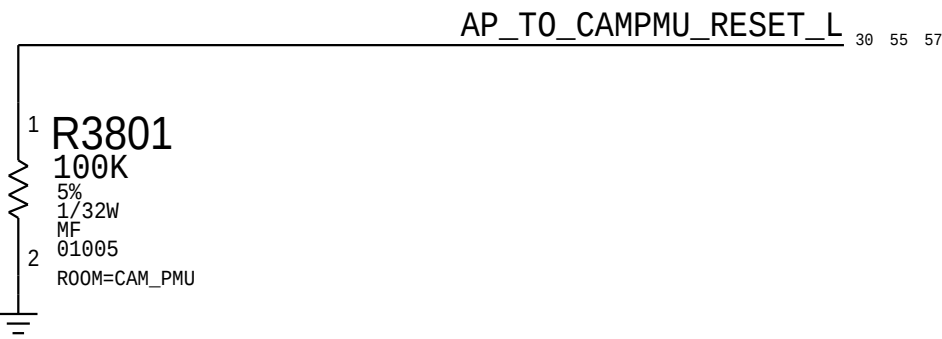


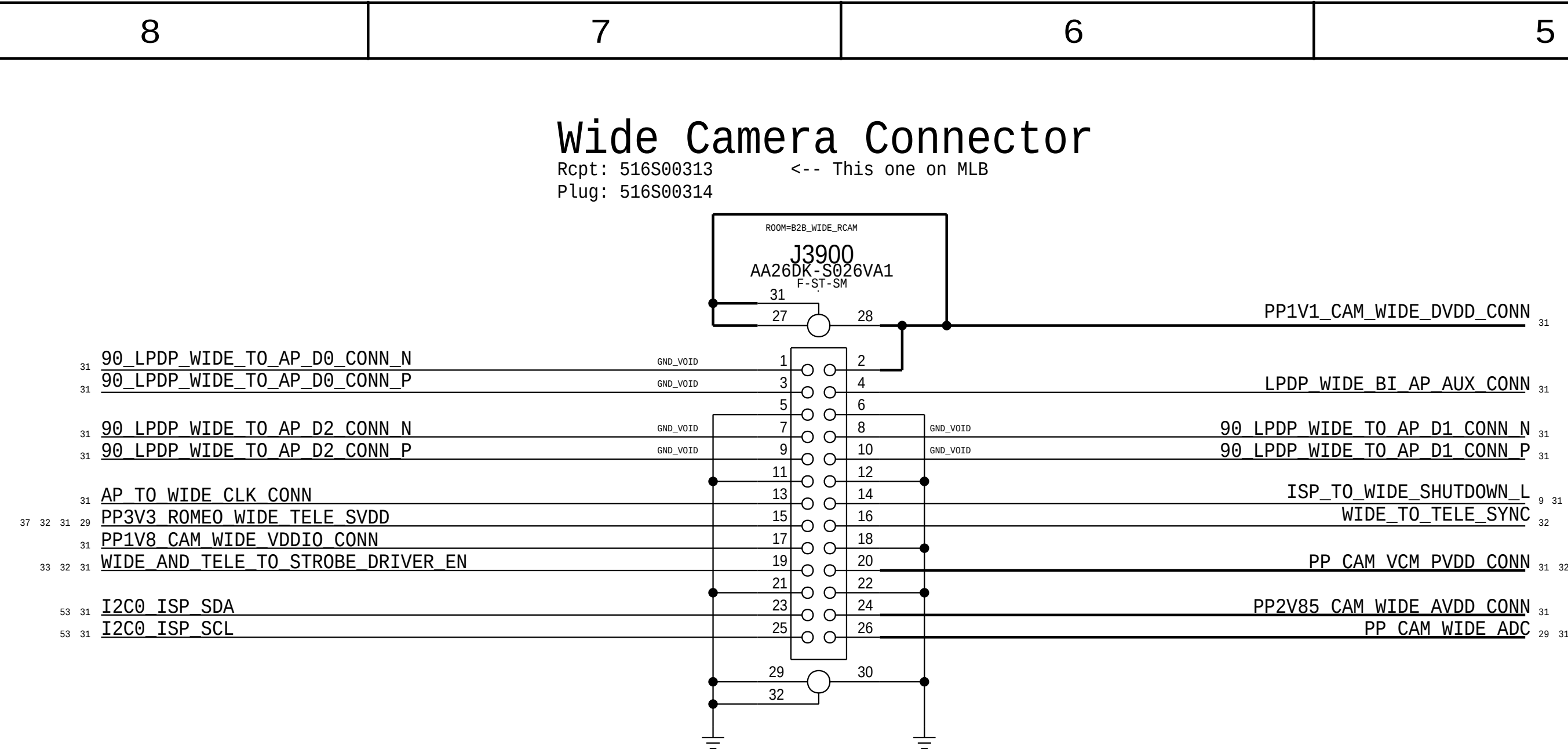


Camera PMU

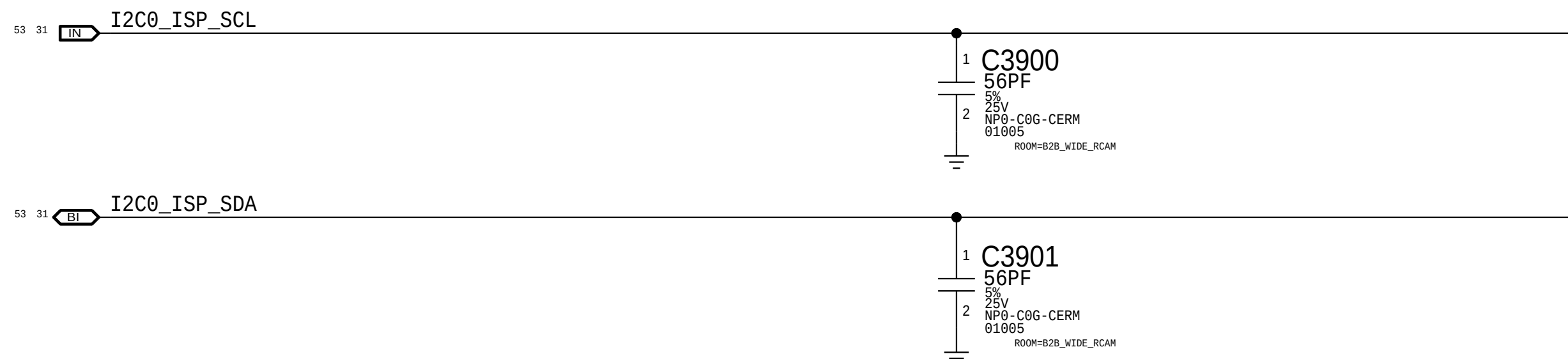


Pull Downs

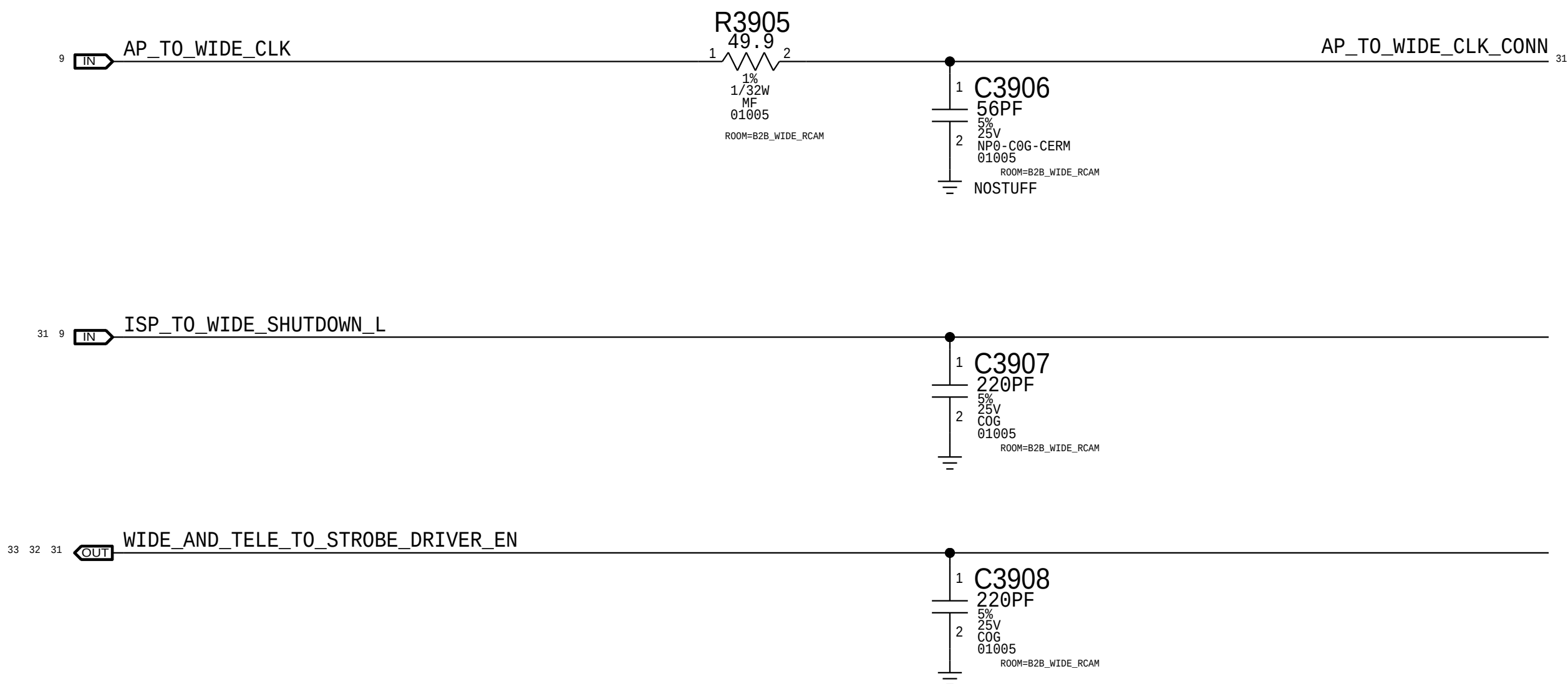




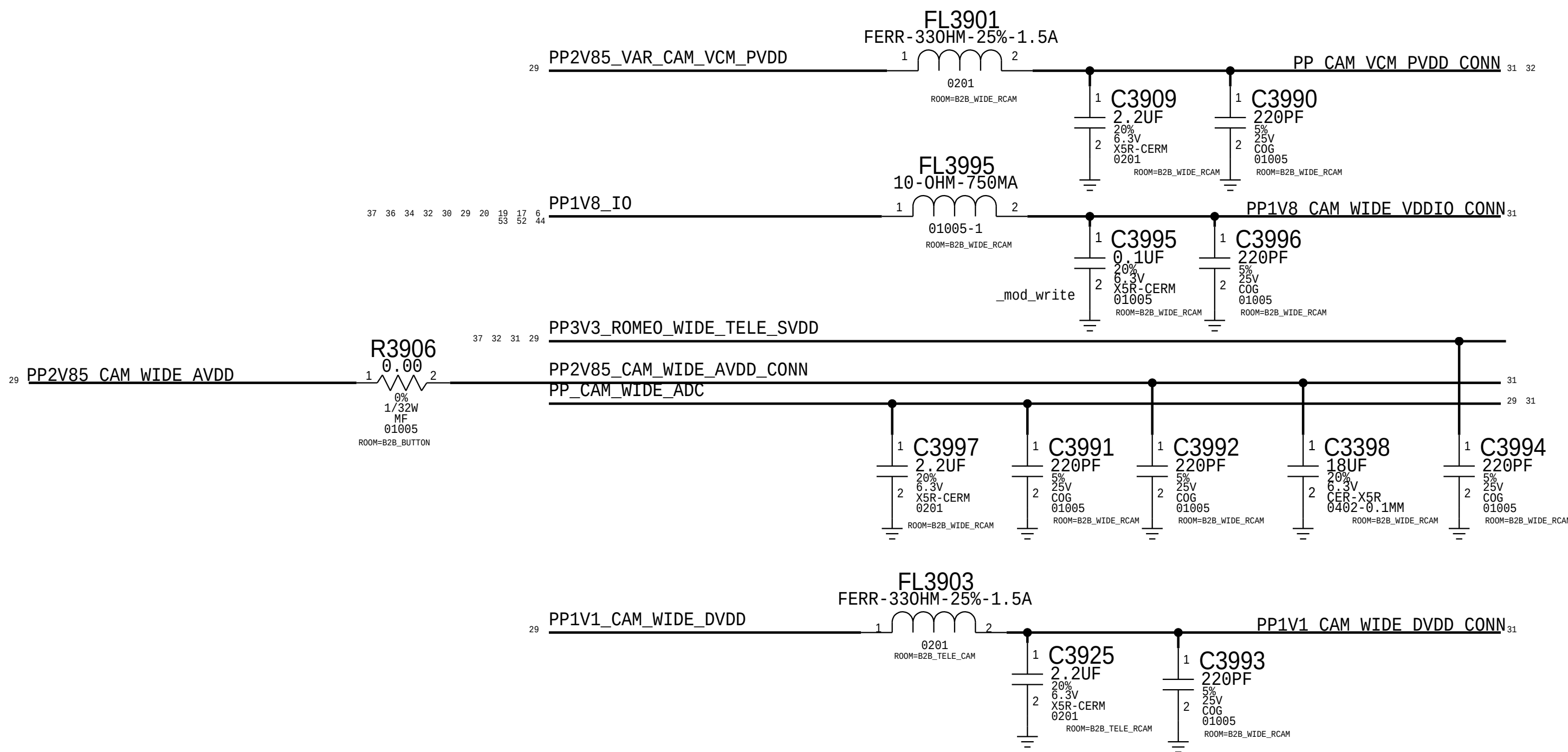
ISP I2C



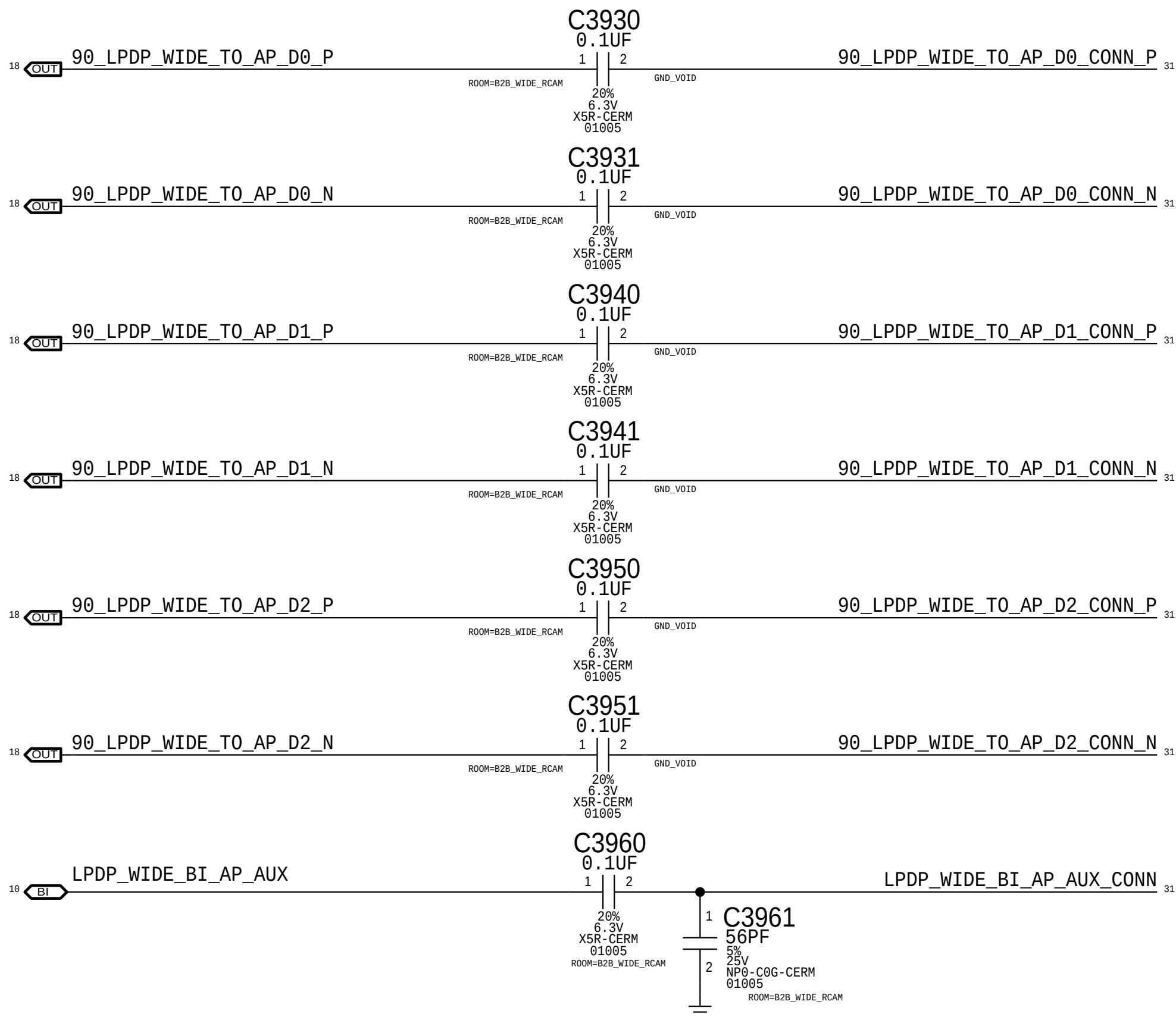
IO Filters



Power Filtering

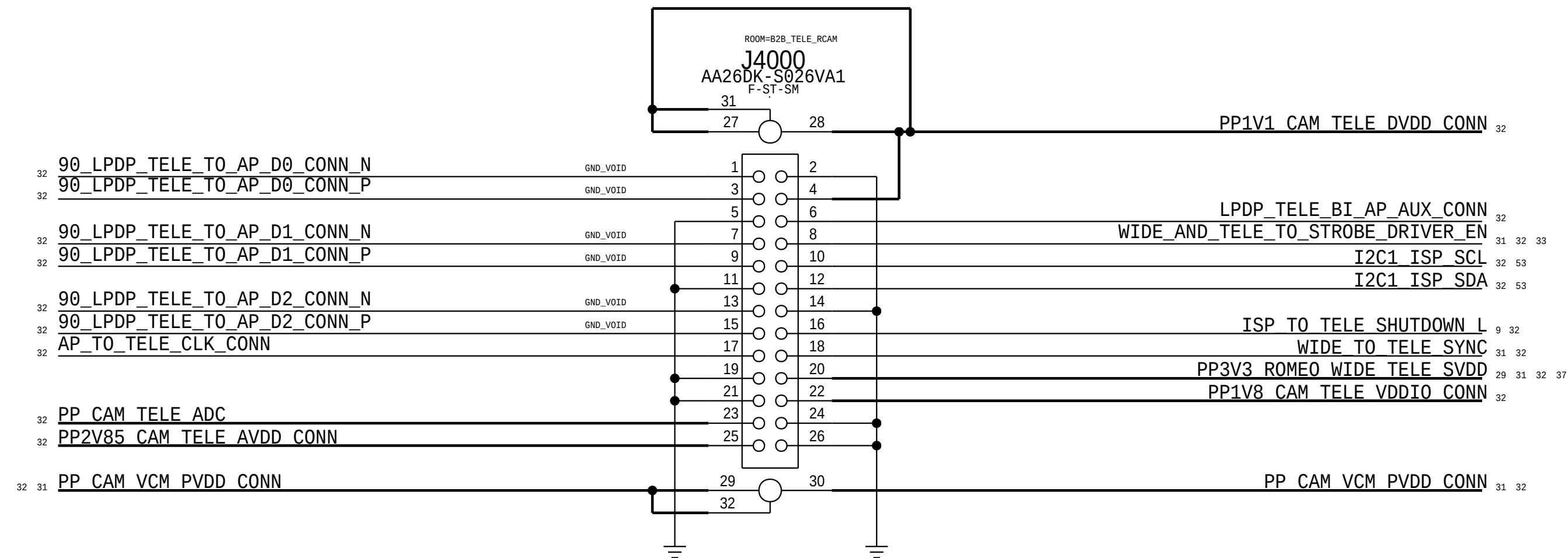


LPDP Filters

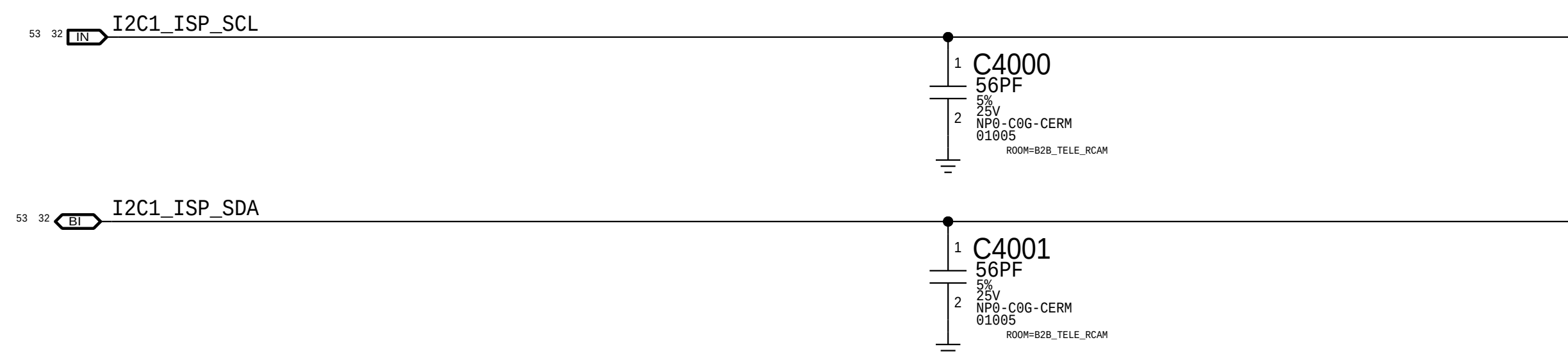




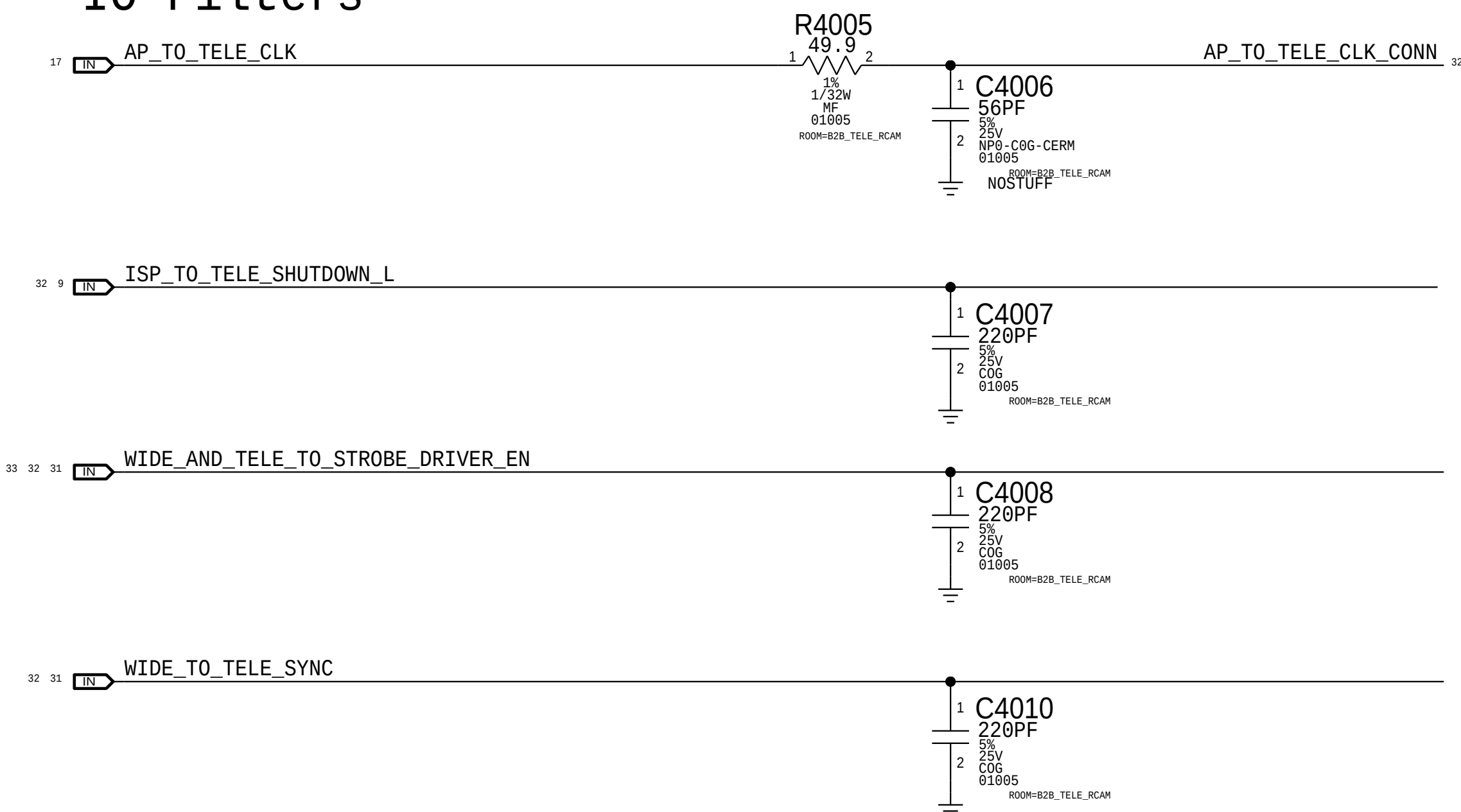
```
Rcpt: 516S00313      <-- This one on MLB
Plug: 516S00314
```



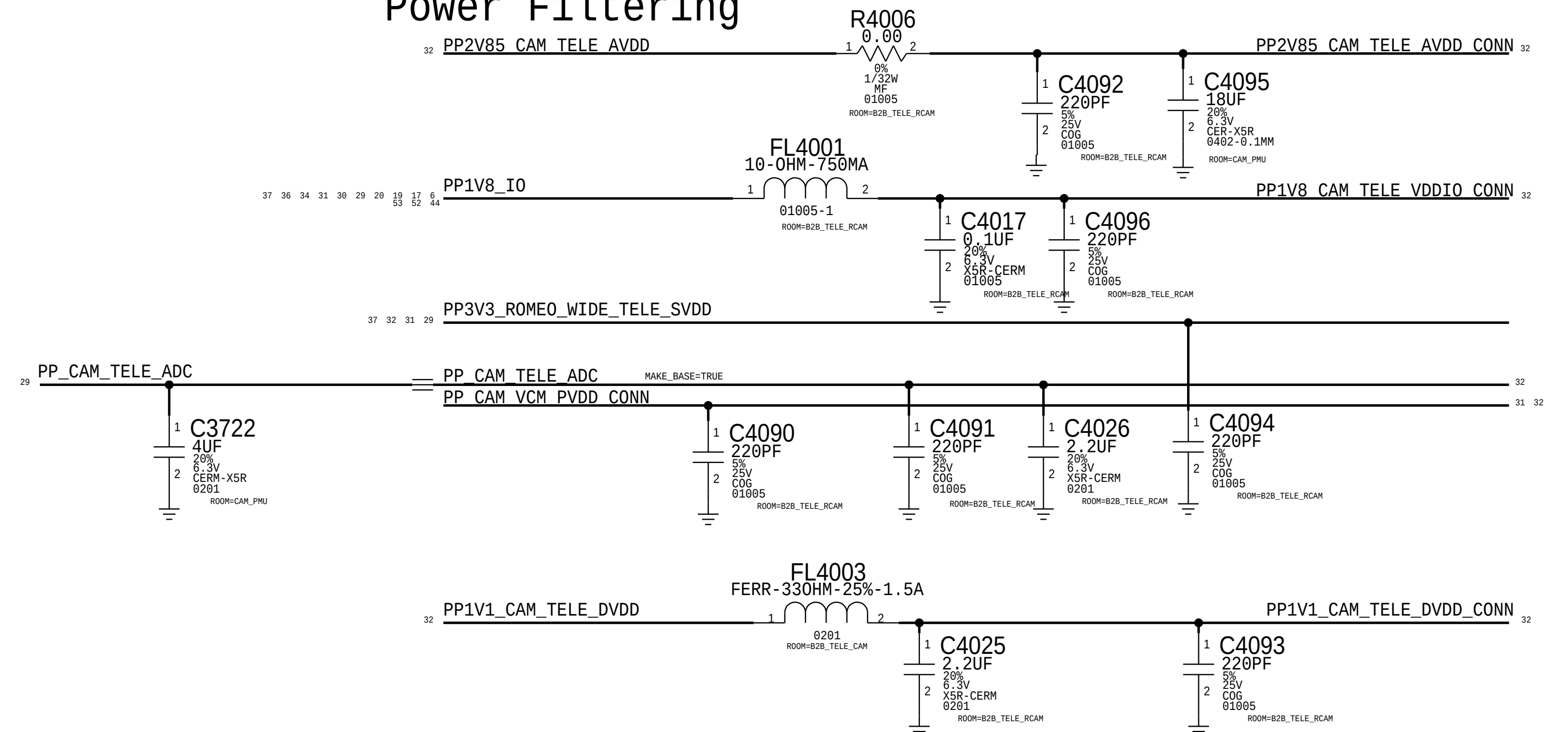
ISP I2C



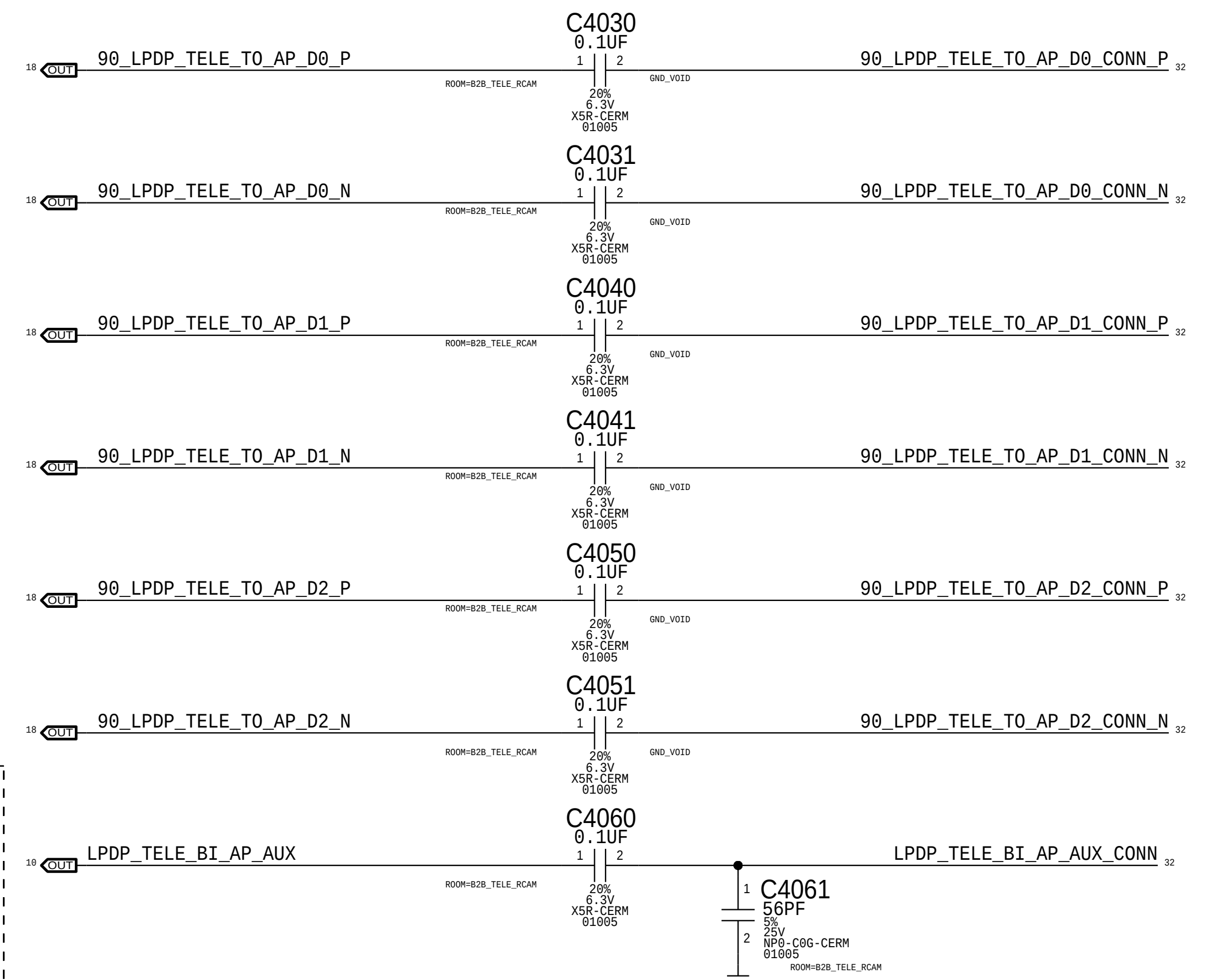
IO Filters



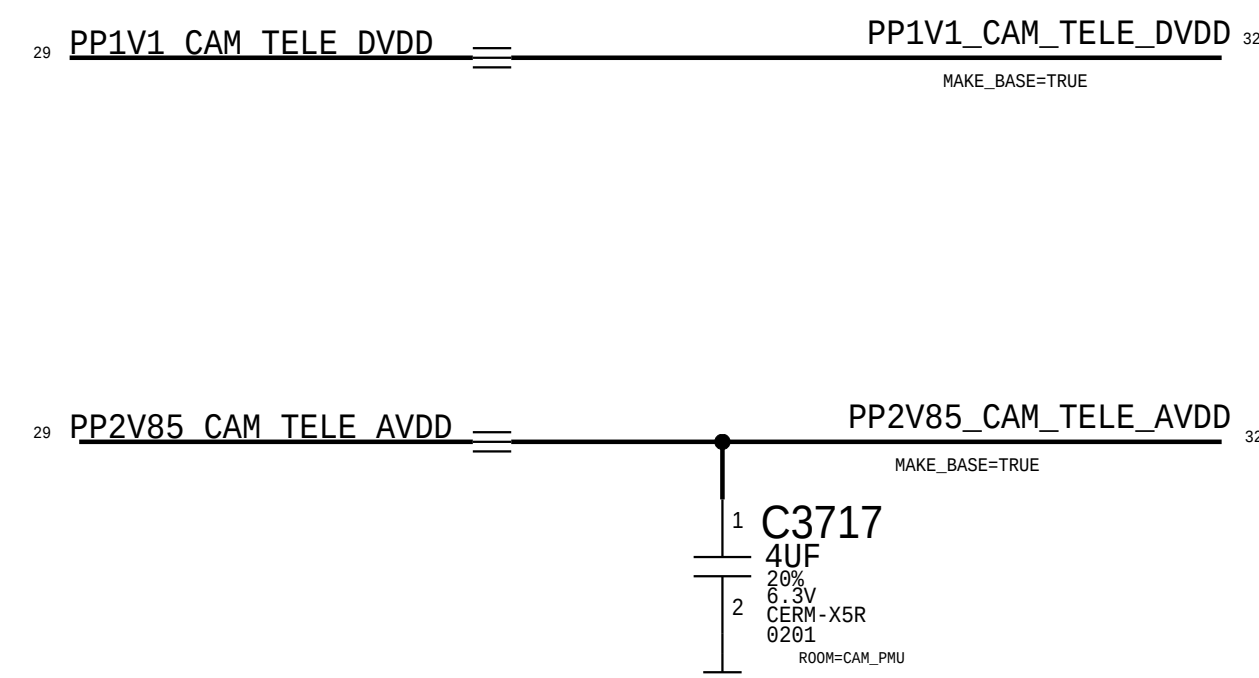
Power Filtering

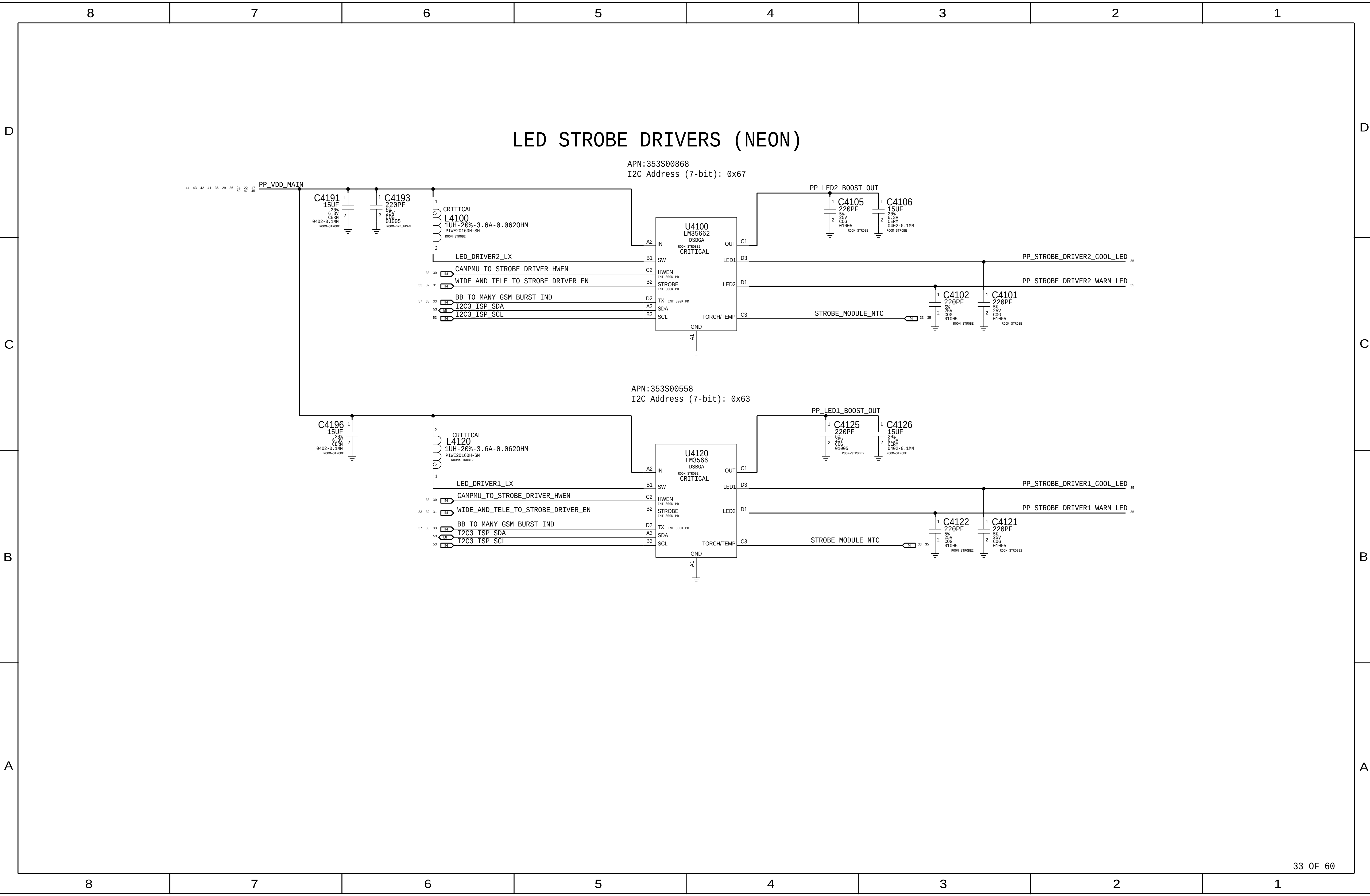


LPDP

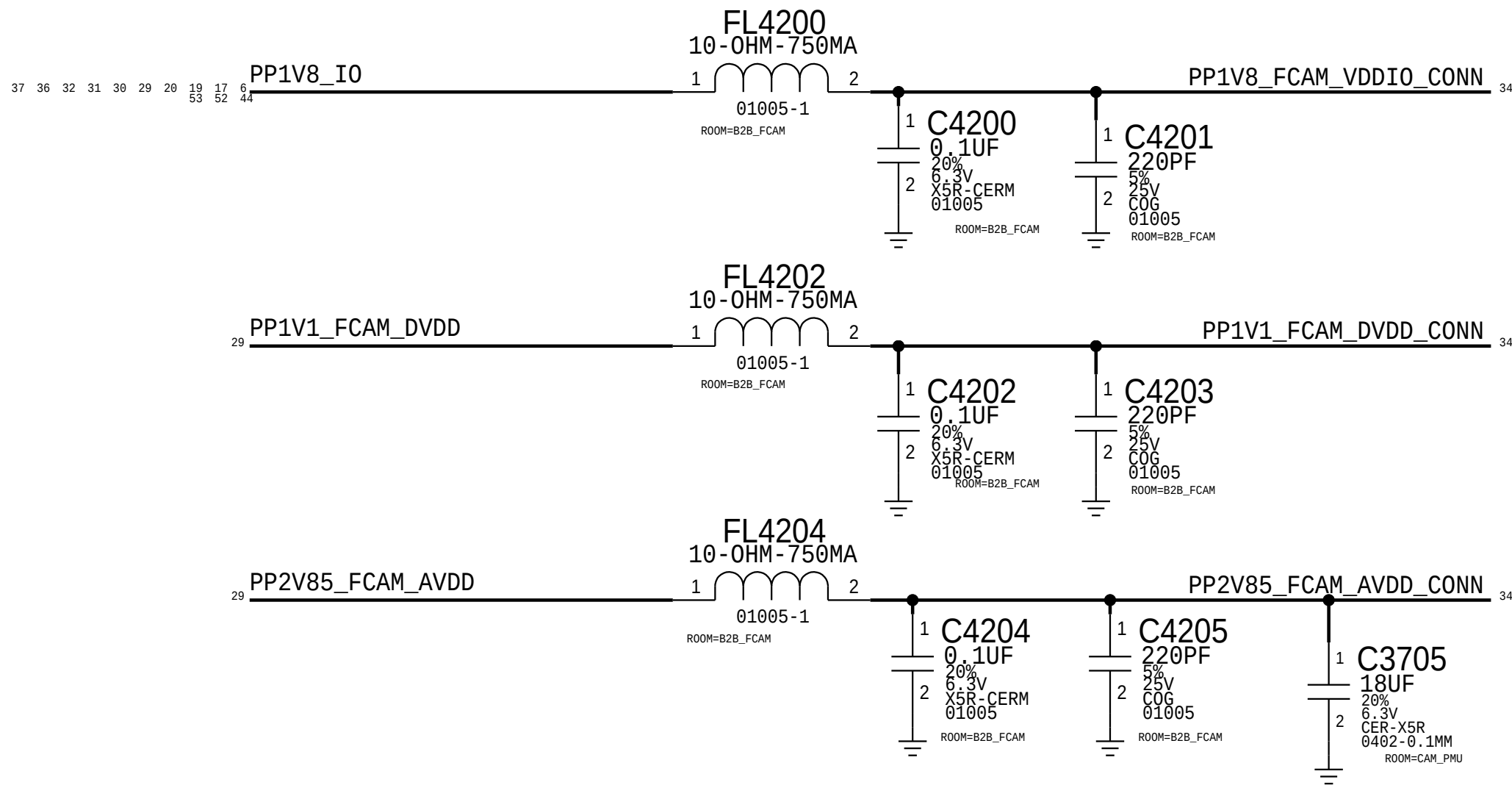


Lives Here For synccking purposes





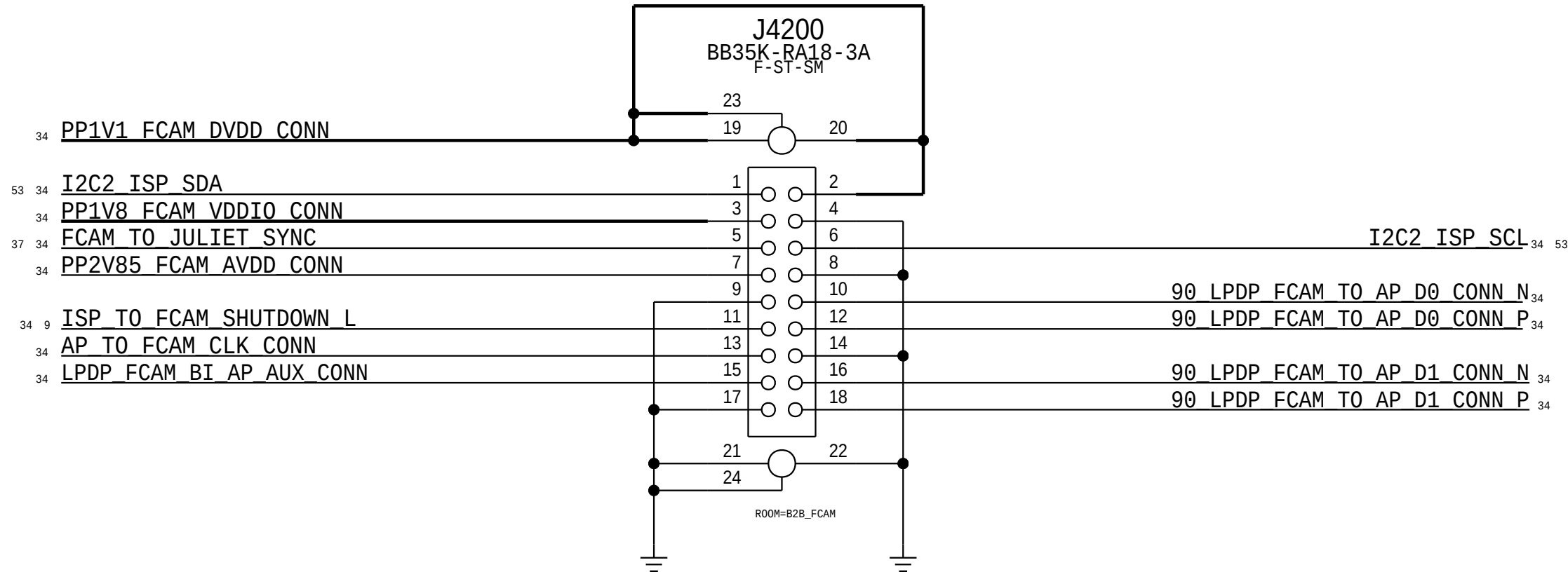
LONG ISLAND POWER



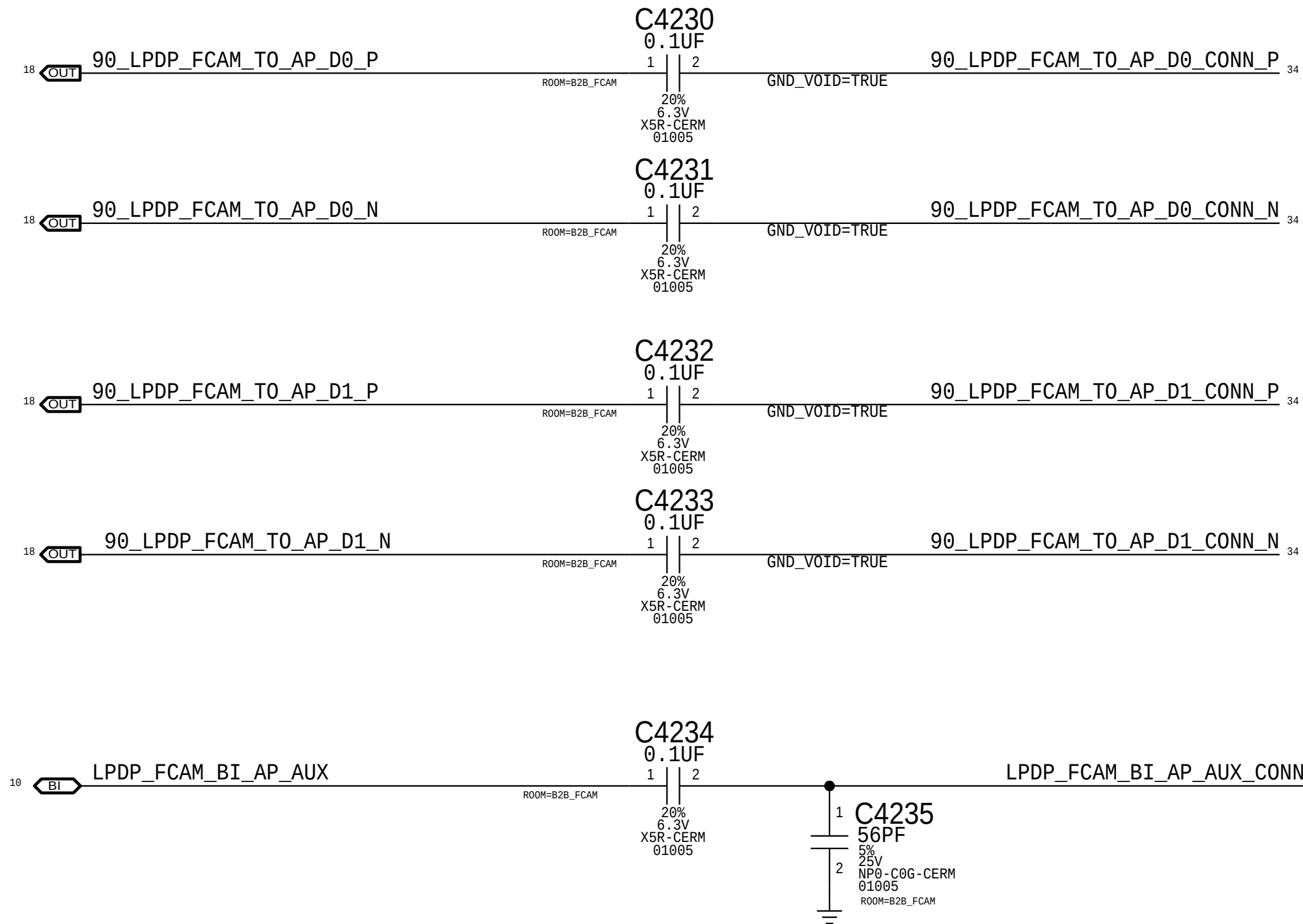
FCAM Connector

Rcpt: 516S00244
Plug: 516S00245

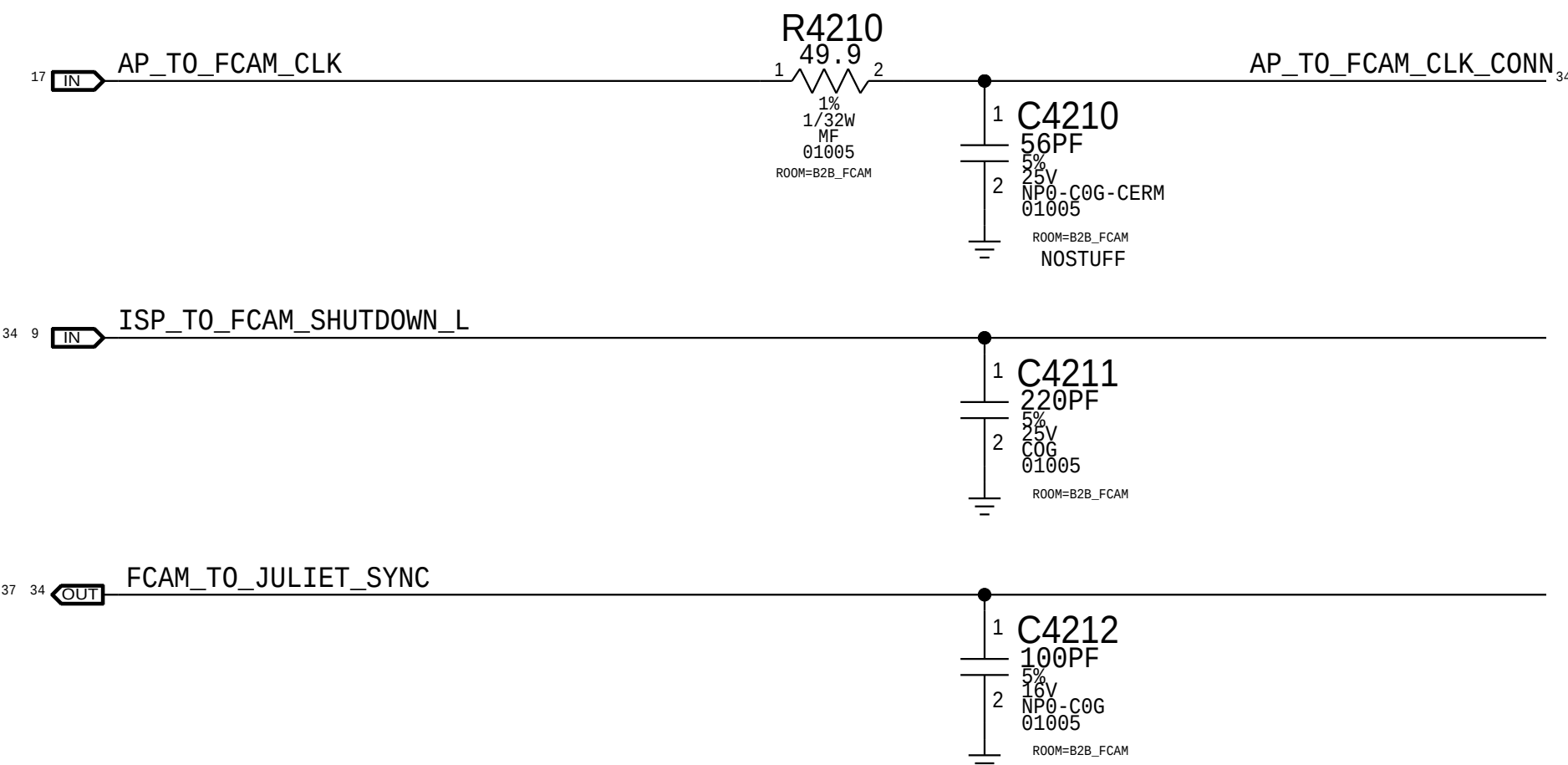
<-- This one on MLB



LPDP FILTERS

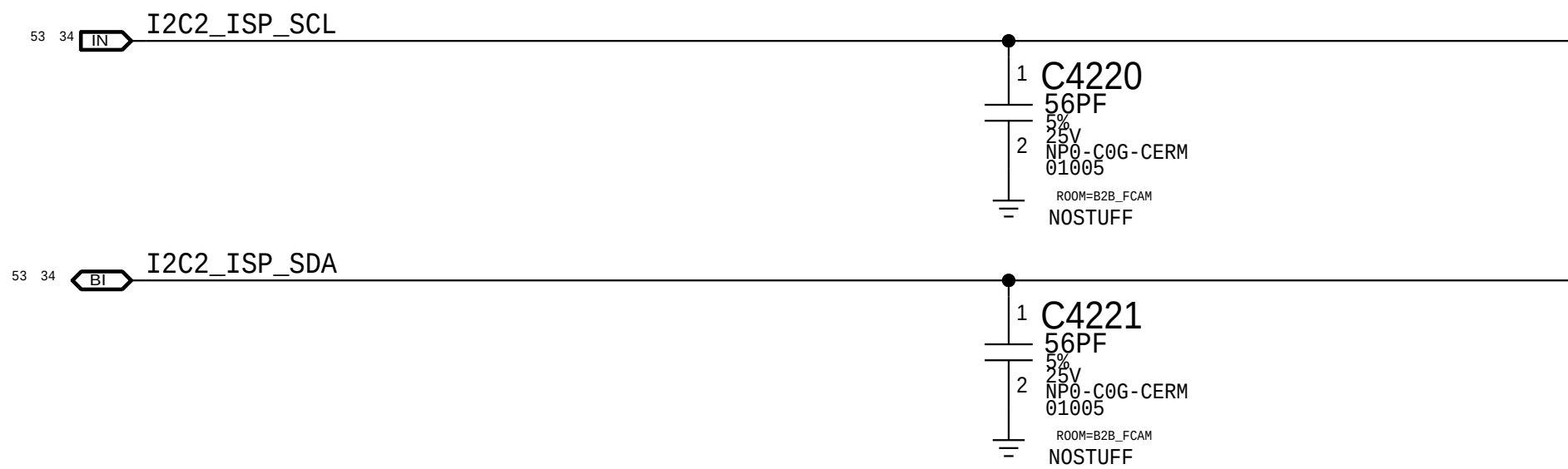


FCAM I/O



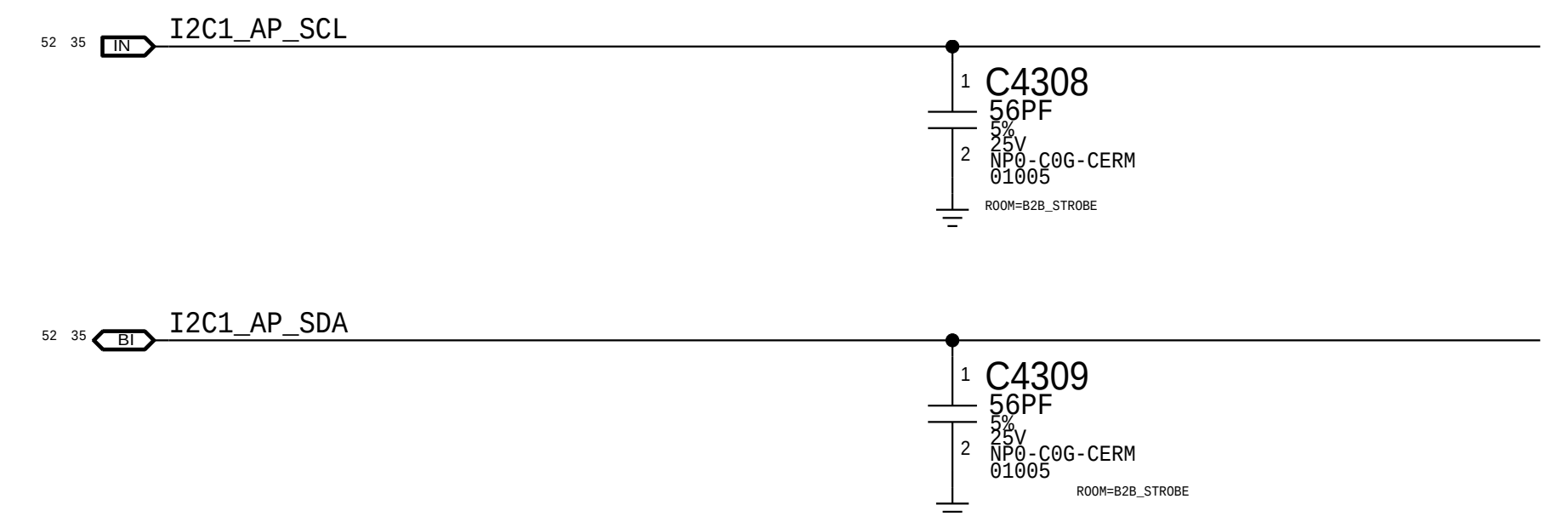
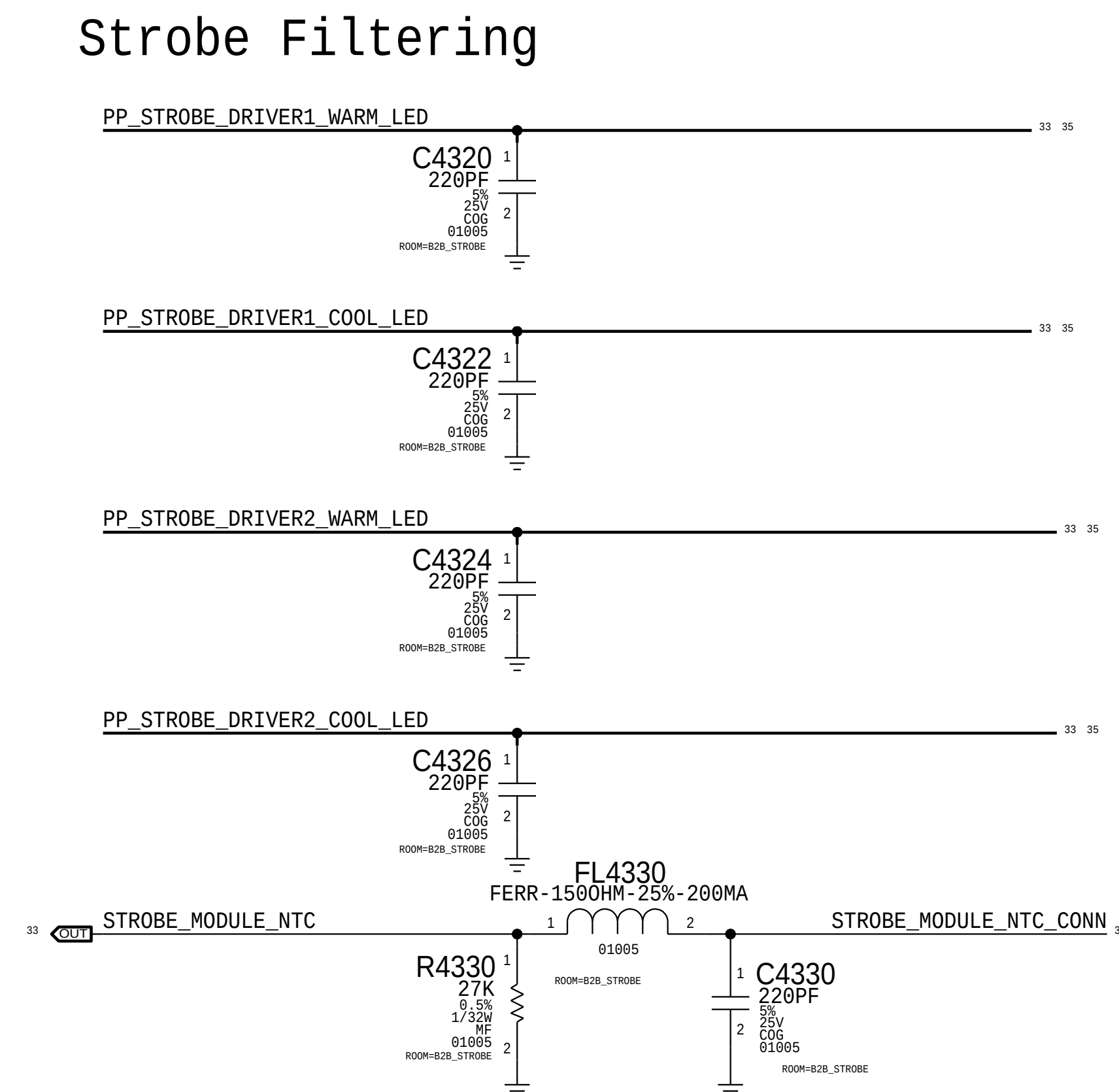
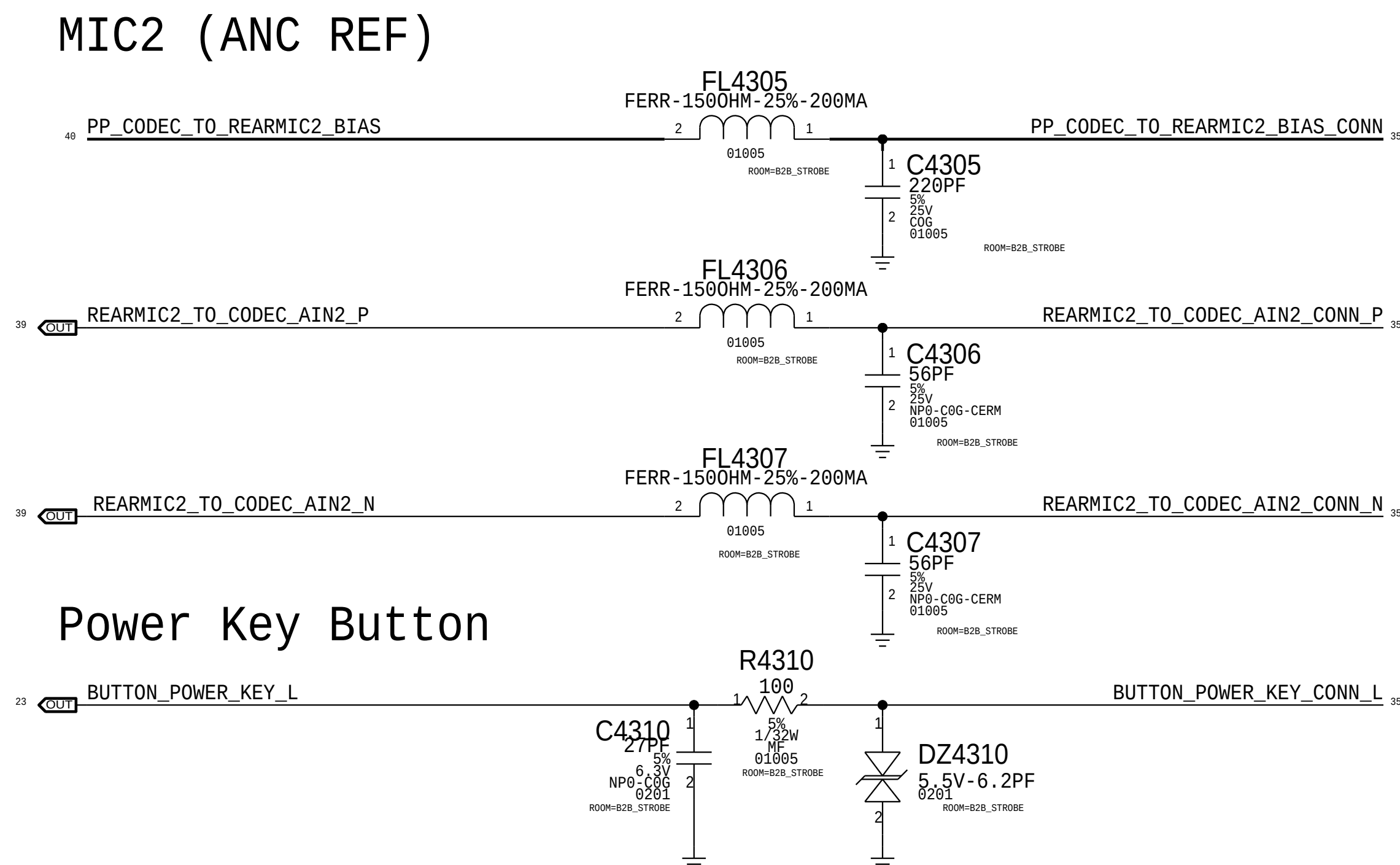
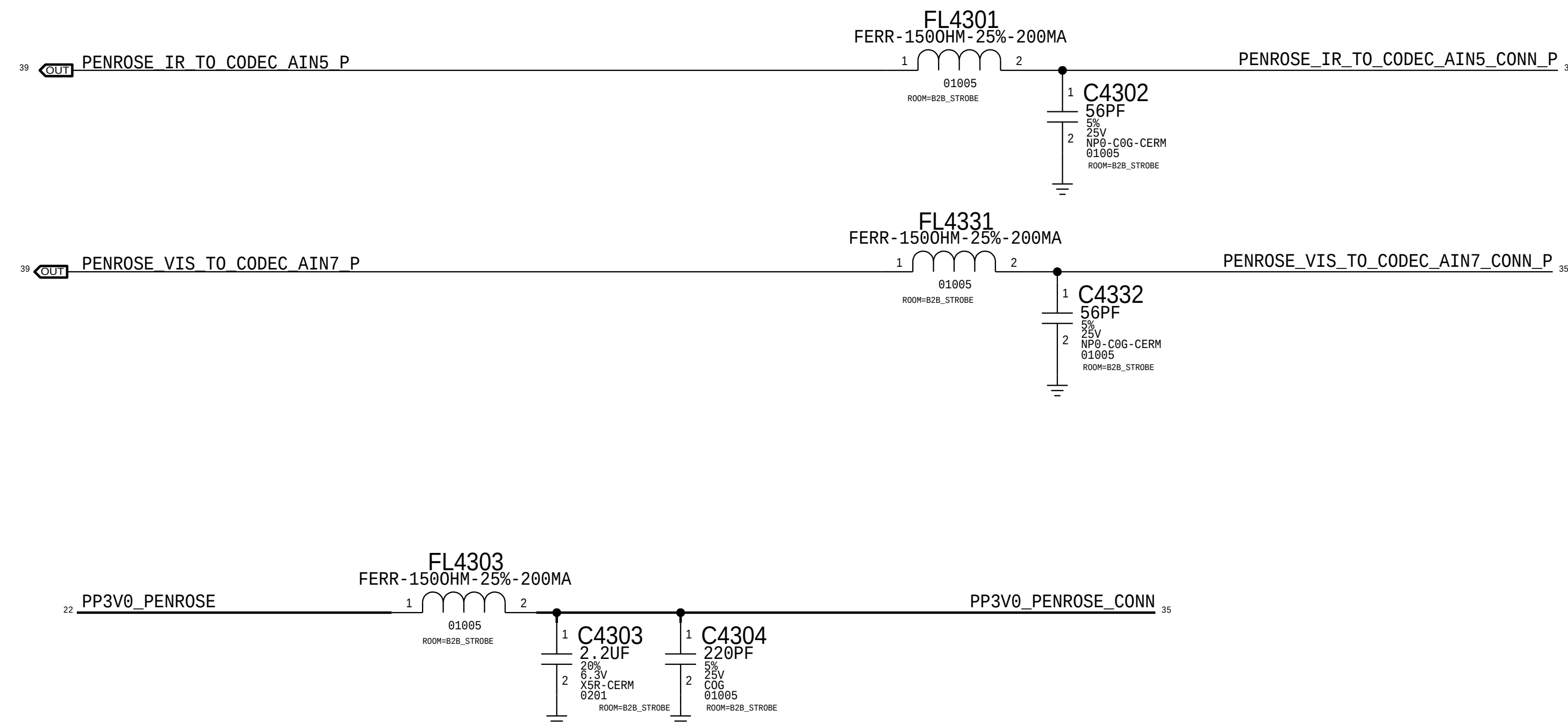
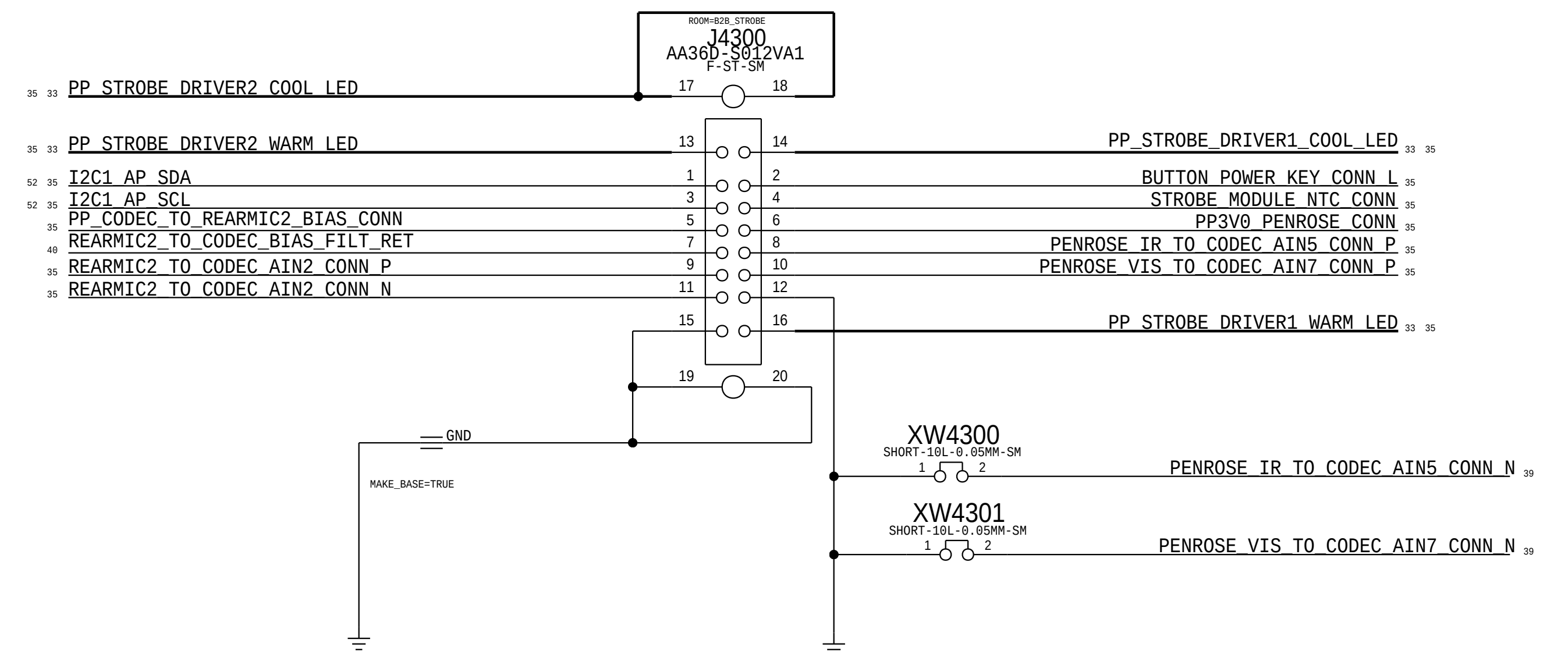
ISP I2C2

NOTE: SAME I2C as FCAM



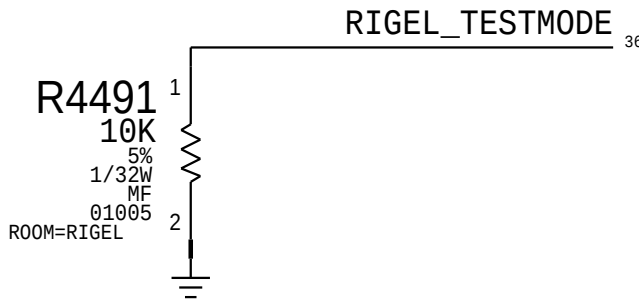


```
Rcpt: 516S00381      <-- This one on MLB
Plug: 516S00382
```



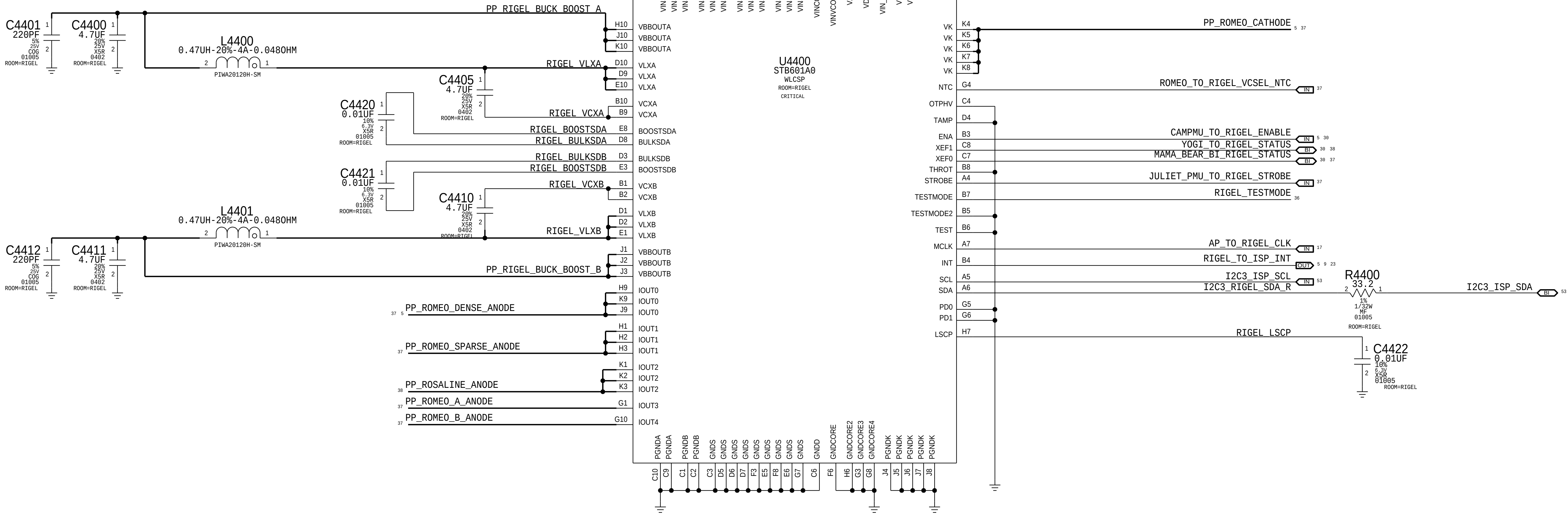
Rigel Driver

Test Mode Debugging

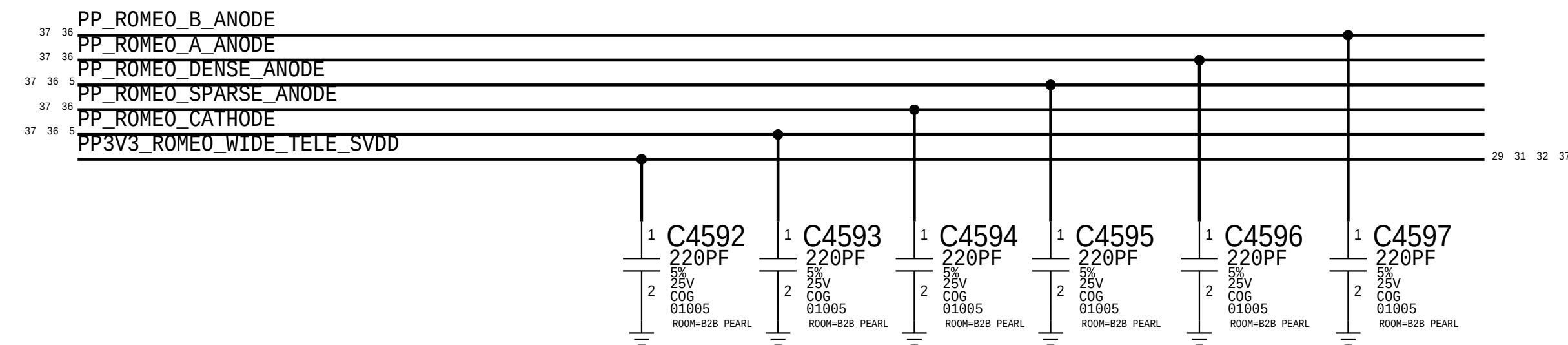


Rigel ALTs

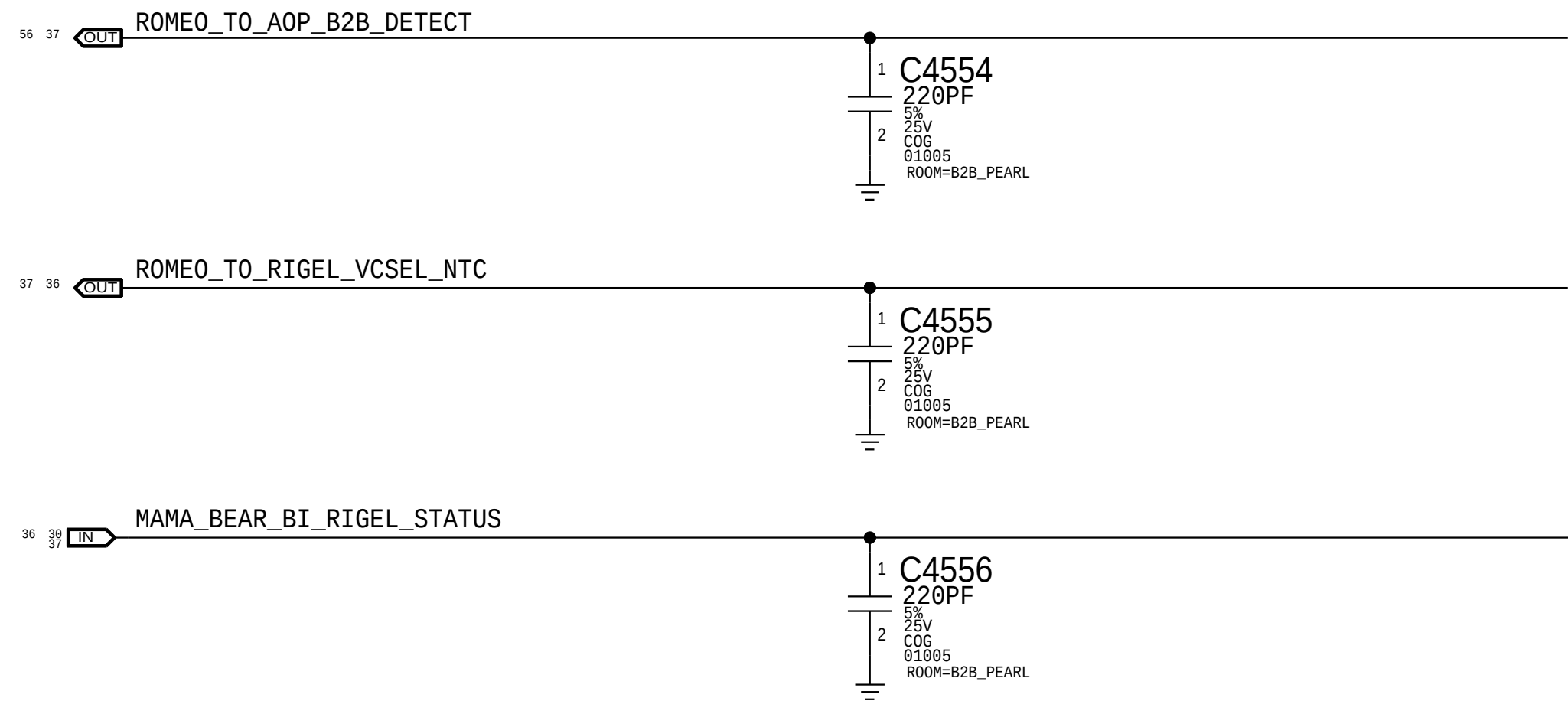
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S90720	152S90640	ALT_PARTS	L4490, L4491	RIGEL Inductors



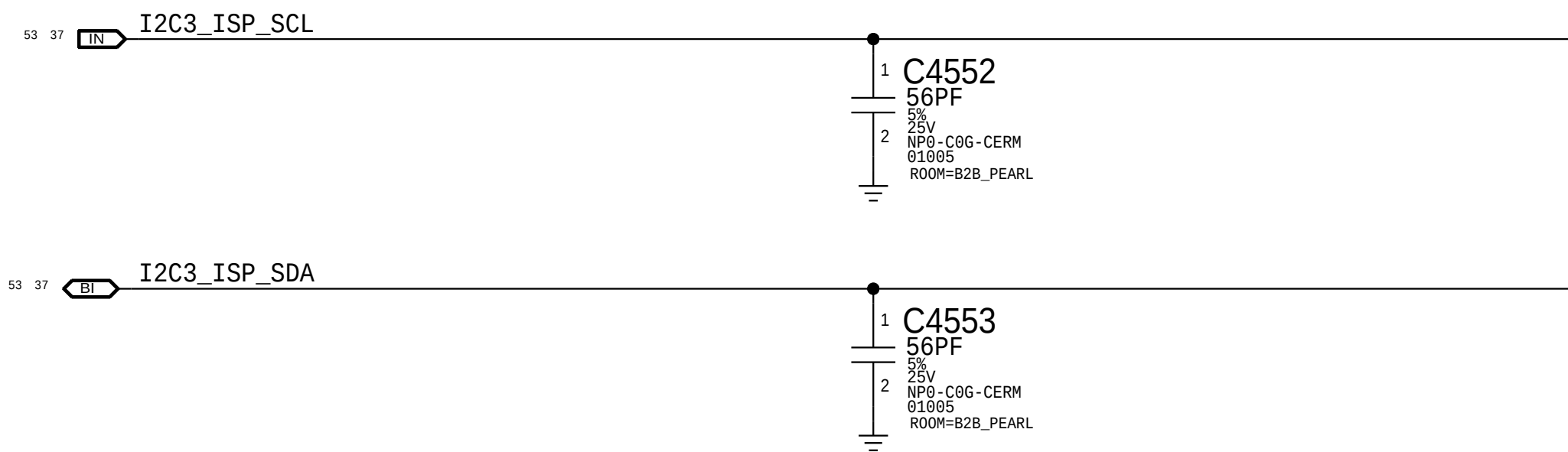
Romeo Power Filtering



Romeo I/O

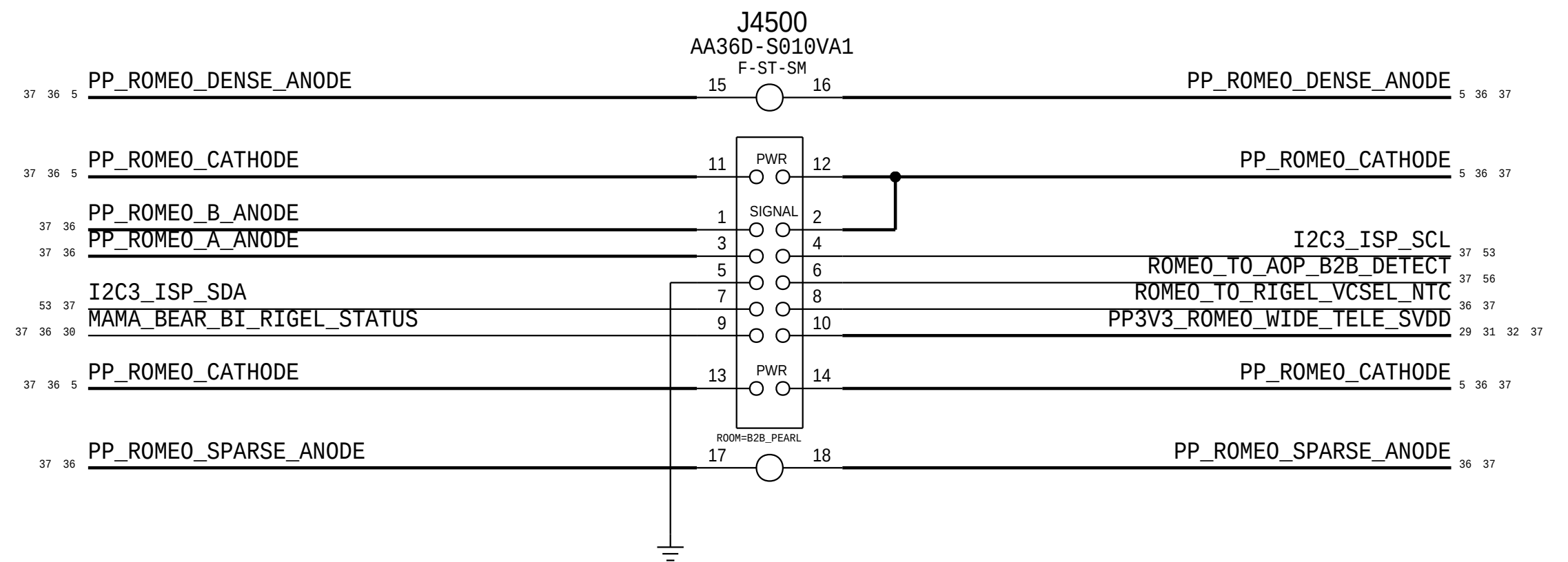


ISP I2C3



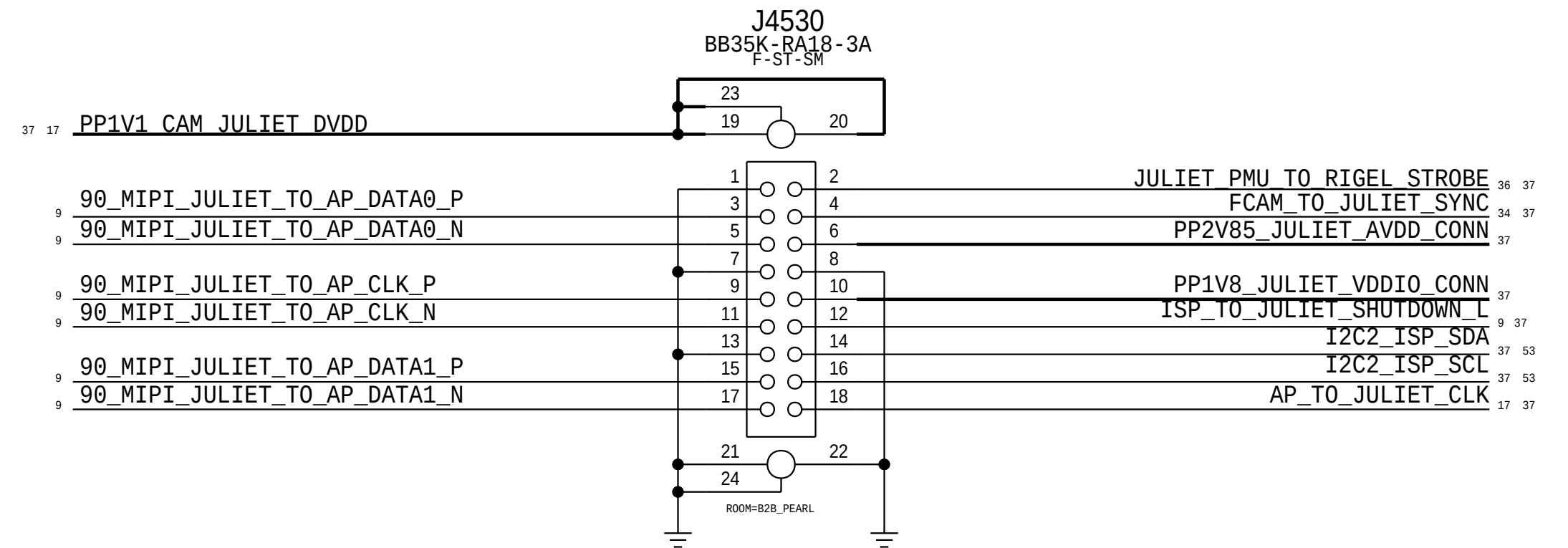
Romeo Connector

Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268

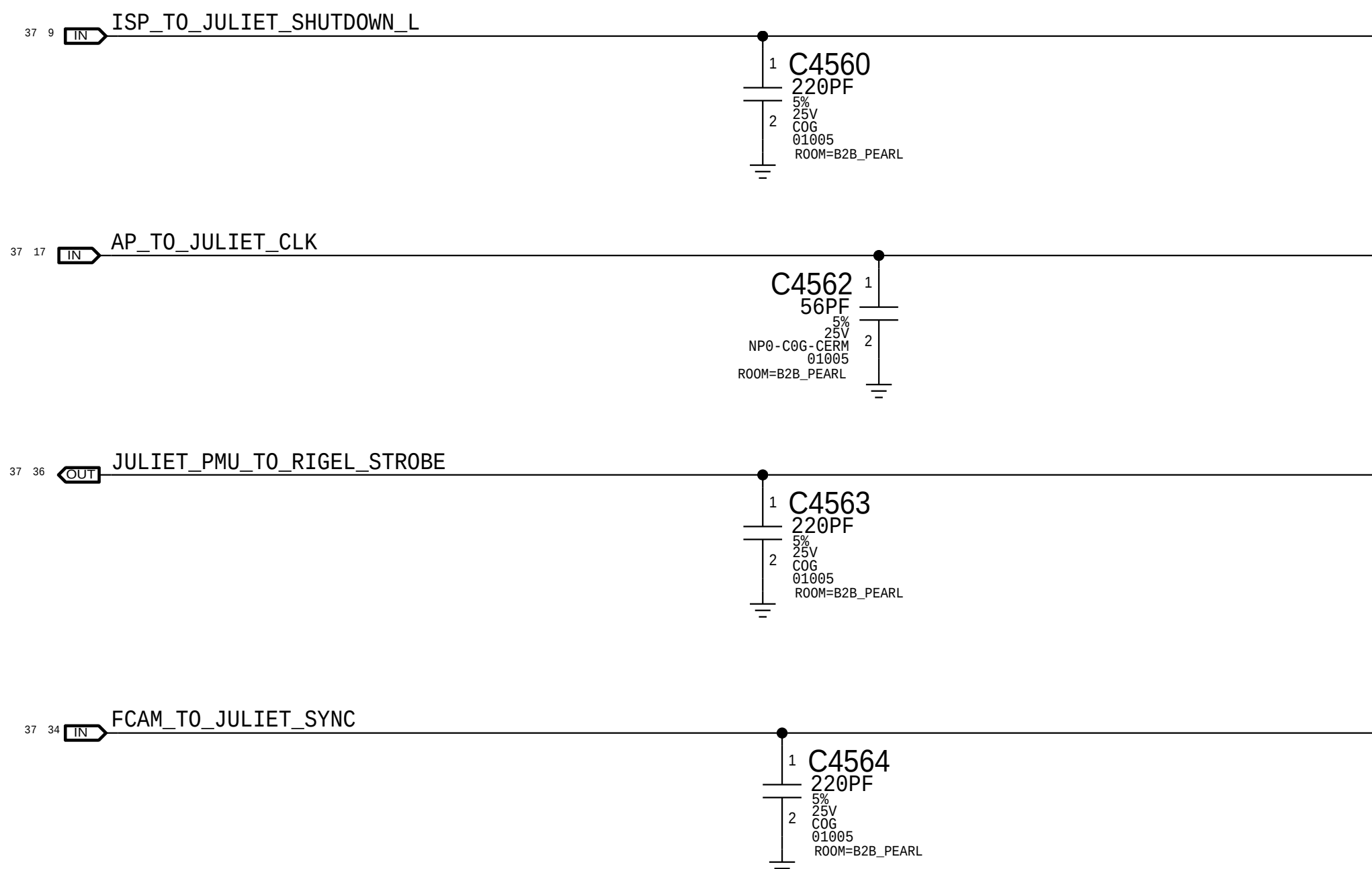
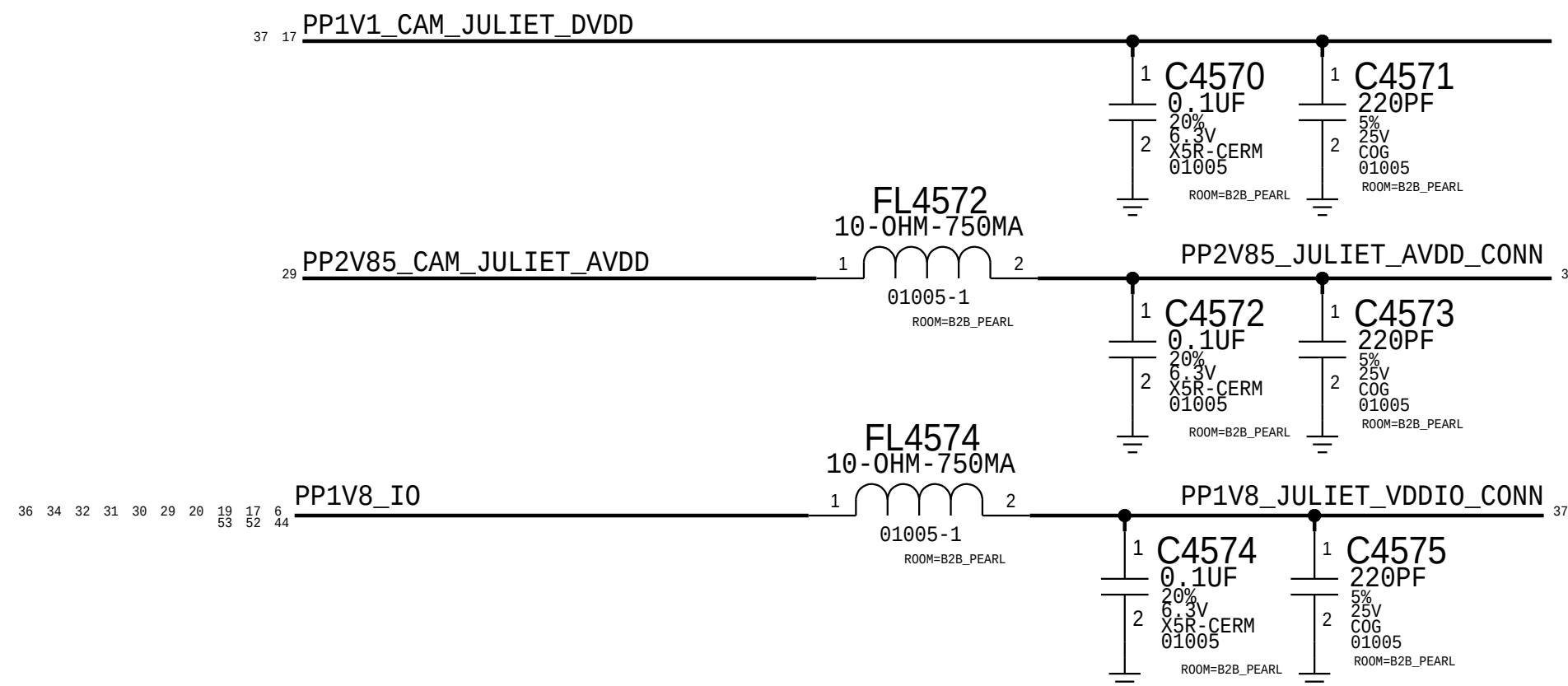


Juliet Connector

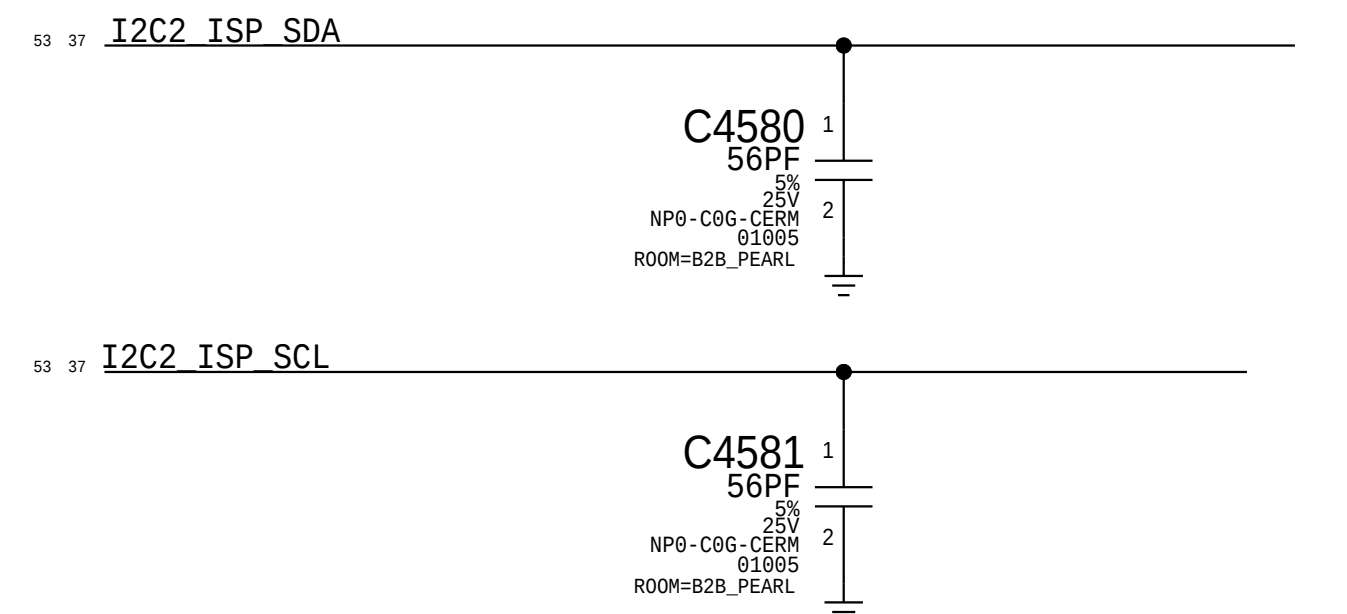
Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245



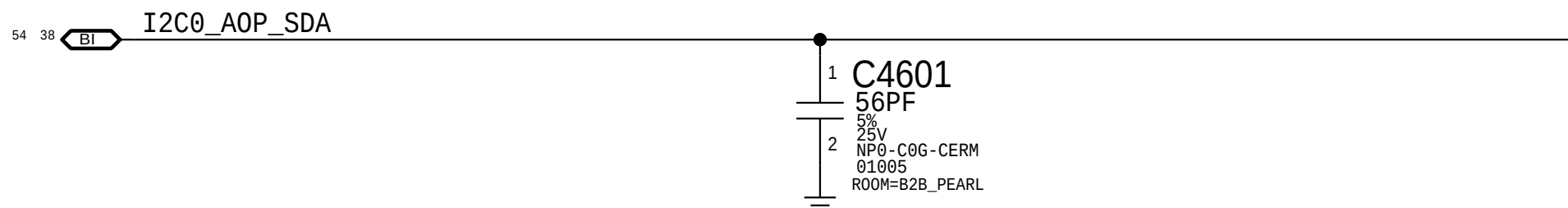
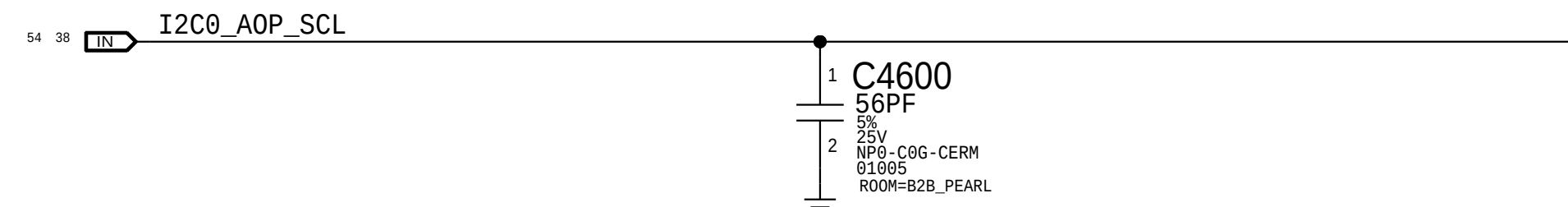
Juliet Power and I/O



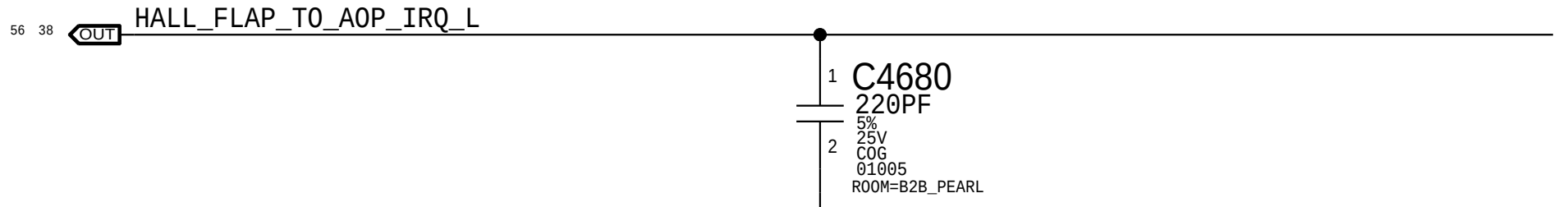
NOTE: SAME I2C as FCAM



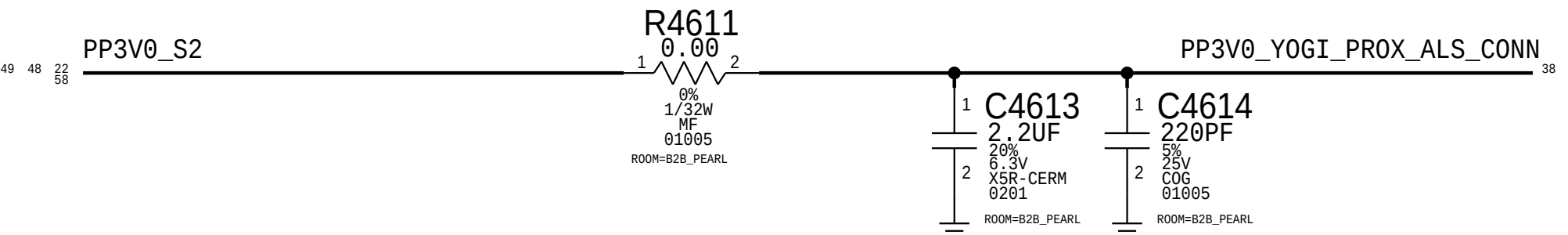
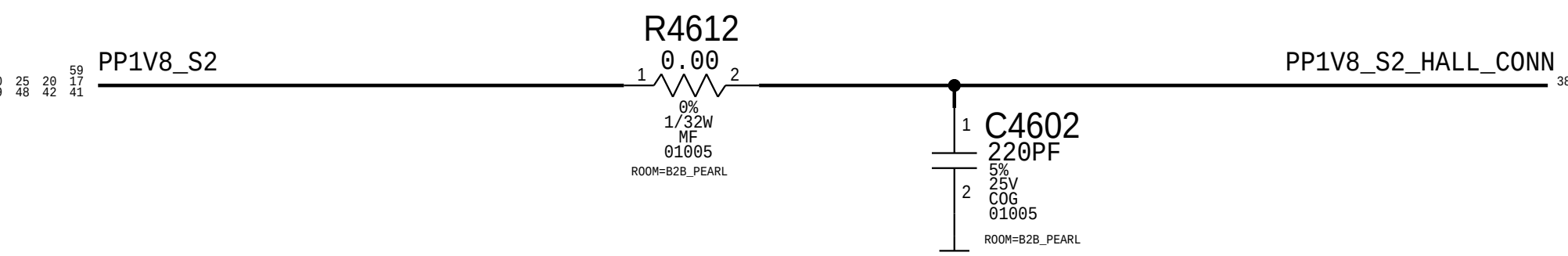
AOP I2C



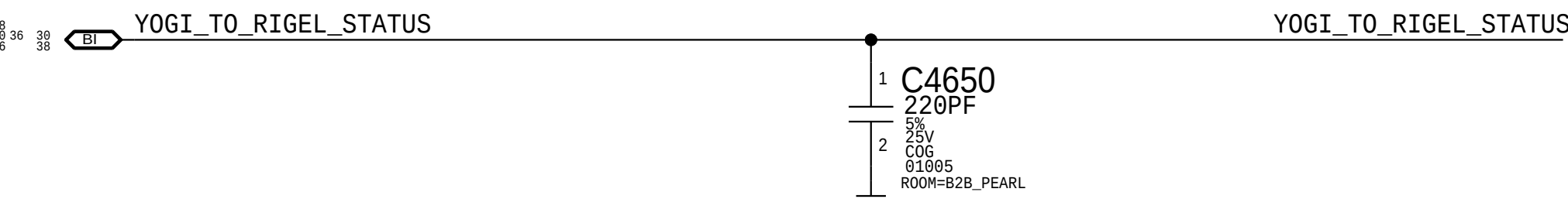
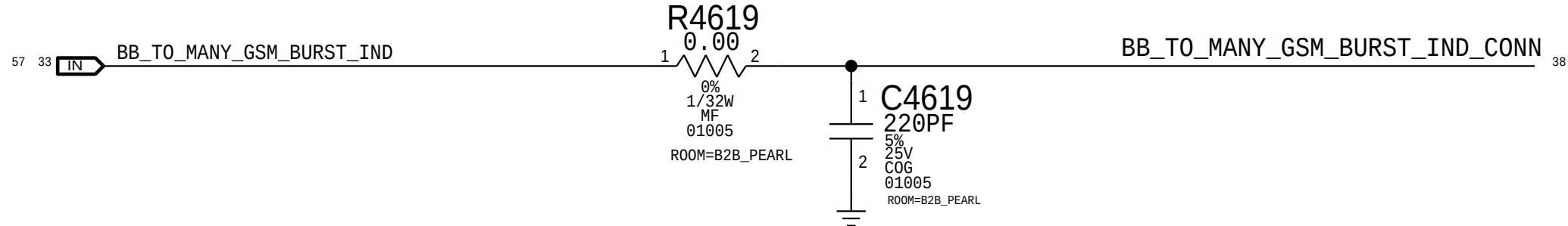
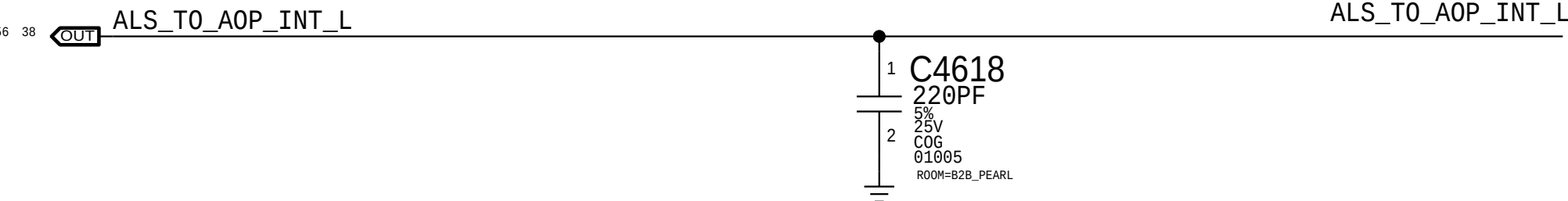
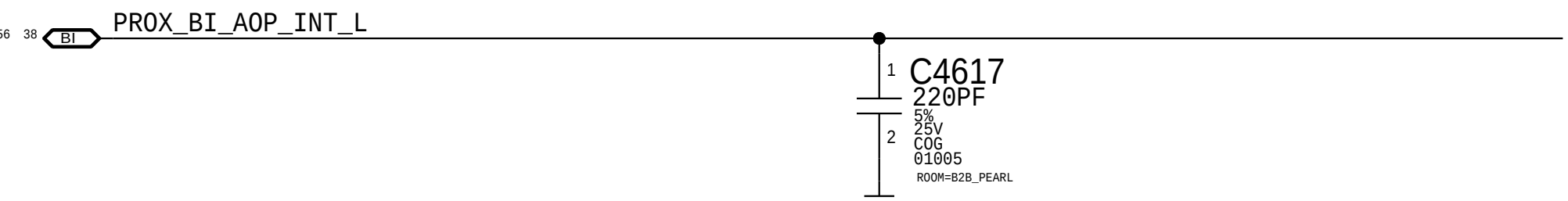
HALL I/Os



PROX & HALL POWER

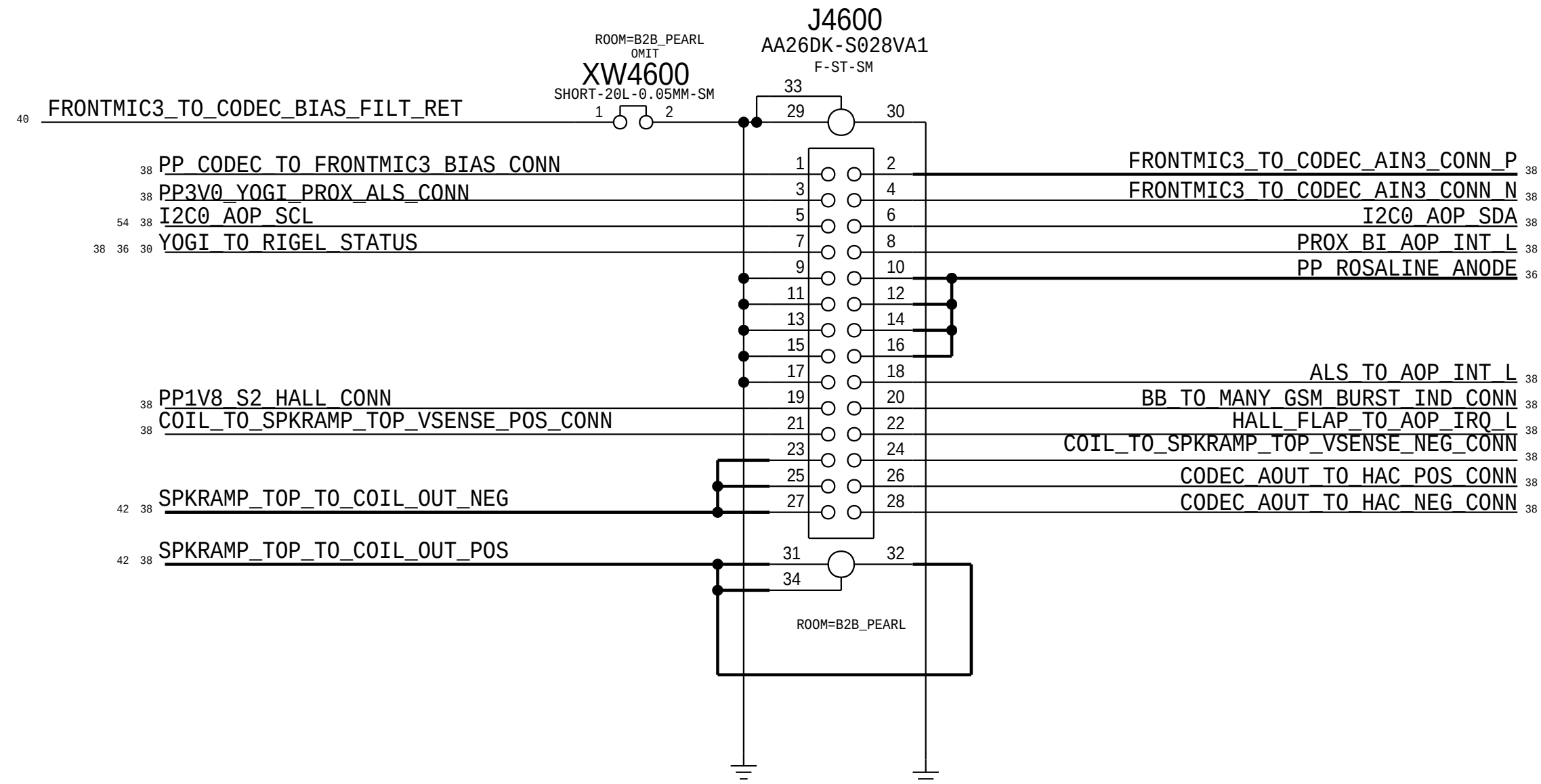


PROX/ALS/YOGI I/O

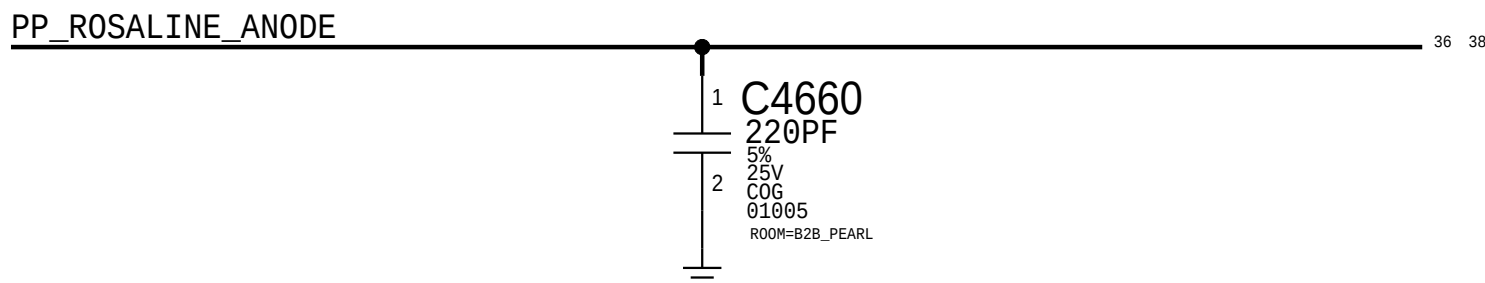


Rosaline + Sensor Connector

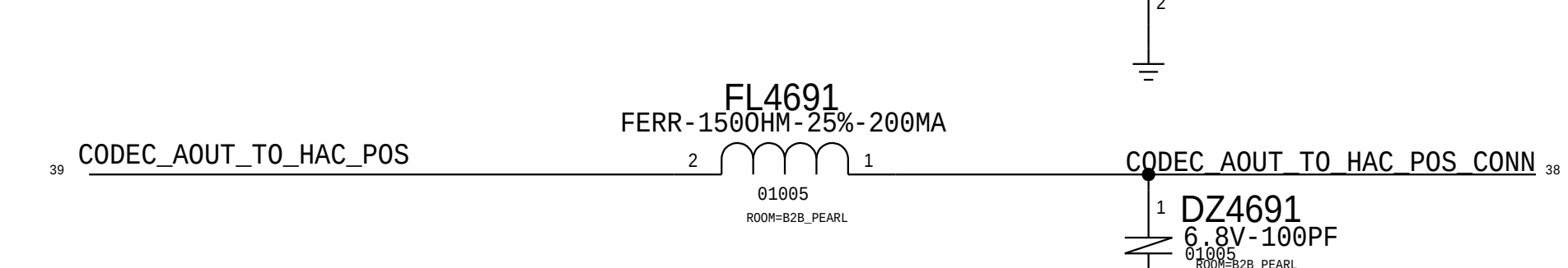
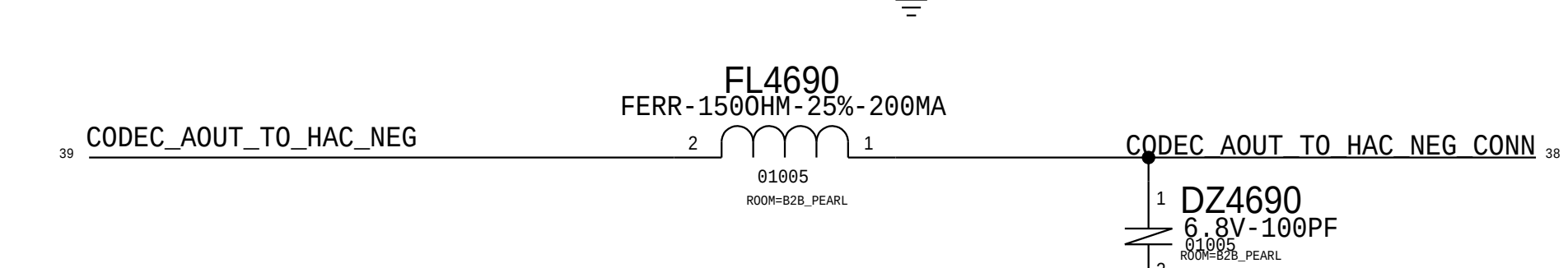
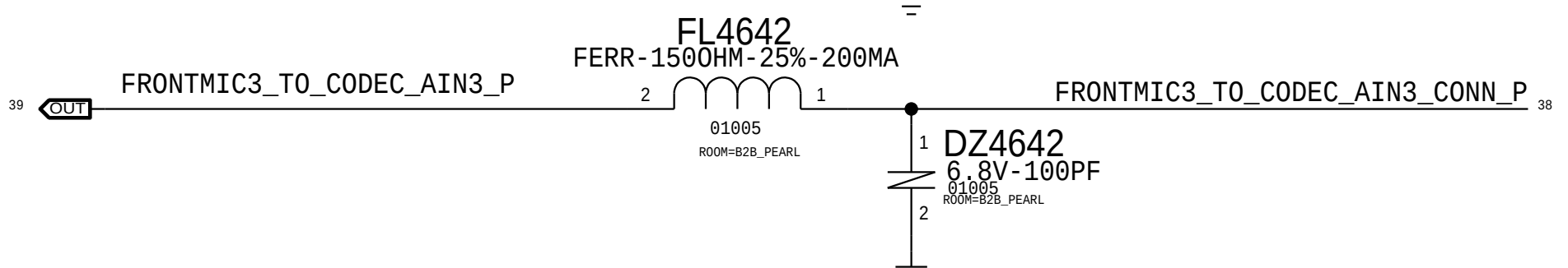
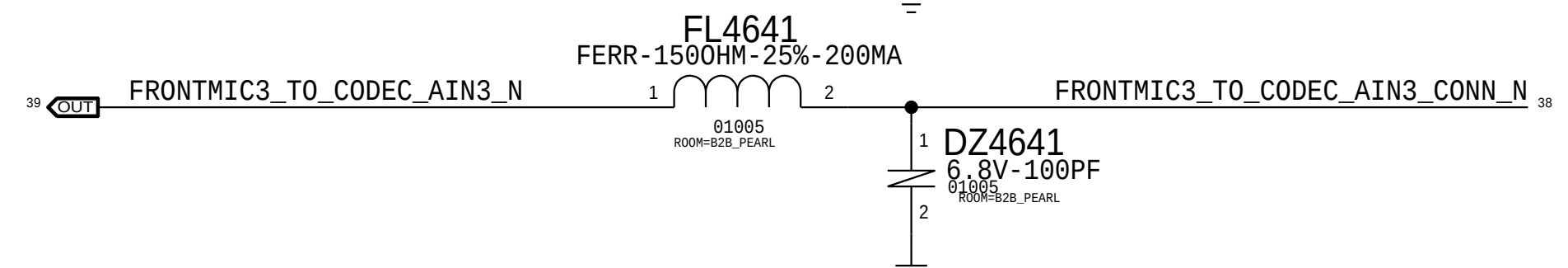
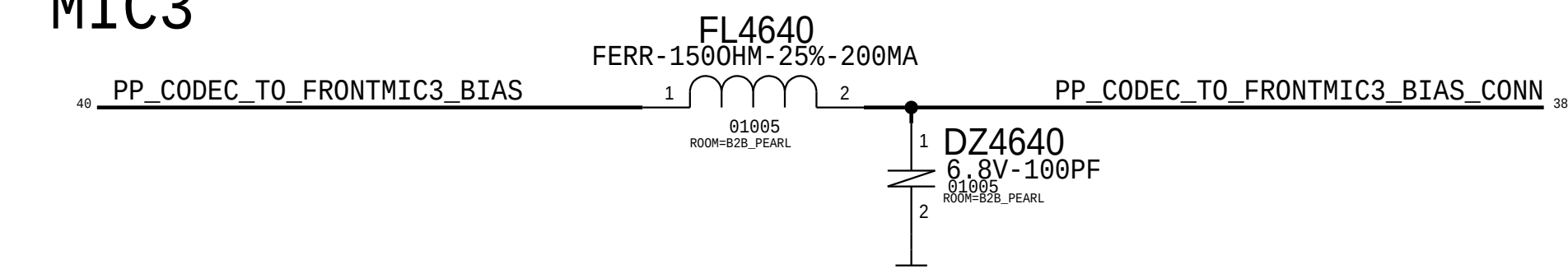
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Rcpt: 516S00325      <-- This one on MLB
Plug: 516S00326
```



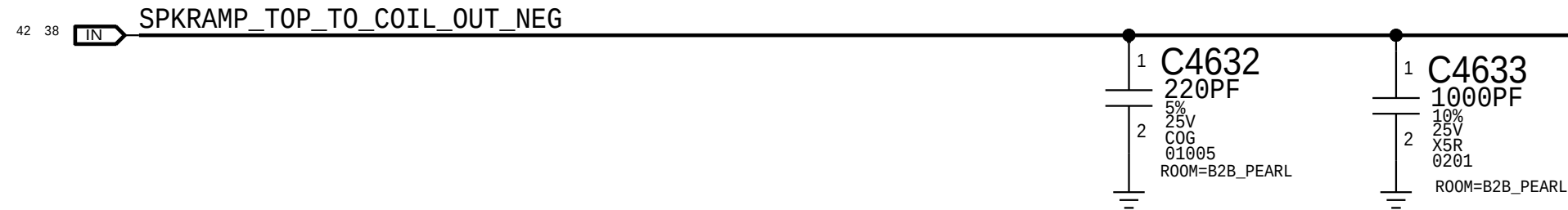
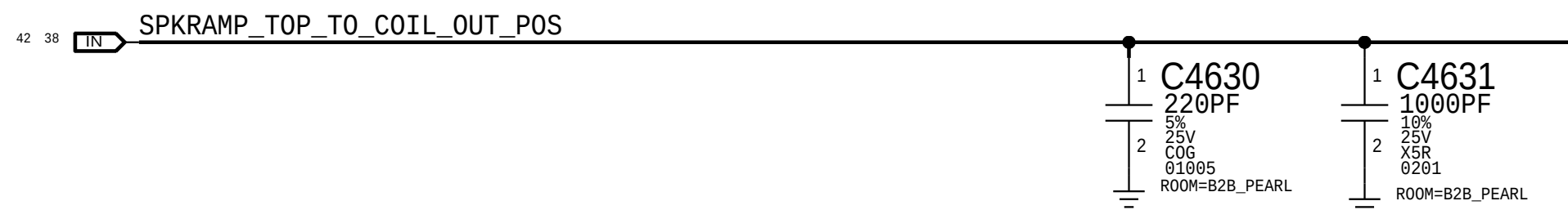
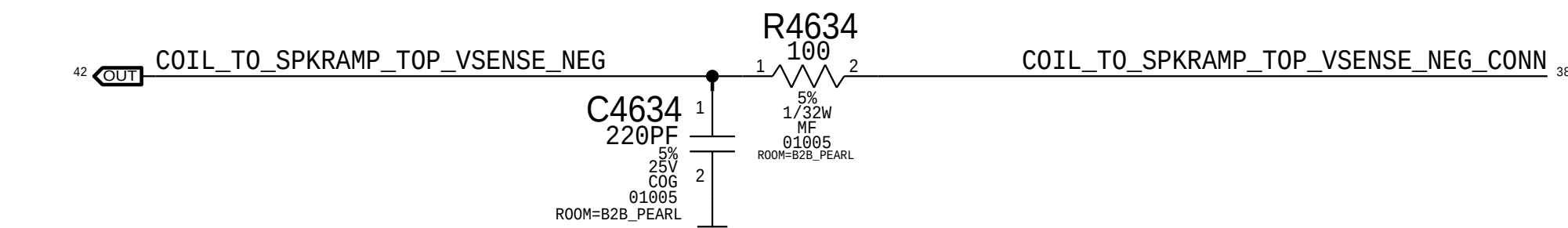
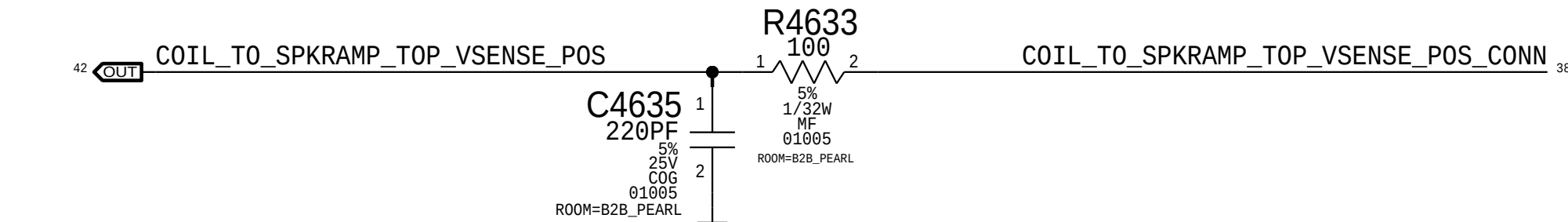
Yogi Signals



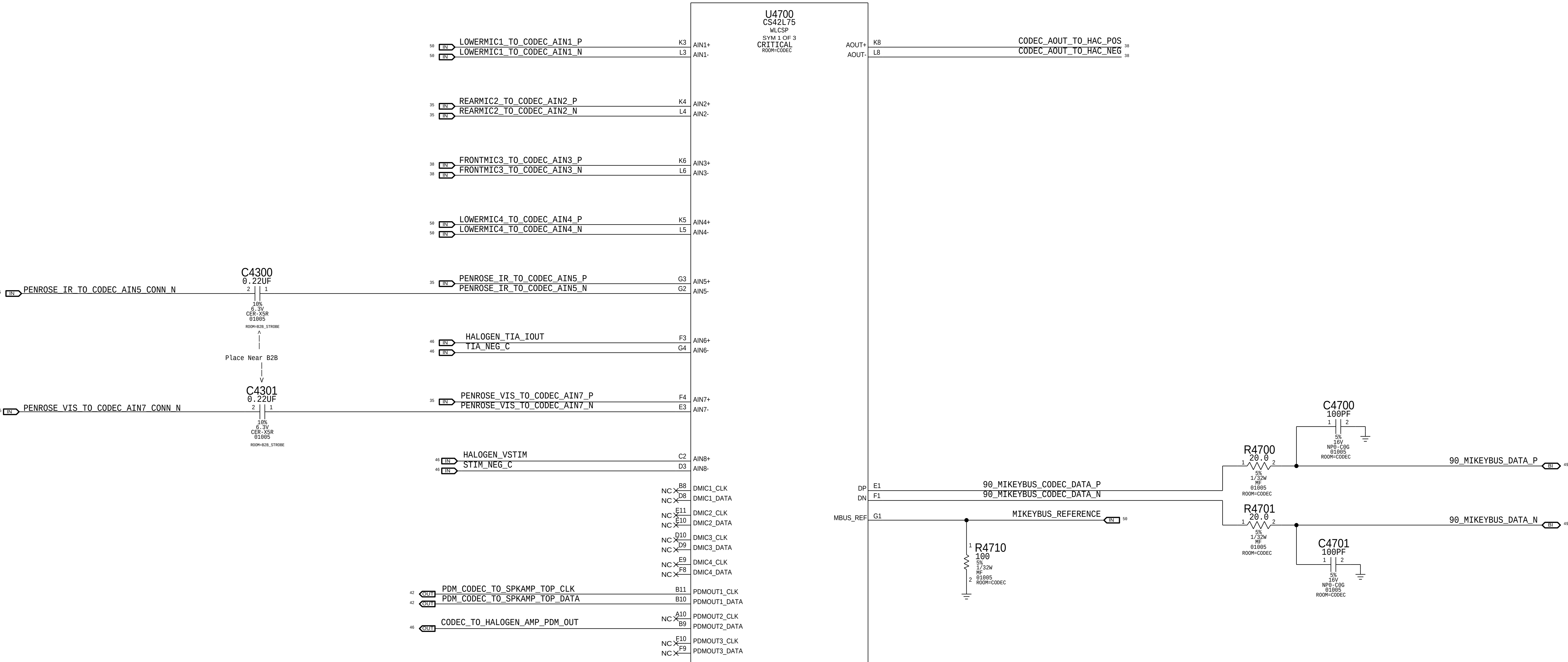
MIC3



SPEAKER2



CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



CALLAN AUDIO CODEC (POWER & I/O)

D

C

B

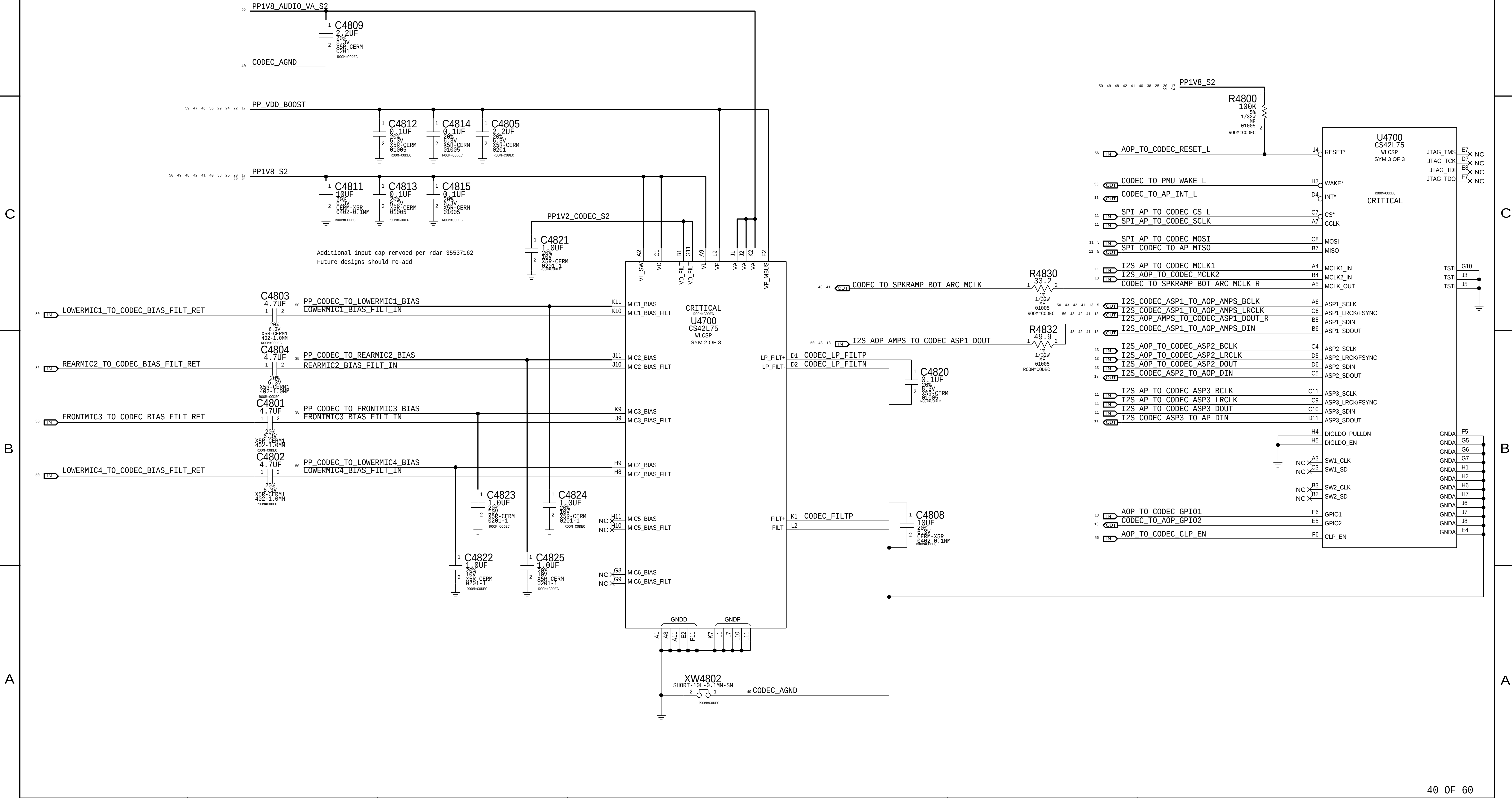
A

D

C

B

A



D

C

B

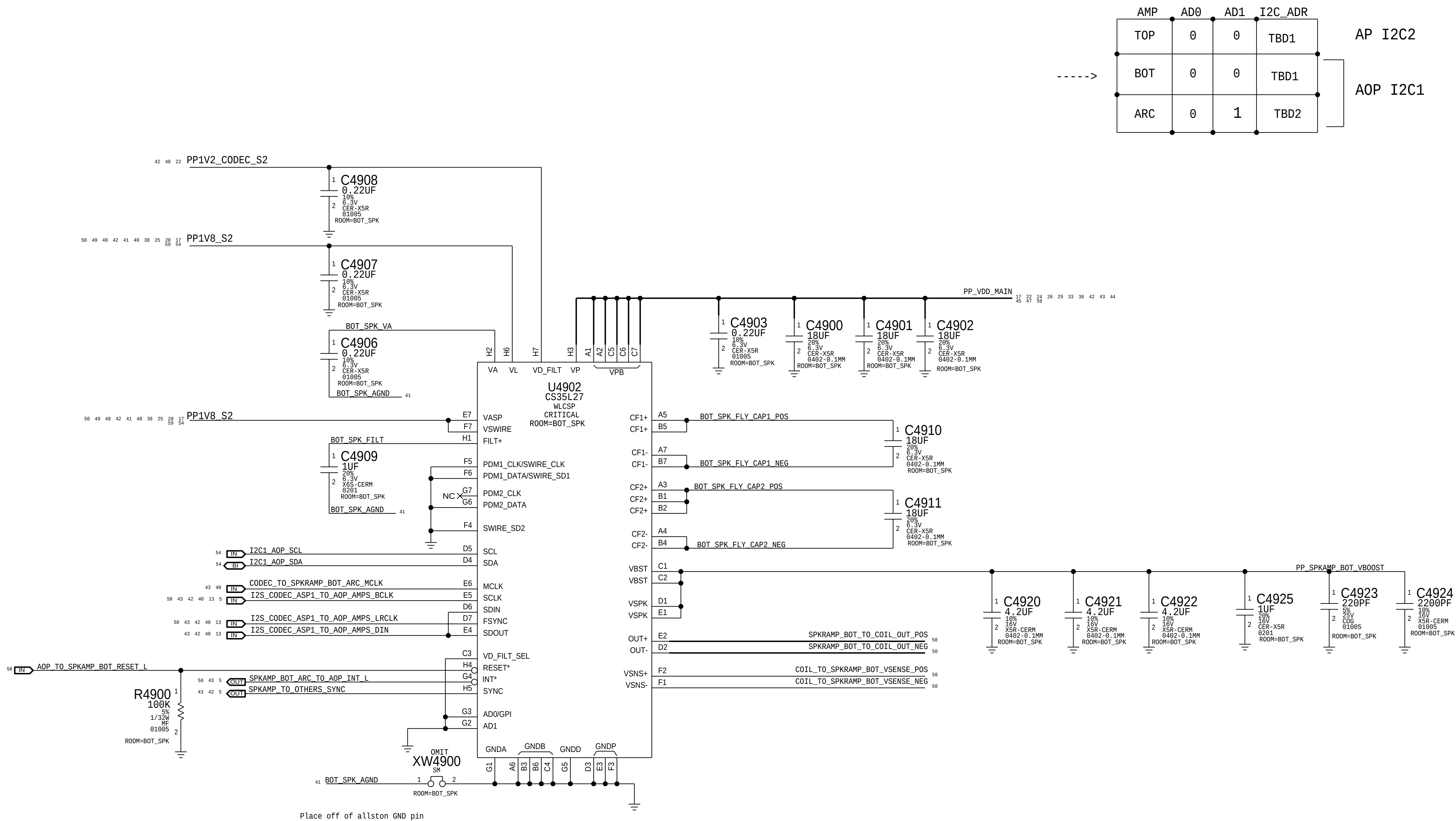
A

D

C

B

A



D

C

B

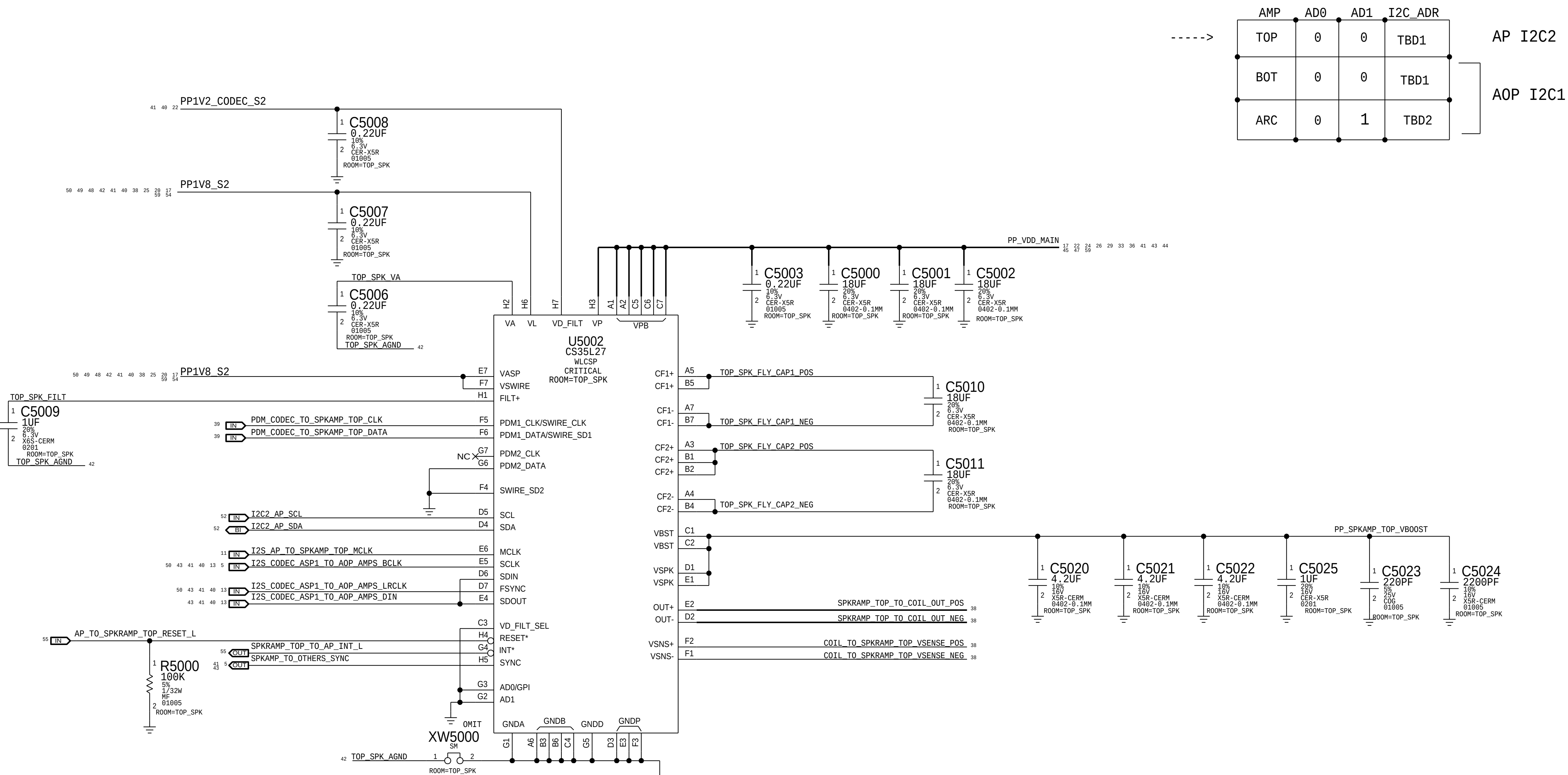
A

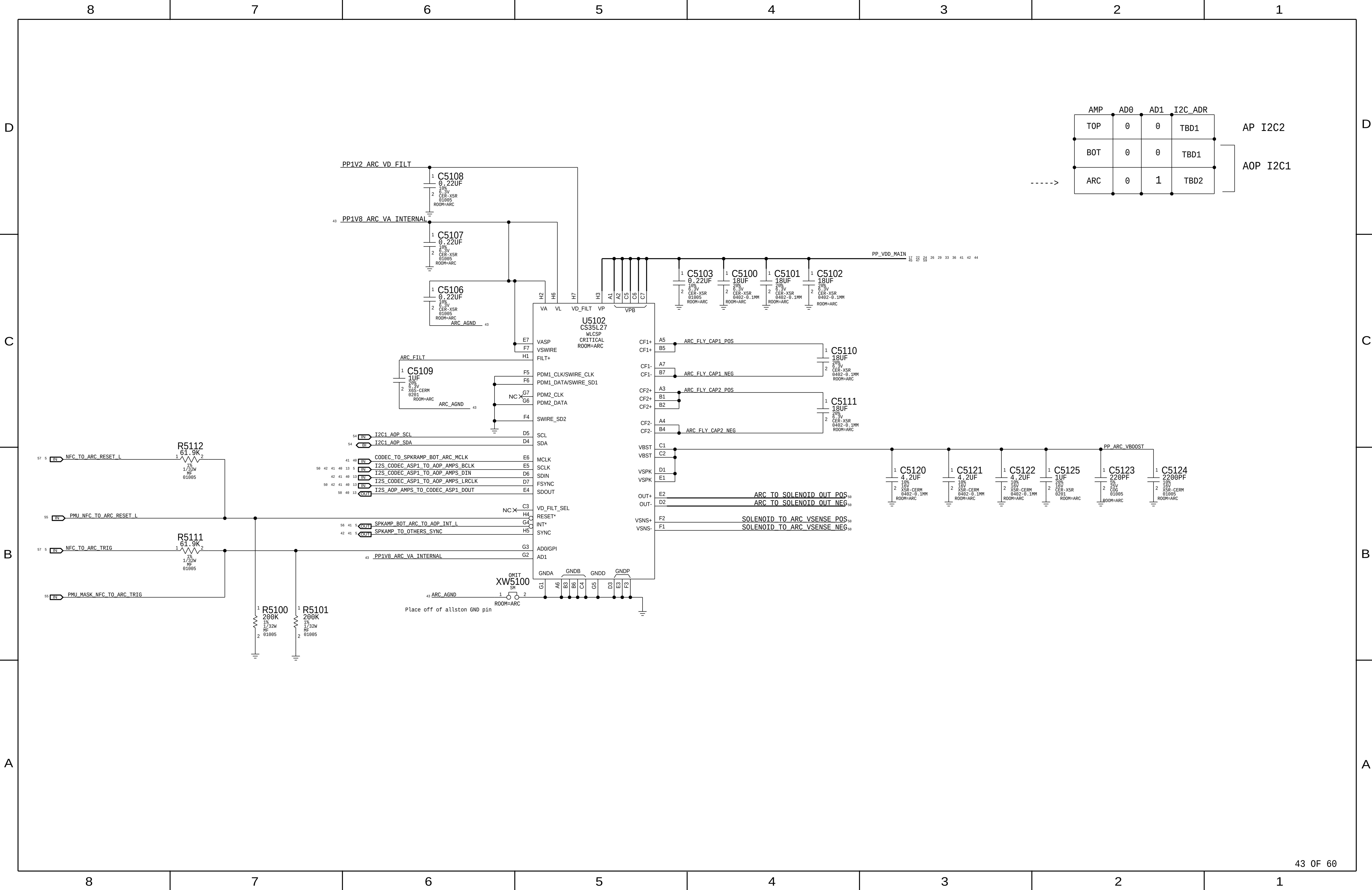
D

C

B

A





----->

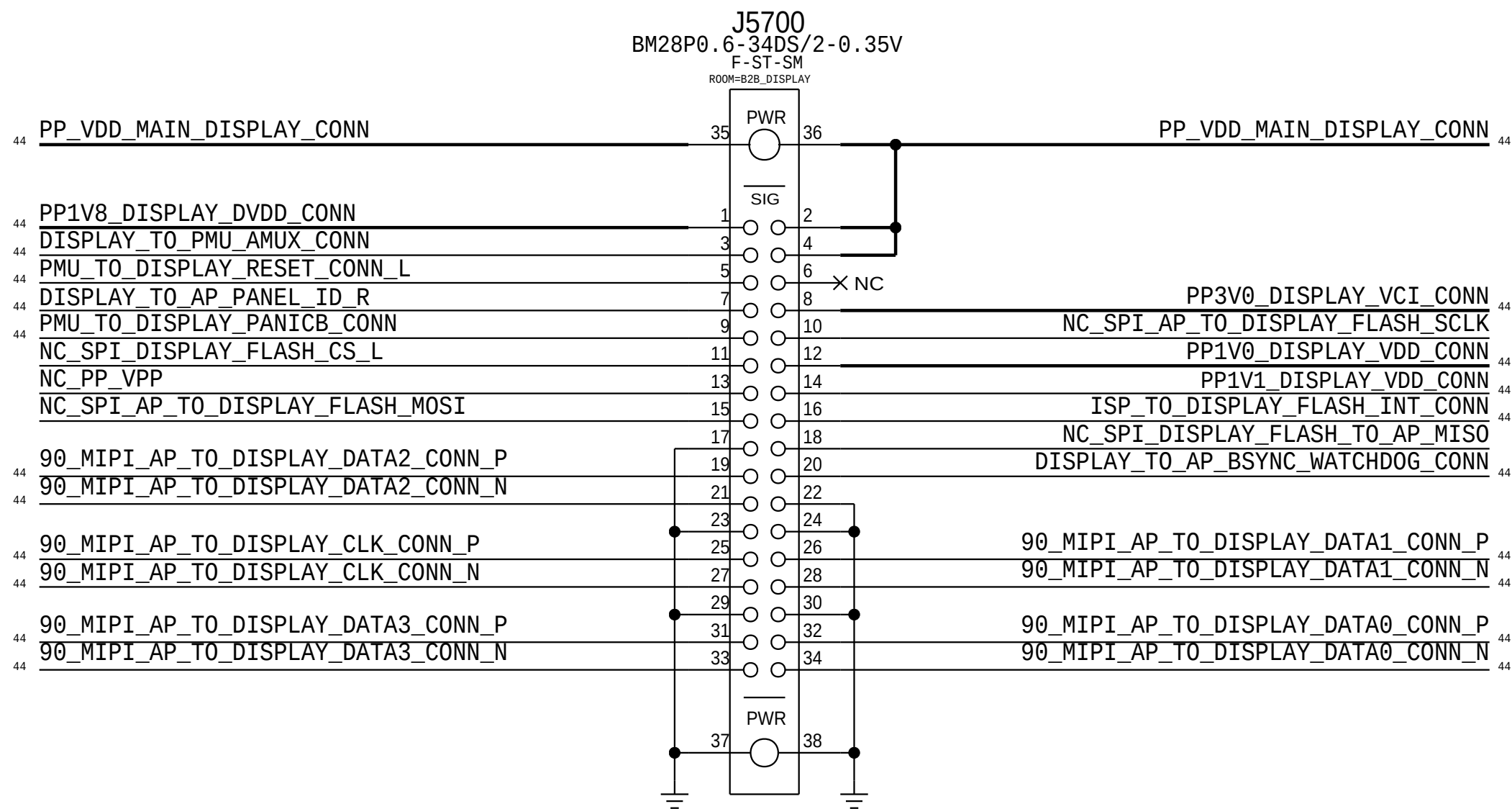
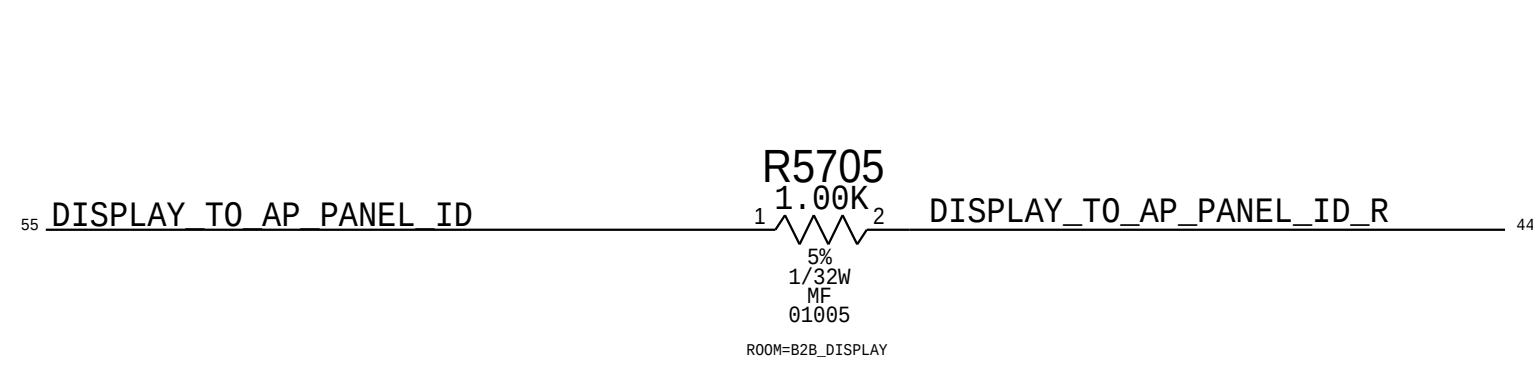
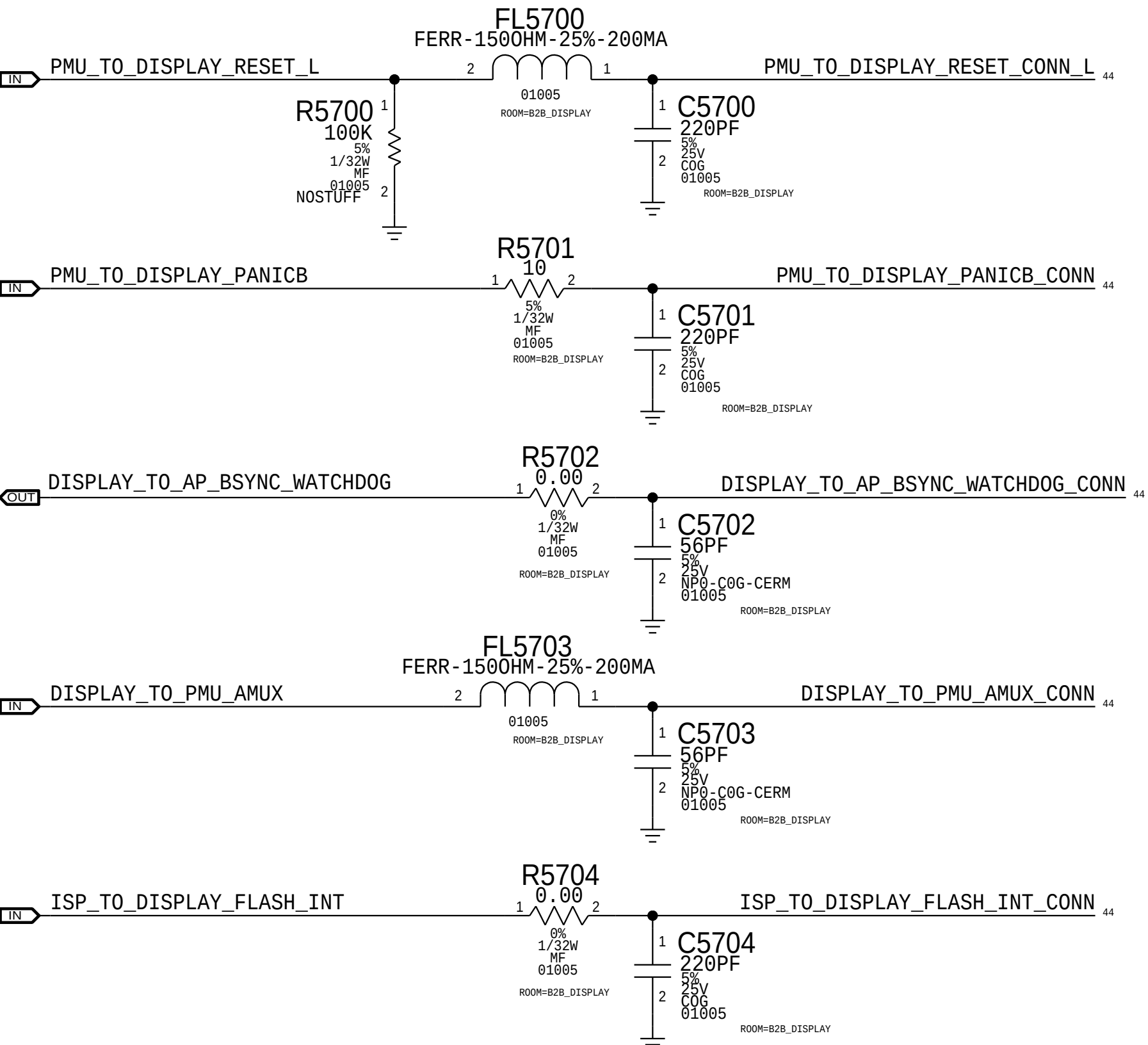
AMP	AD0	AD1	I2C_ADR	
TOP	0	0	TBD1	AP I2C2
BOT	0	0	TBD1	AOP I2C1
ARC	0	1	TBD2	

Display Control Signals

Display 1V0 LD0 for D33 second display vendor
rdar: #29872369

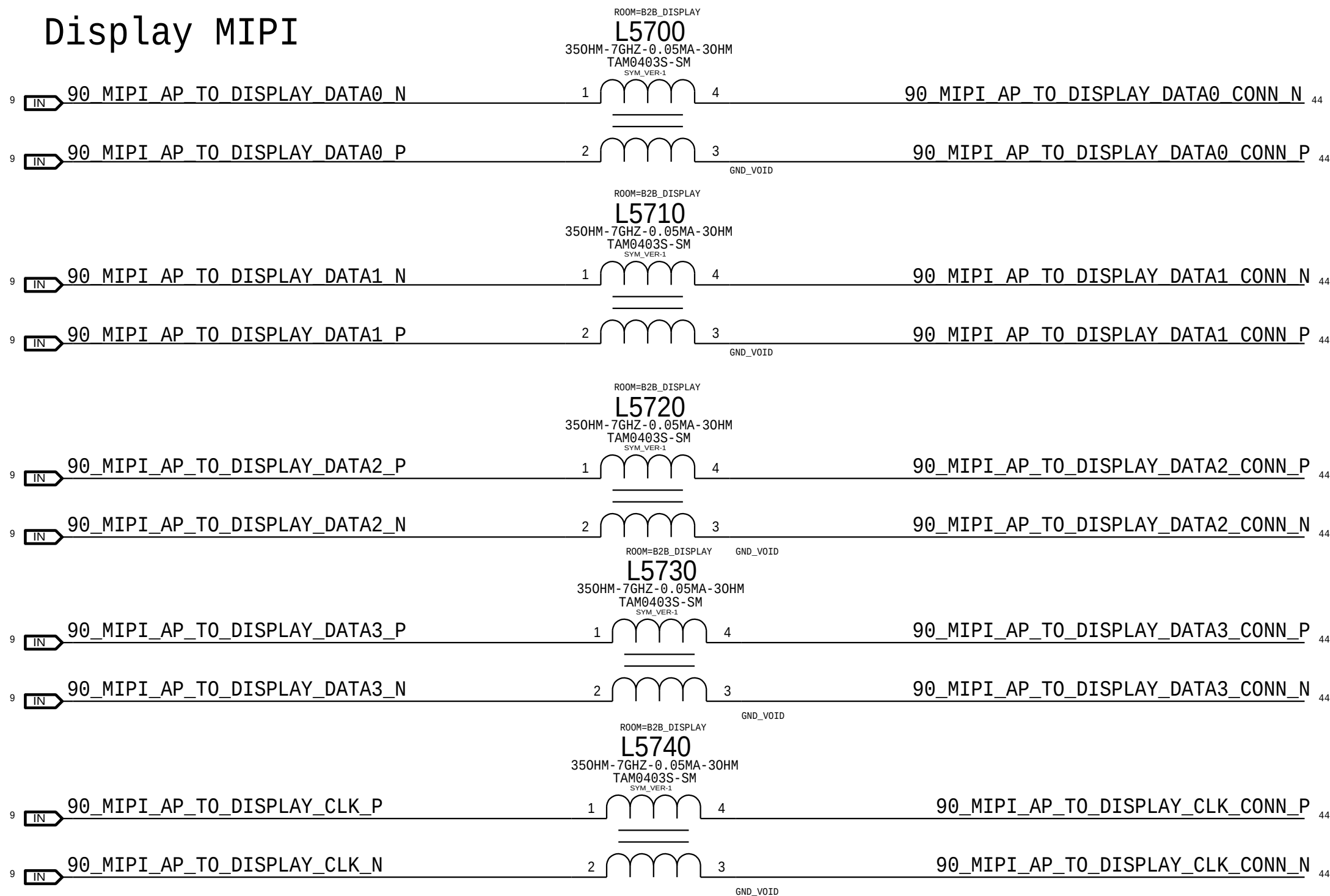
Display Flex Connector

Rcpt: 516S00210 <-- This one on MLB
Plug: 516S00211

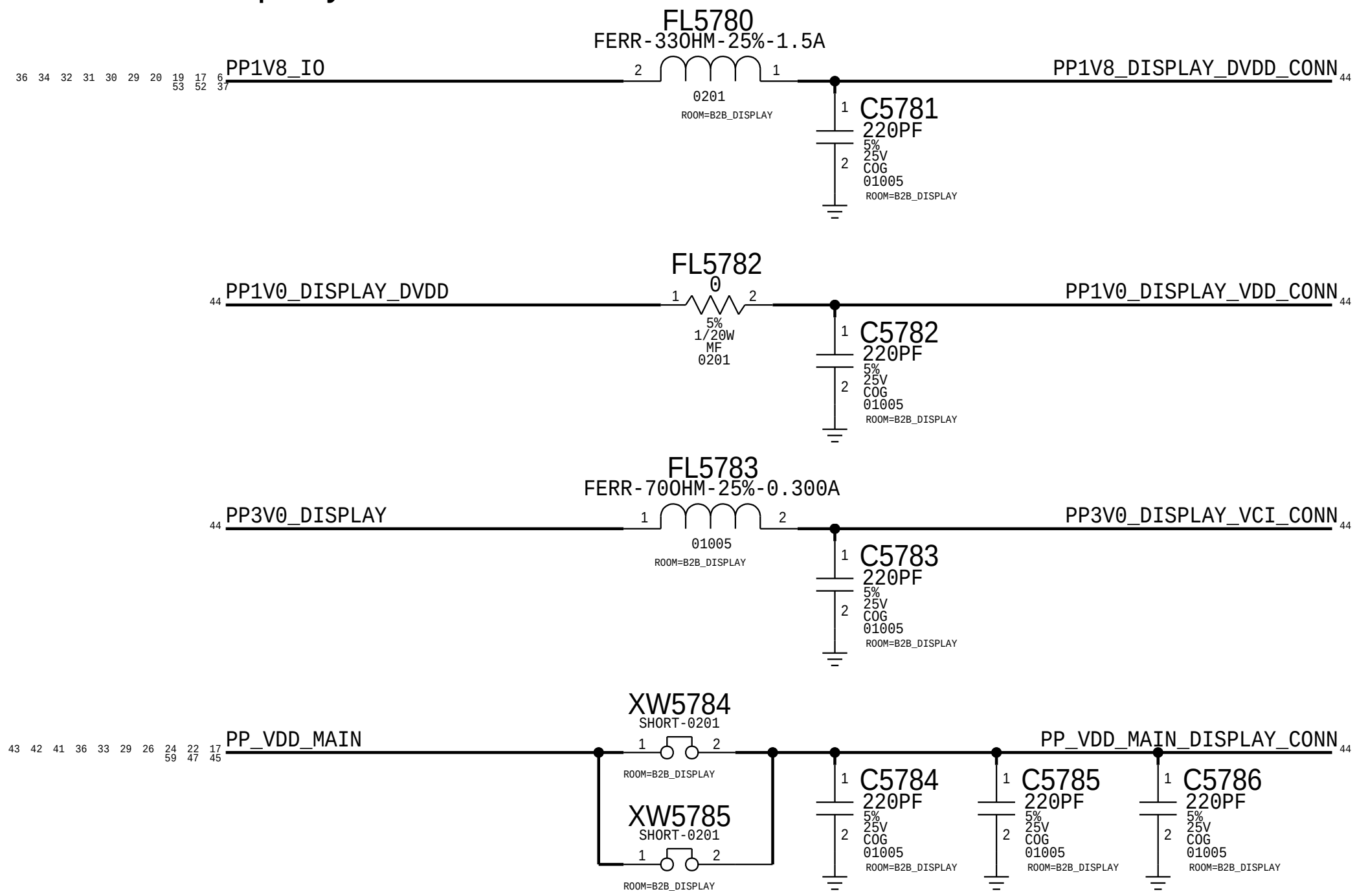


Display MIPI

Display MIPI

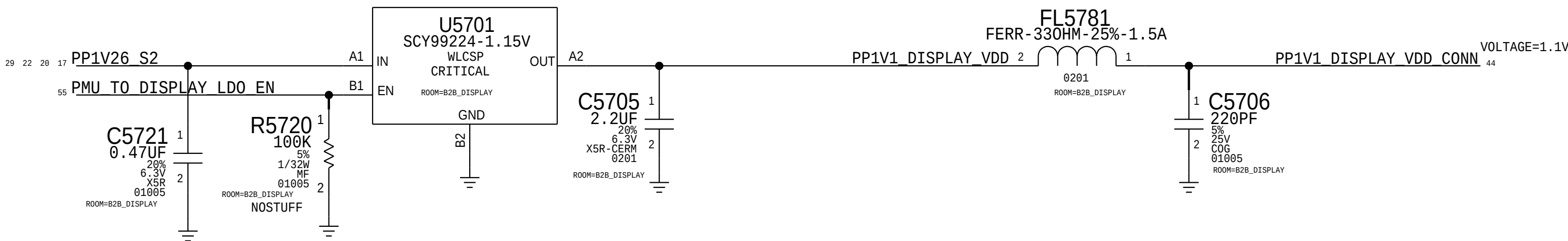


Display Power

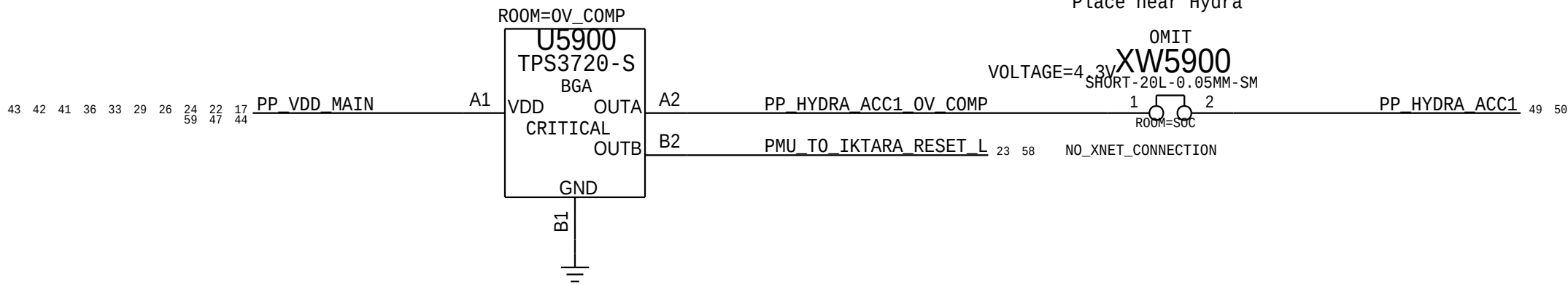


1.2V LD0 is for LGC test chip

Once normal panel is available switch to 1.1V

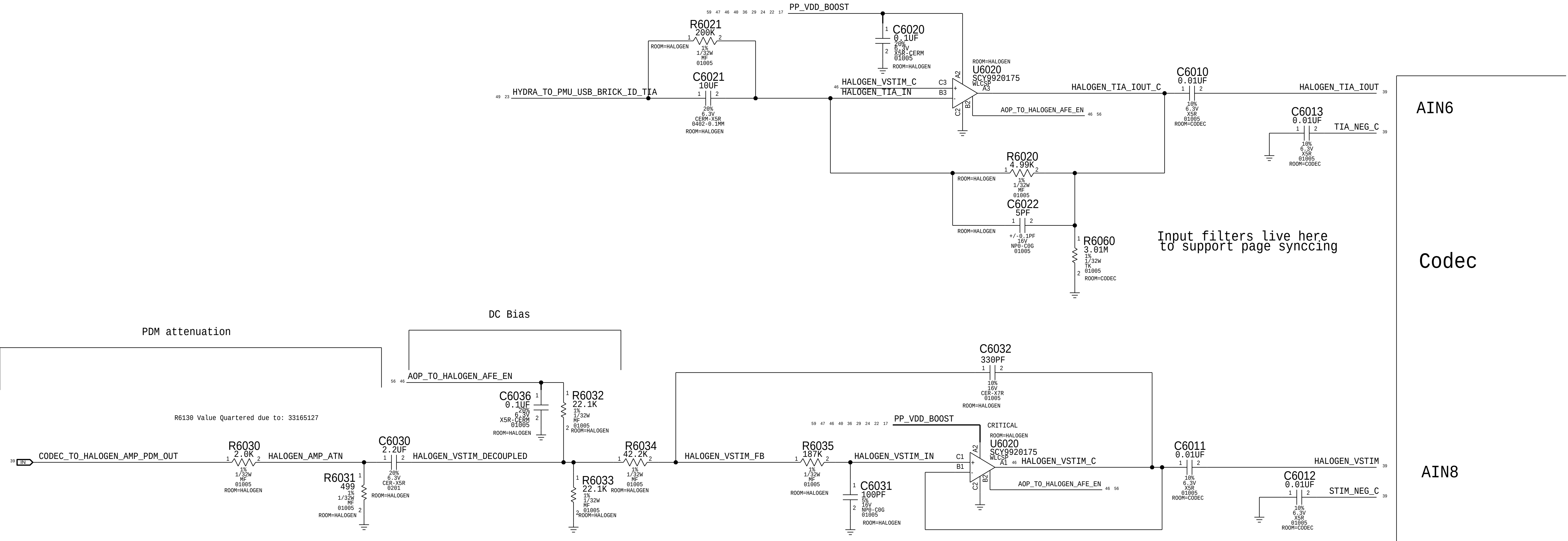


VDD_MAIN 0V CUT-OFF CIRCUIT



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S01375	353S01398	ALT_PARTS	U5900	ON SEMI

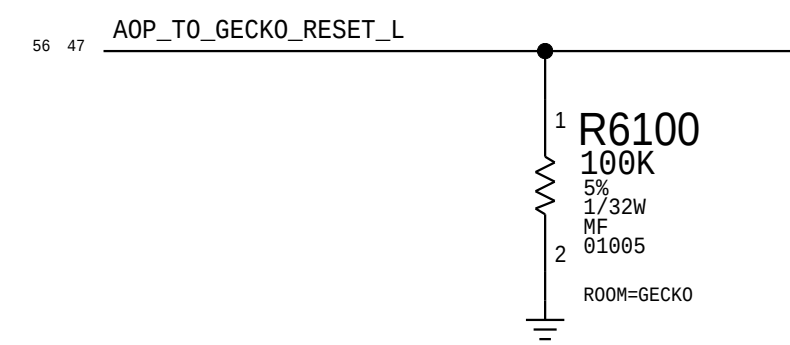
LDCM



AIN6

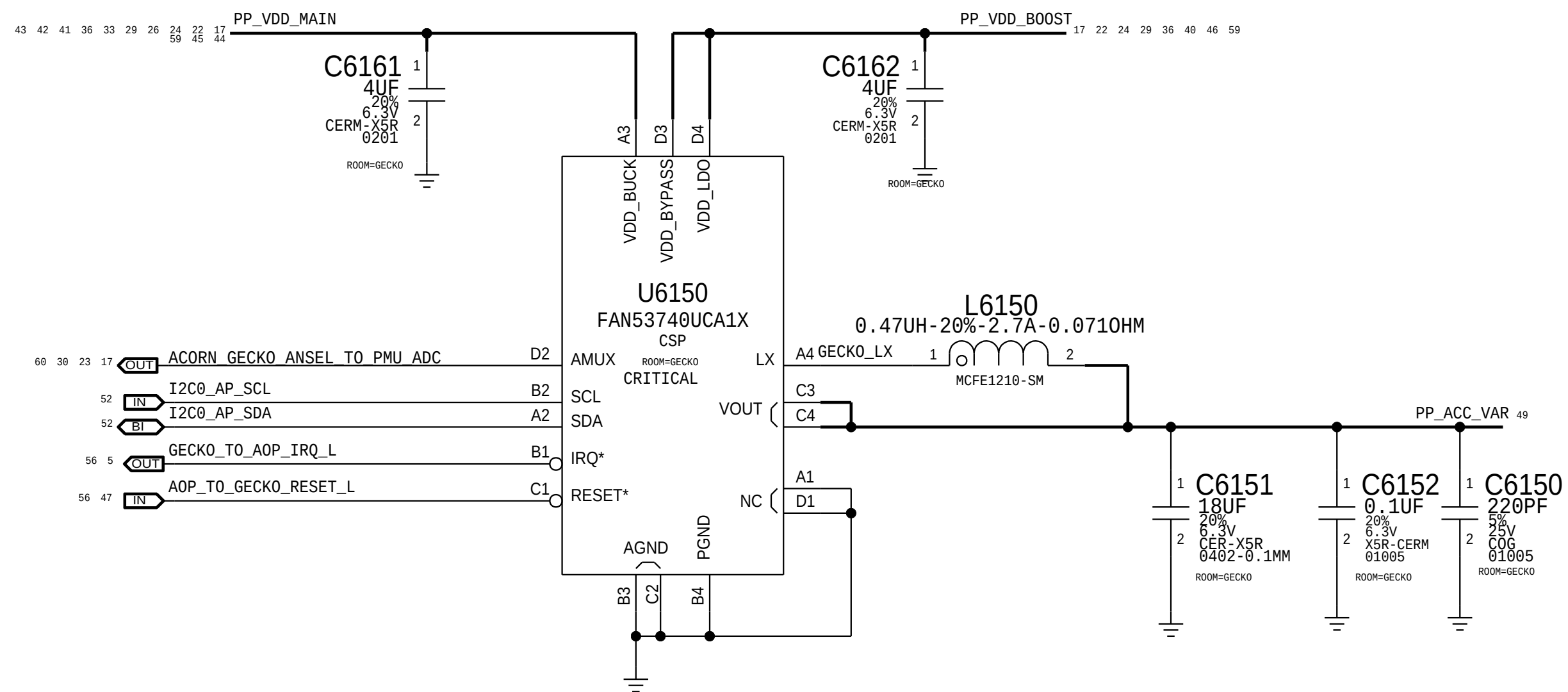
Codec

AIN8



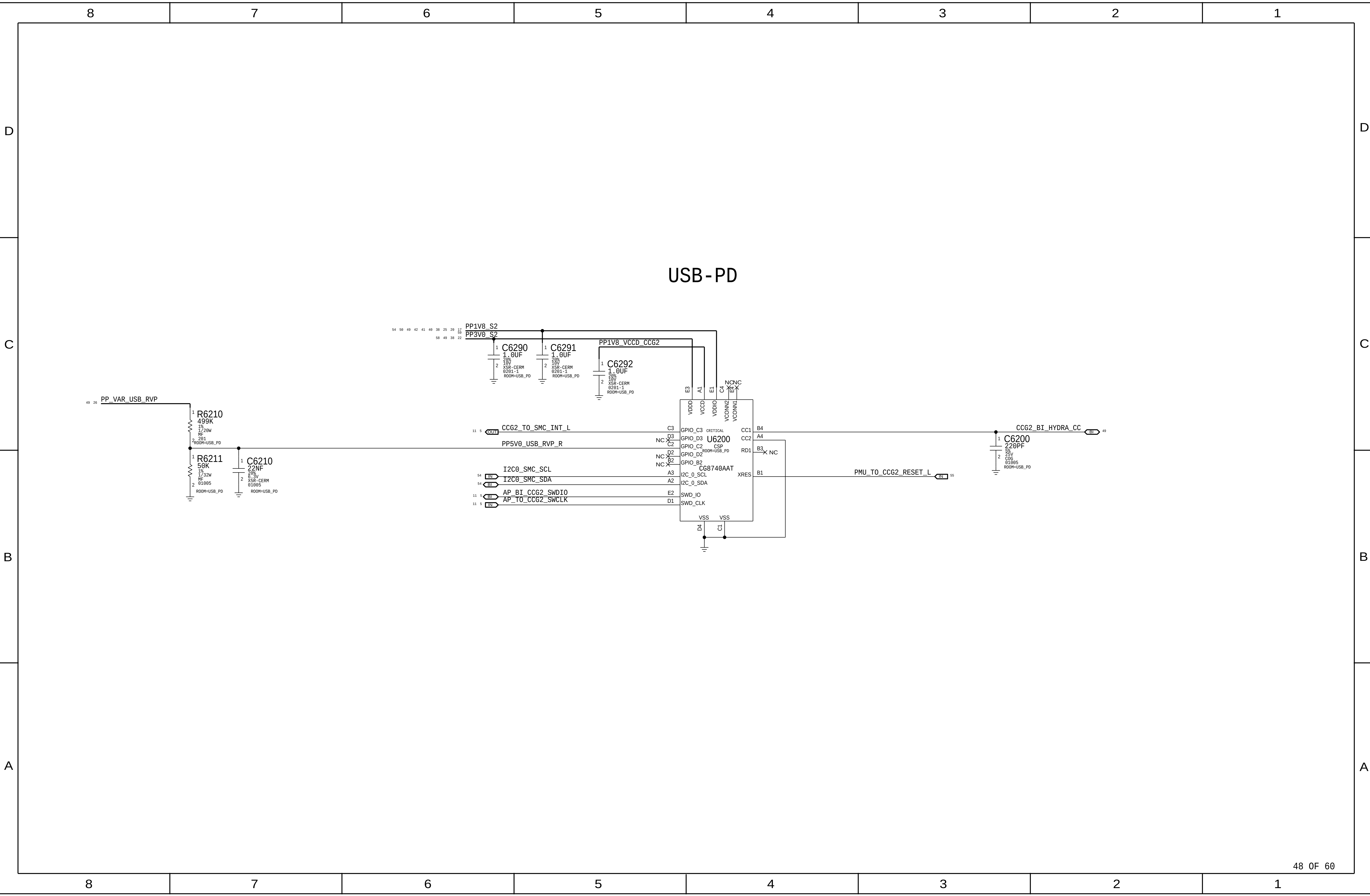
Gecko

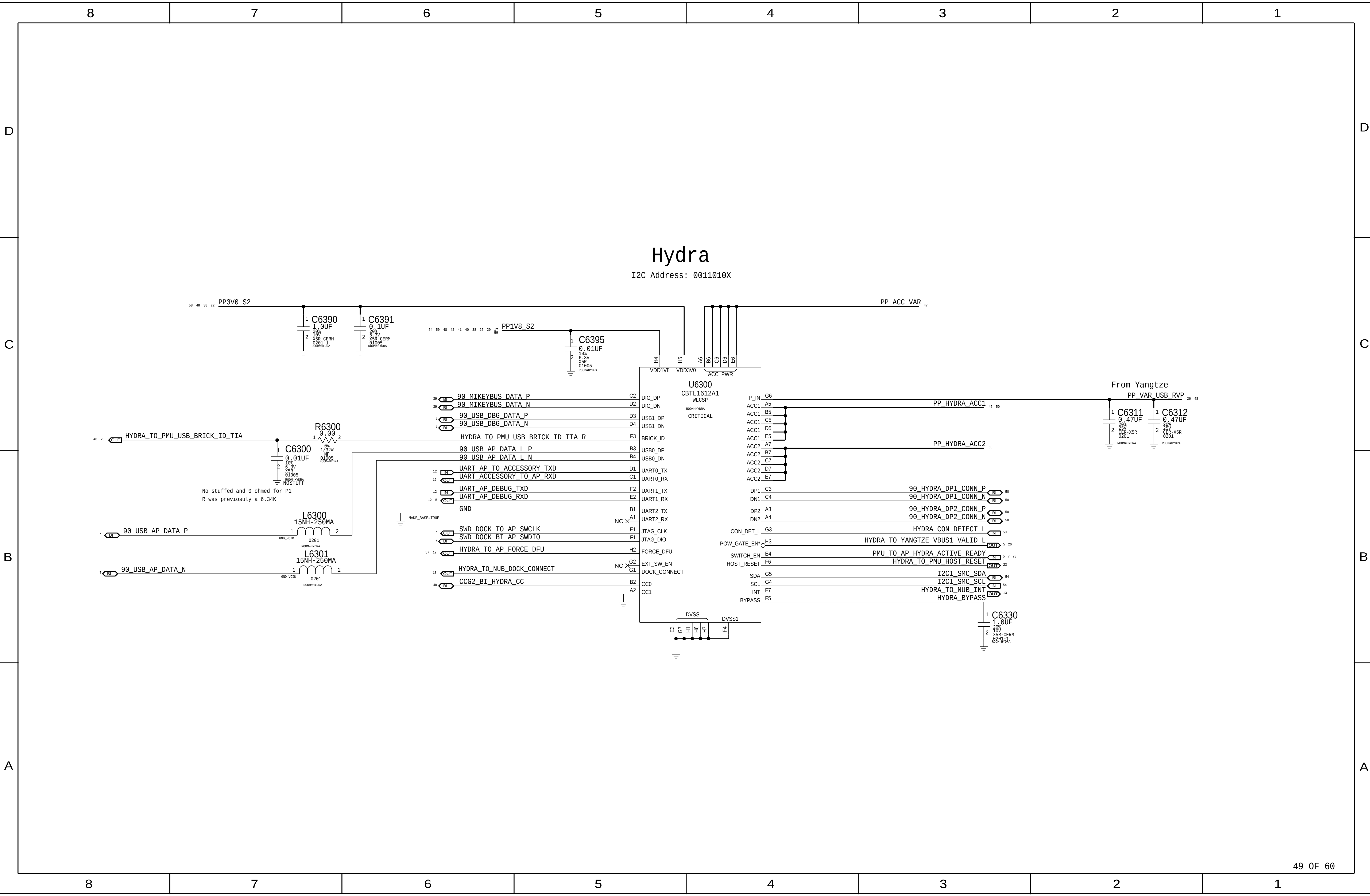
I2C ADDRESS: 0X52



IND Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00854	152S00853	ALT_PARTS	L6150	IND, PAR, 0.47UH, 20K, 2.8A, CY
152S00855	152S00853	ALT_PARTS	L6150	IND, PAR, 0.47UH, 20K, 2.7A, Murata





8		7		6		5		4		3		2		1	

J_INT BOT		
SMT-PAD		
SYM 1 OF 2		
57 GND	1	IO1
57 GND	2	IO2
57 GND	3	IO3
57 GND	4	IO4
57 GND	5	IO5
57 GND	6	IO6
57 PMU_TO_SYSTEM_COLD_RESET_L	7	IO7
57 GND	8	IO8
57 AP_TO_CAMPMU_RESET_L	9	IO9
57 GND	10	IO10
57 BB_TO_MANY_GSM_BURST_IND	11	IO11
57 GND	12	IO12
57 HALL_CASE_TO_AOP_NORTH_L	13	IO13
57 GND	14	IO14
57 AP_TO_TOUCH_SCAN_CLK	15	IO15
57 GND	16	IO16
57 I2S_BB_TO_AP_BCLK	17	IO17
57 GND	18	IO18
57 I2S_BB_TO_AP_DIN	19	IO19
57 GND	20	IO20
57 I2S_AP_TO_BB_DOUT	21	IO21
57 GND	22	IO22
57 I2S_BB_TO_AP_LRCLK	23	IO23
57 GND	24	IO24
57 PP1V8_ALWAYS	25	IO25
57 GND	26	IO26
57 GND	27	IO27
57 GND	28	IO28
57 GND	29	IO29
57 GND	30	IO30
57 GND	31	IO31
57 GND	32	IO32
57 GND	33	IO33
57 GND	34	IO34
57 GND	35	IO35
57 GND	36	IO36
57 GND	37	IO37
57 GND	38	IO38
57 GND	39	IO39
57 GND	40	IO40
57 GND	41	IO41
57 GND	42	IO42
57 GND	43	IO43
57 GND	44	IO44
57 GND	45	IO45
57 GND	46	IO46
57 GND	47	IO47
57 GND	48	IO48
57 GND	49	IO49
57 NFC_TO_ARC_RESET_L	50	IO50
57 GND	51	IO51
57 NFC_TO_ARC_TRIG	52	IO52
57 GND	53	IO53
57 GND	54	IO54
57 GND	55	IO55
57 GND	56	IO56
57 GND	57	IO57
57 GND	58	IO58
57 GND	59	IO59
57 GND	60	IO60
57 GND	61	IO61
57 GND	62	IO62
57 GND	63	IO63
57 GND	64	IO64
57 AP_TO_BB_RESET_L	65	IO65
57 GND	66	IO66
57 SWD_AOP_BT_BB_SWDIO	67	IO67
57 GND	68	IO68
57 UART_GNSS_TO_AP_CTS_L	69	IO69
57 GND	70	IO70
57 UART_AP_TO_GNSS_RTS_L	71	IO71
57 GND	72	IO72
57 PCIE_AP_TO_WLAN_PERST_L	73	IO73
57 GND	74	IO74
57 AP_TO_RACER_RESET_L	75	IO75
57 GND	76	IO76
57 AP_TO_WLAN_TIME_SYNC	77	IO77
57 GND	78	IO78
57 GNSS_TO_AP_LOW_PWR_IND	79	IO79
57 GND	80	IO80
57 HYDRA_TO_AP_FORCE_DFU	81	IO81
57 GND	82	IO82
57 PP1V8_S2	83	IO83
57 PP1V8_S2	84	IO84
57 GND	85	IO85
57 GND	86	IO86
57 GND	87	IO87
57 GND	88	IO88
57 GND	89	IO89
57 INTERPOSER_PIN_90	90	IO90
57 GND	91	IO91
57 AP_TO_BB_COEX	92	IO92
57 BB_TO_AP_COEX	93	IO93
57 GND	94	IO94

IO95	95	AP_TO_NFC_DEV_WAKE	57
IO96	96	AP_TO_NFC_FW_DWLD_REQ	57
IO97	97	GND	57
IO98	98	PMU_TO_NFC_VDD_MAIN_EN	57
IO99	99	UART_AOP_TO_BB_TXD	57
IO100	100	UART_AP_TO_GNSS_TXD	57
IO101	101	GND	57
IO102	102	AP_TO_BB_COREDUMP_TRIG	57
IO103	103	UART_AP_TO_NFC_TXD	57
IO104	104	UART_NFC_TO_AP_RXD	57
IO105	105	BB_TO_AP_RESET_DETECT_L	57
IO106	106	GND	57
IO107	107	BOARD_ID2	57
IO108	108	AP_TO_GNSS_TIME_MARK	57
IO109	109	NC_INTERPOSER_109	57
IO110	110	AP_TO_BB_PEAK_POWER_INDICATOR	57
IO111	111	AP_TO_BBPMU_RADIO_ON_L	57
IO112	112	PP_VDD_MAIN	57
IO113	113	PP_VDD_MAIN	57
IO114	114	PP_VDD_MAIN	57
IO115	115	GND	57
IO116	116	90_PCIE_BB_TO_AP_RXD_N	57
IO117	117	90_PCIE_BB_TO_AP_RXD_P	57
IO118	118	GND	57
IO119	119	90_PCIE_AP_TO_BB_TXD_N	57
IO120	120	90_PCIE_AP_TO_BB_TXD_P	57
IO121	121	GND	57
IO122	122	90_PCIE_AP_TO_BB_REFCLK_P	57
IO123	123	90_PCIE_AP_TO_BB_REFCLK_N	57
IO124	124	GND	57
IO125	125	UART_BB_TO_AOP_RXD	57
IO126	126	UART_GNSS_TO_AP_RXD	57
IO127	127	PCIE_AP_TO_BB_PERST_L	57
IO128	128	UART_NFC_TO_AP_CTS_L	57
IO129	129	GND	57
IO130	130	UART_AP_TO_NFC_RTS_L	57
IO131	131	PMU_AMUX_BY	57
IO132	132	PMU_AMUX_AY	57
IO133	133	GND	57
IO134	134	PCIE_BB_BT_AP_CLKREQ_L	57
IO135	135	NC_INT_135	57
IO136	136	AP_TO_BB_TO_AP_PEAK_POWER_INDICATOR	57
IO137	137	GND	57
IO138	138	PP_VDD_MAIN	57
IO139	139	PP_VDD_MAIN	57
IO140	140	PP_VDD_MAIN	57
IO141	141	GND	57
IO142	142	GND	57
IO143	143	GND	57
IO144	144	GND	57
IO145	145	GND	58
IO146	146	GND	58
IO147	147	GND	58
IO148	148	GND	58
IO149	149	GND	58
IO150	150	GND	58
IO151	151	GND	58
IO152	152	GND	58
IO153	153	GND	58
IO154	154	PP_VDD_MAIN	58
IO155	155	PP_VDD_MAIN	58
IO156	156	GND	58
IO157	157	PP_VDD_MAIN	58
IO158	158	PP_VDD_MAIN	58
IO159	159	GND	58
IO160	160	PMU_TO_NFC_EN	58
IO161	161	GND	58
IO162	162	PMU_TO_BBPMU_RESET_L	58
IO163	163	GND	58
IO164	164	PMU_TO_TOUCH_CLK32K	58
IO165	165	GND	58
IO166	166	PCIE_WLAN_BT_AP_CLKREQ_L	58
IO167	167	GND	58
IO168	168	GND	58
IO169	169	BB_TO_PMU_PCIE_HOST_WAKE_L	58
IO170	170	GND	58
IO171	171	GND	58
IO172	172	WLAN_TO_PMU_HOST_WAKE	58
IO173	173	GND	58
IO174	174	PMU_TO_WLAN_CLK32K	58
IO175	175	GND	58
IO176	176	NFC_TO_AOP_HOST_WAKE	58
IO177	177	GND	58
IO178	178	TOUCH_TO_MANY_FORCE_PWM	58
IO179	179	GND	58
IO180	180	UART_AP_TO_BT_TXD	58
IO181	181	GND	58
IO182	182	UART_AP_TO_BT_RTS_L	58
IO183	183	GND	58
IO184	184	GND	58
IO185	185	GND	58
IO186	186	GND	58
IO187	187	GND	58
IO188	188	GND	58

AP I2C

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C0	PP1V8_I0	400 kHz	GECK0	0x52	1010 010X	0xA4, 0xA5	-	1 MHz	TOP MLB
			SAKONNET	0x08	0001 000X	0x10, 0x11	-	1 MHz	Dock Flex
			BOOST	0x75	1110 101X	0xEA, 0xEB	-	400 KHz	TOP MLB
			ARC EEPROM	0x50	1010 000X	0xA0, 0xA1	-	400 KHz	Dock Flex

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C1	PP1V8_I0	100 kHz	MIC2	0x56	1010 100X	0xA8, 0xA9	-	1 MHz	Strobe Flex

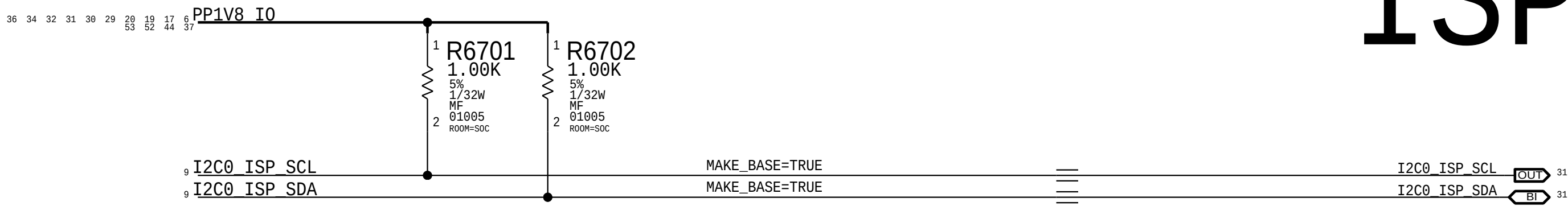
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C2	PP1V8_I0	1 MHz	Top Speaker Amp	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top MLB

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C3	PP1V8_I0	400 kHz	ACORN	0X2A	0101 010X	0x54, 0x55	-	1 MHz	Bot MLB
			TOUCH EEPROM	0x51	1010 001X	0xA2, 0xA3	-	1 MHz	Touch Flex

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Location
AP I2C4	PP1V8_I0	400 kHz	LYNX	0X71	Top MLB

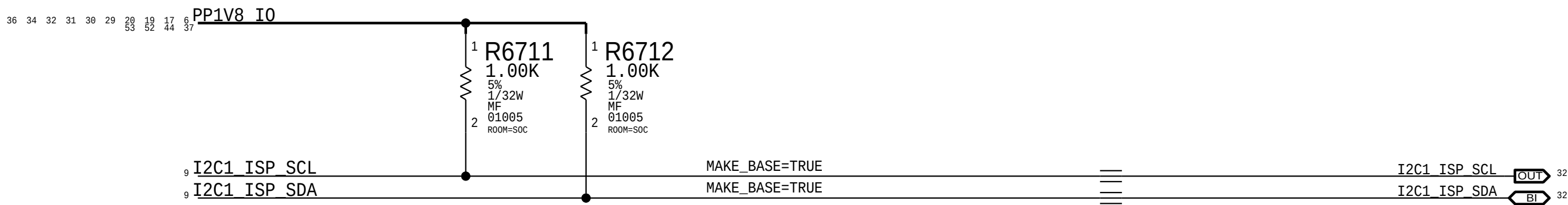
ISP I2C

ISP I2C0



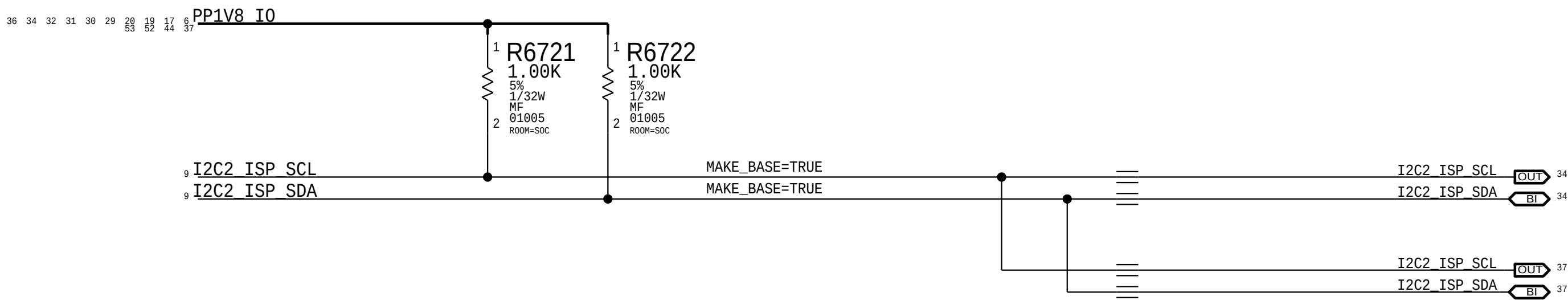
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C0	PP1V8_I0	1 MHz	Austin	0X10	0010 000X	0x20, 0x21	-	1 MHz	Wide Cam
			Raman	0X3C	0111 100X	0x78, 0x79	-	1 MHz	Wide Cam

ISP I2C1



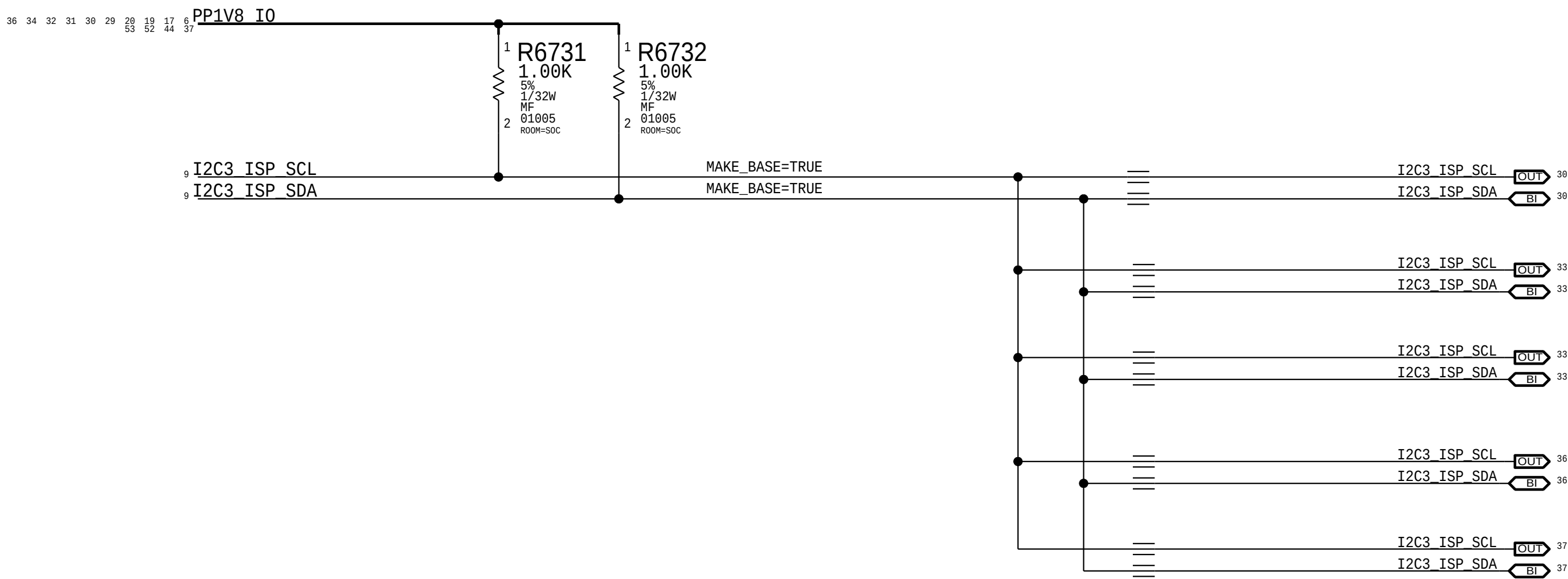
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C1	PP1V8_I0	1 MHz	Billings	0x20	0100 000X	0x40, 0x41	-	1 MHz	Tele Cam
			Grunberg+	0x1C	0011 100X	0x38, 0x39	-	1 MHz	Tele Cam

ISP I2C2



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C2	PP1V8_I0	1 MHz	Yonkers	0x10	0010 000X	0x20, 0x21	-	1 MHz	Fcam
			Flatiron	0x70	1110 000X	0xE0, 0xE1	-	1 MHz	Fcam
			Savage	0x18	0011 000X	0x30, 0x31	-	1 MHz	Juliet Flex

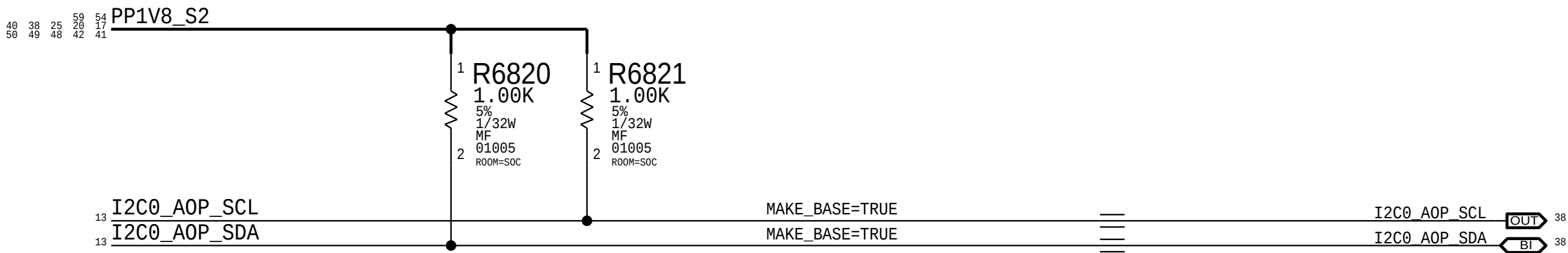
ISP I2C3



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C3	PP1V8_I0	1 MHz	Ansel	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top Board
			Neon	0x63	1100 011X	0xC6, 0xC7	-	1 MHz	Top Board
			Neon	0x67	1100 111X	0xCE, 0xCF	-	1 MHz	Top Board
			Rigel	0x55	1100 011X	0xAA, 0xAB	-	1 MHz	Top Board
			Mama Bear	0x50	1010 000X	0xA0, 0xA1	-	1 MHz	Romeo Flex

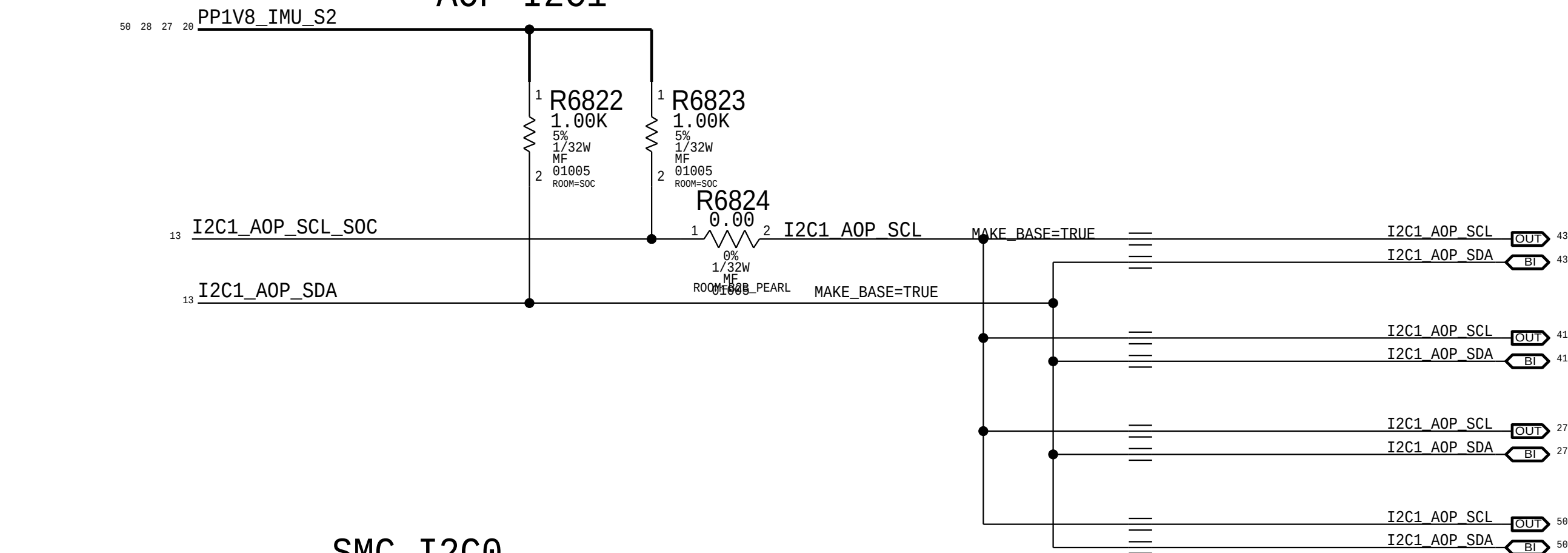
AOP/SMC I2C

AOP I2C0



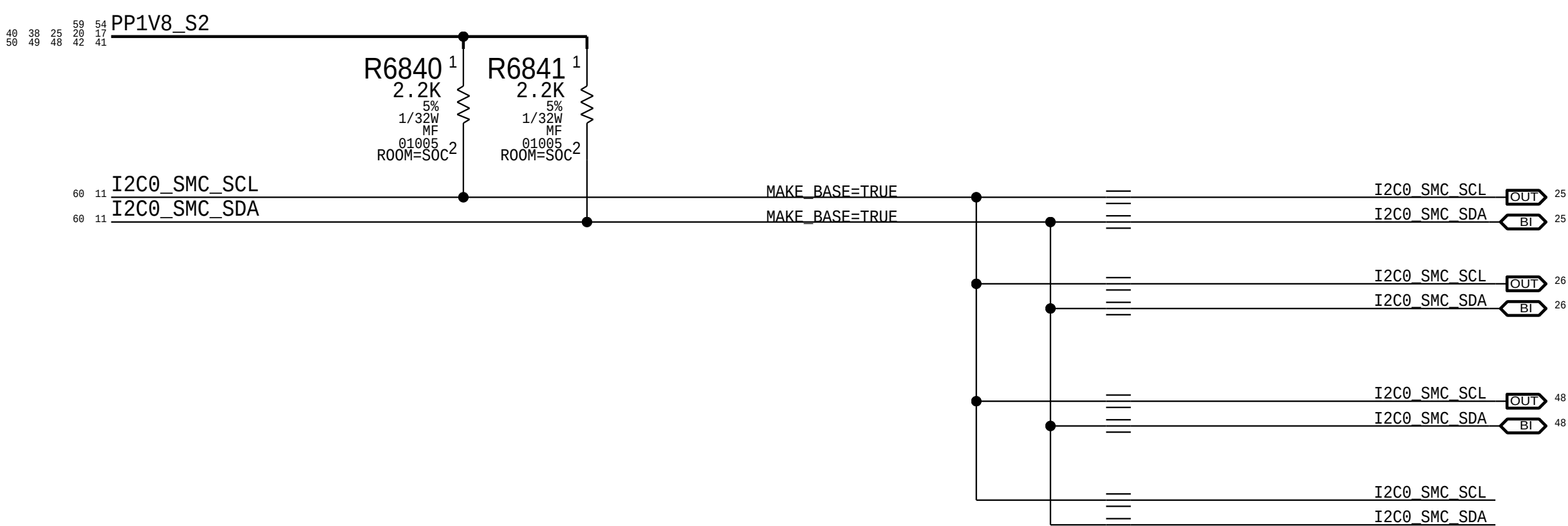
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C0	PP1V8_S2	750 kHz	Doppler	0x58	1011 000X	0xB0, 0xB1	-	1 MHz	Sensor Flex
			Blackbird	0x29	0101 001X	0x52, 0x53	-	1 MHz	Sensor Flex
			Yogi	0x33	0110 011X	0x66, 0x67	-	1 MHz	Sensor Flex

AOP I2C1



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C1	PP1V8_IMU_S2	400 kHz	Arc	0x42	1000 001X	0x82, 0x83	-	1 MHz	Top Board
			Bottom Speaker	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top Board
			Moly	0x0E	0001 110X	0x1C, 0x1D	-	1 MHz	Button Cyclone
			Potassium	0x76	1110 110X	0xEC, 0xED	-	1 MHz	Dock Flex

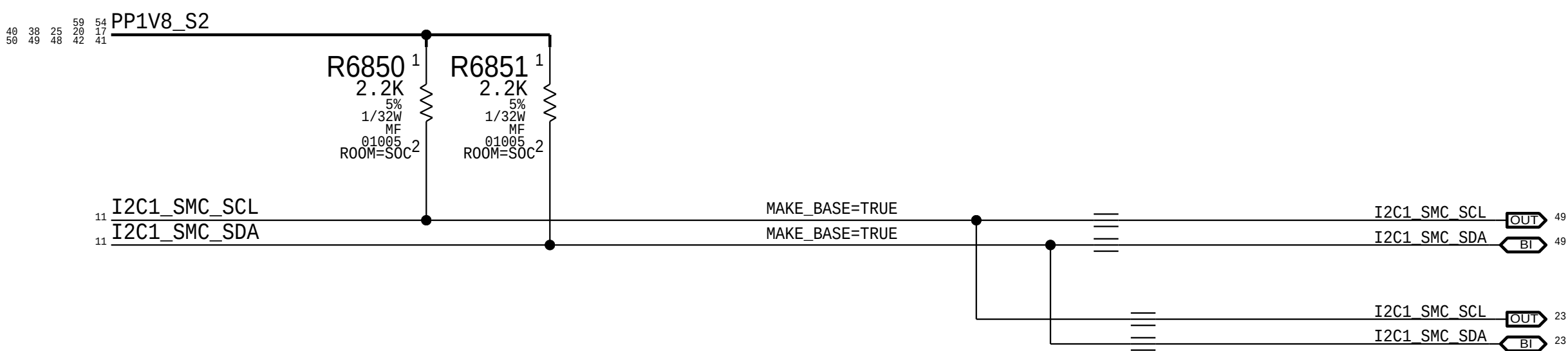
SMC I2C0



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C0	PP1V8_S2	400 kHz	Yangtze	0x71	1110 001X	0xE2, 0xE3	-	400 KHz	Top Board
			Iktara	0x39	0111 001X	0x72, 0x73	-	400 KHz	Bot Board
			CCG2	0x12	0010 010X	0x24, 0x25	-	1 MHz	Top Board
			Gas Guage	0x55	0010 010X	0xAA, 0xAB	-	1 MHz	BMU Flex
			Roswell	0x10	0100 000X	0x20, 0x21	-	400 KHz	BMU Flex

Lives on bottom board

SMC I2C1



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C1	PP1V8_S2	400 kHz	Hydra	0x1A	0011 010X	0x34, 0x35	-	400 KHz	Top Board
			Denali	0x74	1110 100X	0xE8, 0xE9	-	400 KHz	Top Board

AP/PMU GPIOs

SOC

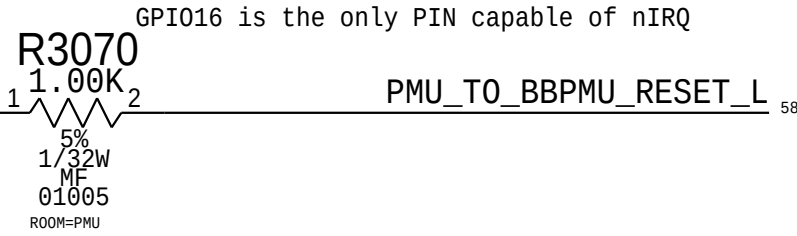
PMU

GPIO_0	12	AP_TO_BT_DEVICE_WAKE	AP_TO_BT_DEVICE_WAKE	58
GPIO_1	12	BOARD_REV0	BOARD_REV0	6
GPIO_2	12	BOARD_REV1	BOARD_REV1	6
GPIO_3	12	BOARD_REV2	BOARD_REV2	6
GPIO_4	12	AP_TO_PMU_AMUX_SYNC	AP_TO_PMU_AMUX_SYNC	23
GPIO_5	12	BOARD_REV3	BOARD_REV3	6
GPIO_6	12	AP_CANARY1	AP_CANARY1	68
GPIO_7	12	AP_BUTTON_VOL_UP_L	PMU_TO_AP_BUTTON_VOL_UP_L	23
GPIO_8	12	NC_AP_GPIO8	NC_AP_GPIO8	
GPIO_9	12	AP_TO_BBPMU_RADIO_ON_L	AP_TO_BBPMU_RADIO_ON_L	57
GPIO_10	12	AP_TO_SPKRAMP_TOP_RESET_L	AP_TO_SPKRAMP_TOP_RESET_L	42
GPIO_11	12	AP_TO_NFC_FW_DWLD_REQ	AP_TO_NFC_FW_DWLD_REQ	57
GPIO_12	15	AP_TO_BB_PEAK_POWER_INDICATOR	AP_TO_BB_PEAK_POWER_INDICATOR	57
GPIO_13	12	AP_TO_NFC_DEV_WAKE	AP_TO_NFC_DEV_WAKE	57
GPIO_14	12	CAMPMU_TO_AP_IRQ_L	CAMPMU_TO_AP_IRQ_L	38
GPIO_15	12	AP_TO_GNSS_TIME_MARK	AP_TO_GNSS_TIME_MARK	57
GPIO_16	15	SPKRAMP_TOP_TO_AP_INT_L	SPKRAMP_TOP_TO_AP_INT_L	42
GPIO_17	12	BB_TO_AP_COEX	BB_TO_AP_COEX	57
GPIO_18	12	BT_TO_AP_TIME_SYNC	BT_TO_AP_TIME_SYNC	58
GPIO_19	12	AP_TO_BB_RESET_L	AP_TO_BB_RESET_L	57
GPIO_20	12	BB_TO_AP_PEAK_POWER_INDICATOR	BB_TO_AP_PEAK_POWER_INDICATOR	57
GPIO_21	12	BB_TO_AP_RESET_DETECT_L	BB_TO_AP_RESET_DETECT_L	57
GPIO_22	12	AP_TO_BB_COREDUMP_TRIG	AP_TO_BB_COREDUMP_TRIG	57
GPIO_23	12	AP_TO_CAMPMU_RESET_L	AP_TO_CAMPMU_RESET_L	57
GPIO_24	12	AP_TO_BB_COEX	AP_TO_BB_COEX	57
GPIO_25	12	DISPLAY_TO_AP_PANEL_ID	DISPLAY_TO_AP_PANEL_ID	44
GPIO_26	12	AP_CANARY2	AP_CANARY2	58
GPIO_27	12	NC_AP_GPIO27	NC_AP_GPIO27	
GPIO_28	12	NC_AP_GPIO28	NC_AP_GPIO28	
GPIO_29	12	AP_TO_RACER_RESET_L	AP_TO_RACER_RESET_L	57
GPIO_30	15	GNSS_TO_AP_LOW_PWR_IND	GNSS_TO_AP_LOW_PWR_IND	57

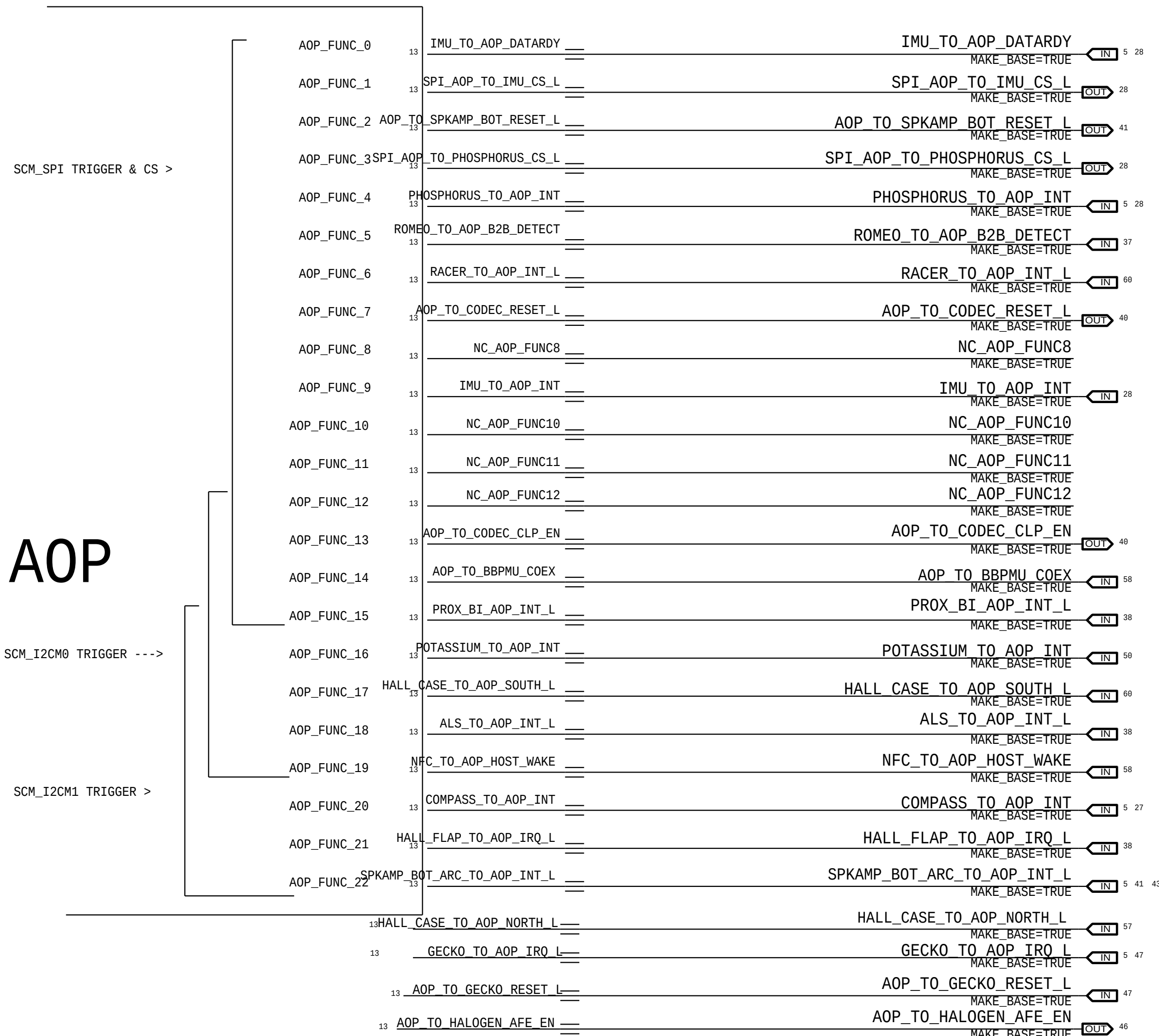
GPIO_1	23	PMU_TO_CCG2_RESET_L	PMU_TO_CCG2_RESET_L	48
GPIO_2	23	PMU_TO_AP_THROTTLE_GPU1_L	PMU_TO_AP_THROTTLE_GPU1_L	7
GPIO_3	23	NC_BT_TO_PMU_HOST_WAKE	NC_BT_TO_PMU_HOST_WAKE	
GPIO_4	23	WLAN_TO_PMU_HOST_WAKE	WLAN_TO_PMU_HOST_WAKE	58
GPIO_5	23	BB_TO_PMU_PCIE_HOST_WAKE_L	BB_TO_PMU_PCIE_HOST_WAKE_L	58
GPIO_6	23	PMU_NFC_TO_ARC_RESET_L	PMU_NFC_TO_ARC_RESET_L	43
GPIO_7	23	PMU_TO_GNSS_EN	PMU_TO_GNSS_EN	58
GPIO_8	23	PMU_TO_WLAN_CLK32K	PMU_TO_WLAN_CLK32K	23 58
GPIO_9	23	PMU_TO_BT_REG_ON	PMU_TO_BT_REG_ON	58
GPIO_10	23	PMU_TO_PHALANX2	PMU_TO_PHALANX2	50
GPIO_11	23	YANGTZE_TO_PMU_INT_L	YANGTZE_TO_PMU_INT_L	26
GPIO_12	23	CODEC_TO_PMU_WAKE_L	CODEC_TO_PMU_WAKE_L	46
GPIO_13	23	PMU_MASK_NFC_TO_ARC_TRIG	PMU_MASK_NFC_TO_ARC_TRIG	43
GPIO_14	23	PMU_TO_WLAN_REG_ON	PMU_TO_WLAN_REG_ON	58
GPIO_15	23	PMU_TO_NFC_VDD_MAIN_EN	PMU_TO_NFC_VDD_MAIN_EN	57
GPIO_16	23	PMU_TO_NAND_LOW_BATT_BOOT_L	PMU_TO_NAND_LOW_BATT_BOOT_L	19
GPIO_17	23	PMU_TO_PHALANX1	PMU_TO_PHALANX1	50
GPIO_18	23	PMU_TO_DISPLAY_RESET_L	PMU_TO_DISPLAY_RESET_L	44
GPIO_19	23	PMU_TO_BBPMU_RESET_R_L	PMU_TO_BBPMU_RESET_R_L	
GPIO_20	23	PMU_TO_NFC_EN	PMU_TO_NFC_EN	58
GPIO_21	23	NC_PMU_GPIO21	NC_PMU_GPIO21	
GPIO_22	23	PMU_TO_IKTARA_EN_EXT_1V8	PMU_TO_IKTARA_EN_EXT_1V8	68
GPIO_23	23	PMU_TO_BOOST_EN	PMU_TO_BOOST_EN	24
GPIO_24	23	PMU_TO_DISPLAY_PANICB	PMU_TO_DISPLAY_PANICB	44
GPIO_25	23	PMU_TO_DISPLAY_LDO_EN	PMU_TO_DISPLAY_LDO_EN	44

Held Through 1 Reset

Sequenced GPIOs



AOP GPIOs



Interposer Aliases: Pins 1-144

THIS SIDE HAS ATTRIBUTE			
MAKE_BASE=TRUE			
51	GND	=====	GND 57 58 59 68
51	GND	=====	GND 57 58 59 68
51	GND	=====	GND 57 58 59 68
51	GND	=====	GND 57 58 59 68
51	GND	=====	GND 57 58 59 68
51	GND	=====	GND 57 58 59 68
51	PMU_TO_SYSTEM_COLD_RESET_L	=====	PMU_TO_SYSTEM_COLD_RESET_L 7 15 23
51	GND	=====	GND 57 58 59 68
51	AP_TO_CAMPMU_RESET_L	=====	AP_TO_CAMPMU_RESET_L 39 55
51	GND	=====	GND 57 58 59 68
51	BB_TO_MANY_GSM_BURST_IND	=====	BB_TO_MANY_GSM_BURST_IND 33 38
51	GND	=====	GND 57 58 59 68
51	HALL_CASE_TO_AOP_NORTH_L	=====	HALL_CASE_TO_AOP_NORTH_L 56
51	GND	=====	GND 57 58 59 68
51	AP_TO_TOUCH_SCAN_CLK	=====	AP_TO_TOUCH_SCAN_CLK 9
51	GND	=====	GND 57 58 59 68
51	I2S_BB_TO_AP_BCLK	=====	I2S_BB_TO_AP_BCLK 11
51	GND	=====	GND 57 58 59 68
51	I2S_BB_TO_AP_DIN	=====	I2S_BB_TO_AP_DIN 11
51	GND	=====	GND 57 58 59 68
51	I2S_AP_TO_BB_DOUT	=====	I2S_AP_TO_BB_DOUT 11
51	GND	=====	GND 57 58 59 68
51	I2S_BB_TO_AP_LRCLK	=====	I2S_BB_TO_AP_LRCLK 11
51	GND	=====	GND 57 58 59 68
51	PP1V8_ALWAYS	=====	PP1V8_ALWAYS 17 22 23 28
51	GND	=====	GND 57 58 59 68
51	GND	=====	GND 58
51	GND	=====	GND 59
51	GND	=====	GND 59
51	GND	=====	GND 57 59
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51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	NFC_TO_ARC_RESET_L	=====	NFC_TO_ARC_RESET_L 5 43
51	GND	=====	GND 57 59
51	NFC_TO_ARC_TRIG	=====	NFC_TO_ARC_TRIG 5 43
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	GND	=====	GND 57 59
51	AP_TO_BB_RESET_L	=====	AP_TO_BB_RESET_L 55
51	GND	=====	GND 57 59
51	SWD_AOP_BI_BB_SWDIO	=====	SWD_AOP_BI_BB_SWDIO 13
51	GND	=====	GND 57 59
51	UART_GNSS_TO_AP_CTS_L	=====	UART_GNSS_TO_AP_CTS_L 12
51	GND	=====	GND 57 59
51	UART_AP_TO_GNSS_RTS_L	=====	UART_AP_TO_GNSS_RTS_L 12
51	GND	=====	GND 57 59

THIS SIDE HAS ATTRIBUTE			
MAKE_BASE=TRUE			
51	PCIE_AP_TO_WLAN_PERST_L	=====	PCIE_AP_TO_WLAN_PERST_L 8
51	GND	=====	GND 57 59
51	AP_TO_RACER_RESET_L	=====	AP_TO_RACER_RESET_L 55
51	GND	=====	GND 57 59
51	AP_TO_WLAN_TIME_SYNC	=====	AP_TO_WLAN_TIME_SYNC 12
51	GND	=====	GND 57 59
51	GNSS_TO_AP_LOW_PWR_IND	=====	GNSS_TO_AP_LOW_PWR_IND 55
51	GND	=====	GND 57 59
51	HYDRA_TO_AP_FORCE_DFU	=====	HYDRA_TO_AP_FORCE_DFU 12 49
51	GND	=====	GND 57 59
51	PP1V8_S2	=====	PP1V8_S2 59
51	PP1V8_S2	=====	PP1V8_S2 59
51	GND	=====	GND 57 59
51	GND	=====	GND 59
51	GND	=====	GND 59
51	GND	=====	GND 59
51	GND	=====	GND 57 58 59
51	INTERPOSER_PIN_90	=====	INTERPOSER_PIN_90
51	GND	=====	GND 59
51	AP_TO_BB_COEX	=====	AP_TO_BB_COEX 55
51	BB_TO_AP_COEX	=====	BB_TO_AP_COEX 55
51	GND	=====	GND 59
51	AP_TO_NFC_DEV_WAKE	=====	AP_TO_NFC_DEV_WAKE 55
51	AP_TO_NFC_FW_DWLD_REQ	=====	AP_TO_NFC_FW_DWLD_REQ 55
51	GND	=====	GND 59
51	PMU_TO_NFC_VDD_MAIN_EN	=====	PMU_TO_NFC_VDD_MAIN_EN 55
51	UART_AOP_TO_BB_TXD	=====	UART_AOP_TO_BB_TXD 13
51	UART_AP_TO_GNSS_TXD	=====	UART_AP_TO_GNSS_TXD 12
51	GND	=====	GND 59
51	AP_TO_BB_COREDUMP_TRIG	=====	AP_TO_BB_COREDUMP_TRIG 55
51	UART_AP_TO_NFC_TXD	=====	UART_AP_TO_NFC_TXD 12
51	UART_NFC_TO_AP_RXD	=====	UART_NFC_TO_AP_RXD 12
51	BB_TO_AP_RESET_DETECT_L	=====	BB_TO_AP_RESET_DETECT_L 55
51	GND	=====	GND 59
51	BOARD_ID2	=====	BOARD_ID2 8 12
51	AP_TO_GNSS_TIME_MARK	=====	AP_TO_GNSS_TIME_MARK 55
51	NC_INTERPOSER_109	=====	NC_INTERPOSER_109
51	AP_BB_PEAK_POWER_INDICATOR	=====	AP_TO_BB_PEAK_POWER_INDICATOR 55
51	AP_TO_BBPMU_RADIO_ON_L	=====	AP_TO_BBPMU_RADIO_ON_L 55
51	PP_VDD_MAIN	=====	PP_VDD_MAIN 59
51	PP_VDD_MAIN	=====	PP_VDD_MAIN 59
51	PP_VDD_MAIN	=====	PP_VDD_MAIN 59
51	GND	=====	GND 57 58 59
51	90_PCIE_BB_TO_AP_RXD_N	=====	90_PCIE_BB_TO_AP_RXD_N 17
51	90_PCIE_BB_TO_AP_RXD_P	=====	90_PCIE_BB_TO_AP_RXD_P 17
51	GND	=====	GND 57 59
51	90_PCIE_AP_TO_BB_TXD_N	=====	90_PCIE_AP_TO_BB_TXD_N 17
51	90_PCIE_AP_TO_BB_TXD_P	=====	90_PCIE_AP_TO_BB_TXD_P 17
51	GND	=====	GND 57 59
51	90_PCIE_AP_TO_BB_REFCLK_P	=====	90_PCIE_AP_TO_BB_REFCLK_P 8
51	90_PCIE_AP_TO_BB_REFCLK_N	=====	90_PCIE_AP_TO_BB_REFCLK_N 8
51	GND	=====	GND 57 59
51	UART_BB_TO_AOP_RXD	=====	UART_BB_TO_AOP_RXD 13
51	UART_GNSS_TO_AP_RXD	=====	UART_GNSS_TO_AP_RXD 12
51	PCIE_AP_TO_BB_PERST_L	=====	PCIE_AP_TO_BB_PERST_L 8
51	UART_NFC_TO_AP_CTS_L	=====	UART_NFC_TO_AP_CTS_L 12
51	GND	=====	GND 57 59
51	UART_AP_TO_NFC_RTS_L	=====	UART_AP_TO_NFC_RTS_L 12
51	PMU_AMUX_BY	=====	PMU_AMUX_BY 23
51	PMU_AMUX_AY	=====	PMU_AMUX_AY 23
51	GND	=====	GND 57 59
51	PCIE_BB_BI_AP_CLKREQ_L	=====	PCIE_BB_BI_AP_CLKREQ_L 8
51	NC_INT_135	=====	NC_INT_135
51	BB_AP_PEAK_POWER_INDICATOR	=====	BB_TO_AP_PEAK_POWER_INDICATOR 55
51	GND	=====	GND 57 59
51	PP_VDD_MAIN	=====	PP_VDD_MAIN 59
51	PP_VDD_MAIN	=====	PP_VDD_MAIN 59
51	PP_VDD_MAIN	=====	PP_VDD_MAIN 59
51	GND	=====	GND 57 58 59
51	GND	=====	GND 57 58 59
51	GND	=====	GND 57 58 59
51	GND	=====	GND 57 58 59

Interposer Aliases: Pins 145-285

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	59	60
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59		
51		PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59		
51		PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		PMU_TO_NFC_EN	=====	MAKE_BASE=TRUE	PMU_TO_NFC_EN	55		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		PMU_TO_BBPMU_RESET_L	=====	MAKE_BASE=TRUE	PMU_TO_BBPMU_RESET_L	55		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		PMU_TO_TOUCH_CLK32K	=====	MAKE_BASE=TRUE	PMU_TO_TOUCH_CLK32K	17		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		PCIE_WLAN_BT_AP_CLKREQ_L	=====	MAKE_BASE=TRUE	PCIE_WLAN_BT_AP_CLKREQ_L	9		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		BB_TO_PMU_PCIE_HOST_WAKE_L	=====	MAKE_BASE=TRUE	BB_TO_PMU_PCIE_HOST_WAKE_L	55		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		WLAN_TO_PMU_HOST_WAKE	=====	MAKE_BASE=TRUE	WLAN_TO_PMU_HOST_WAKE	55		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		PMU_TO_WLAN_CLK32K	=====	MAKE_BASE=TRUE	PMU_TO_WLAN_CLK32K	23	55	
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		NFC_TO_AOP_HOST_WAKE	=====	MAKE_BASE=TRUE	NFC_TO_AOP_HOST_WAKE	56		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		TOUCH_TO_MANY_FORCE_PWM	=====	MAKE_BASE=TRUE	TOUCH_TO_MANY_FORCE_PWM	23	24	38
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		UART_AP_TO_BT_TXD	=====	MAKE_BASE=TRUE	UART_AP_TO_BT_TXD	12		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		UART_AP_TO_BT_RTS_L	=====	MAKE_BASE=TRUE	UART_AP_TO_BT_RTS_L	12		
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	57	58	59
51		GND	=====	MAKE_BASE=TRUE	GND	59		
51		GND	=====	MAKE_BASE=TRUE	GND	59		
51		GND	=====	MAKE_BASE=TRUE	GND	59		
51		AP_CANARY2	=====	MAKE_BASE=TRUE	AP_CANARY2	55		
51		GND	=====	MAKE_BASE=TRUE	GND	59		
51		GND	=====	MAKE_BASE=TRUE	GND	59		
51		PP1V8_NFC_S2	=====	MAKE_BASE=TRUE	PP1V8_NFC_S2	20		
51		PMU_TO_GNSS_EN	=====	MAKE_BASE=TRUE	PMU_TO_GNSS_EN	55		
51		PMU_TO_BT_REG_ON	=====	MAKE_BASE=TRUE	PMU_TO_BT_REG_ON	55		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		90_PCIE_AP_TO_WLAN_REFCLK_N	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_WLAN_REFCLK_N	9		
51		90_PCIE_AP_TO_WLAN_REFCLK_P	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_WLAN_REFCLK_P	9		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		90_PCIE_AP_TO_WLAN_TXD_P	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_WLAN_TXD_P	17		
51		90_PCIE_AP_TO_WLAN_TXD_N	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_WLAN_TXD_N	17		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		90_PCIE_WLAN_TO_AP_RXD_N	=====	MAKE_BASE=TRUE	90_PCIE_WLAN_TO_AP_RXD_N	17		
51		90_PCIE_WLAN_TO_AP_RXD_P	=====	MAKE_BASE=TRUE	90_PCIE_WLAN_TO_AP_RXD_P	17		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		PP3V0_S2	=====	MAKE_BASE=TRUE	PP3V0_S2	22	38	48
51		PP1V8_TOUCH_RACER_S2	=====	MAKE_BASE=TRUE	PP1V8_TOUCH_RACER_S2	59		
51		PP1V8_TOUCH_RACER_S2	=====	MAKE_BASE=TRUE	PP1V8_TOUCH_RACER_S2	59		
51		PMU_TO_WLAN_REG_ON	=====	MAKE_BASE=TRUE	PMU_TO_WLAN_REG_ON	55		
51		RADIO_PA_NTC	=====	MAKE_BASE=TRUE	RADIO_PA_NTC	23		
51		BT_TO_AP_TIME_SYNC	=====	MAKE_BASE=TRUE	BT_TO_AP_TIME_SYNC	55		
51		UART_BT_TO_AP_RXD	=====	MAKE_BASE=TRUE	UART_BT_TO_AP_RXD	12		

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		UART_BT_TO_AP_CTS_L	=====	MAKE_BASE=TRUE	UART_BT_TO_AP_CTS_L	12		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		PP_VBUS1_E75	=====	MAKE_BASE=TRUE	PP_VBUS1_E75	26	58	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		PP_GPU_LVCC	=====	MAKE_BASE=TRUE	PP_GPU_LVCC	5		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		PP_CPU_PCORE_LVCC	=====	MAKE_BASE=TRUE	PP_CPU_PCORE_LVCC	5		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		PP_BATT_VCC	=====	MAKE_BASE=TRUE	PP_BATT_VCC	59		
51		PP_BATT_VCC	=====	MAKE_BASE=TRUE	PP_BATT_VCC	59		
51		GND	=====	MAKE_BASE=TRUE	GND	59		
51		AP_TO_BT_DEVICE_WAKE	=====	MAKE_BASE=TRUE	AP_TO_BT_DEVICE_WAKE	55		
51		AOP_TO_WLAN_CONTEXT_A	=====	MAKE_BASE=TRUE	AOP_TO_WLAN_CONTEXT_A	13		
51		UART_AOP_TO_RACER_TXD	=====	MAKE_BASE=TRUE	UART_AOP_TO_RACER_TXD	13		
51		SWD_AOP_TO_MANY_SWCLK	=====	MAKE_BASE=TRUE	SWD_AOP_TO_MANY_SWCLK	5	13	19
51		SPI_AP_TO_RACER_MOSI	=====	MAKE_BASE=TRUE	SPI_AP_TO_RACER_MOSI	11		
51		SPI_AP_TO_RACER_SCLK	=====	MAKE_BASE=TRUE	SPI_AP_TO_RACER_SCLK	11		
51		PP1V1_RACER_S2	=====	MAKE_BASE=TRUE	PP1V1_RACER_S2	59		
51		PP1V1_RACER_S2	=====	MAKE_BASE=TRUE	PP1V1_RACER_S2	59		
51		PP1V1_RACER_S2	=====	MAKE_BASE=TRUE	PP1V1_RACER_S2	59		
51		AP_TO_RACER_REF_CLK	=====	MAKE_BASE=TRUE	AP_TO_RACER_REF_CLK	17		
51		GND	=====	MAKE_BASE=TRUE	GND	59		
51		AOP_TO_BBPMU_COEX	=====	MAKE_BASE=TRUE	AOP_TO_BBPMU_COEX	56		
51		PP_VBUS2_IKTARA	=====	MAKE_BASE=TRUE	PP_VBUS2_IKTARA	59		
51		PP_VBUS2_IKTARA	=====	MAKE_BASE=TRUE	PP_VBUS2_IKTARA	59		
51		PP_VBUS2_IKTARA	=====	MAKE_BASE=TRUE	PP_VBUS2_IKTARA	59		
51		PP_VBUS2_IKTARA	=====	MAKE_BASE=TRUE	PP_VBUS2_IKTARA	59		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		AOP_TO_WLAN_CONTEXT_B	=====	MAKE_BASE=TRUE	AOP_TO_WLAN_CONTEXT_B	13		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		UART_RACER_TO_AOP_RXD	=====	MAKE_BASE=TRUE	UART_RACER_TO_AOP_RXD	13		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		SPI_RACER_TO_AP_MISO	=====	MAKE_BASE=TRUE	SPI_RACER_TO_AP_MISO	11		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		SPI_AP_TO_RACER_CS_L	=====	MAKE_BASE=TRUE	SPI_AP_TO_RACER_CS_L	11		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		PMU_TO_IKTARA_RESET_L	=====	MAKE_BASE=TRUE	PMU_TO_IKTARA_RESET_L	23	45	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		SWD_AOP_BT_RACER_SWDIO	=====	MAKE_BASE=TRUE	SWD_AOP_BT_RACER_SWDIO	13		
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		I2C3_AP_SDA	=====	MAKE_BASE=TRUE	I2C3_AP_SDA	11	52	CKPLUS_WAIVE=I2C_PULLUP
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		I2C3_AP_SCL	=====	MAKE_BASE=TRUE	I2C3_AP_SCL	11	52	CKPLUS_WAIVE=I2C_PULLUP
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	
51		GND	=====	MAKE_BASE=TRUE	GND	58	59	

D



B



D

D

C

C

B

B

A

A

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

[illegible]

51 _____ GND ——— MAKE_RACE=TRUE _____ GND 57 58 59 60

51 _____ GND ——— MAKE_RACE=TRUE _____ GND 57 58 59 60